



**KTH Energy and
Environmental Technology**

Thermophysical Properties of Aqueous Solutions Used as Secondary Working Fluids

Doctoral Thesis

By

Åke Melinder

Division of Applied Thermodynamics and Refrigeration
Dept. of Energy Technology
School of Industrial Engineering and Management
Royal Institute of Technology, KTH
Stockholm, Sweden 2007

TRITA REFR Report No 07/60
ISSN 1102-0245
ISRN KTH/REFR/07/60-SE
ISBN 978-91-7178-707-1

Abstract

Secondary working fluids (secondary refrigerants, heat transfer fluids, antifreezes, brines) have long been used in various indirect refrigeration and heat pump systems. Aqueous solutions (water solutions) have long been used as single phase (liquid only) secondary working fluids for cooling in supermarkets, ice rinks, heat recovery systems, heat pumps and other applications. However, aqueous solutions are increasingly used also for freezers in supermarkets and other applications in low temperature refrigeration.

Of importance when comparing different secondary working fluids for indirect systems are the following basic thermo-physical properties: freezing point, density, viscosity, specific heat, and thermal conductivity. Reliable data of these properties are needed to choose suitable fluid and to make technical calculations of an indirect refrigeration or heat pump system.

The main intention of this work has been to select thermophysical property data with good or acceptable technical accuracy of a number of aqueous solutions that can be used by the refrigeration and heat pump industry, rather than focusing on a limited number of property values or scientifically very accurate measuring techniques.

A thorough literature search was in view of this made to find the most reliable property values of aqueous solutions. Detailed literature references are given for thermo-physical properties of the following aqueous solutions, without other additives: Ethylene and propylene glycol, ethyl and methyl alcohol, glycerol, ammonia, potassium carbonate, calcium, lithium, magnesium and sodium chlorides as well as potassium acetate and potassium formate. Some laboratory measurements were made of most of the fluid types when literature values were incomplete or deemed unreliable. Methods used are briefly described and results are given.

Much of the work was reported on in the Engineering Licentiate Thesis: *Thermo-physical properties of liquid secondary refrigerants, A Critical Review on Literature References and Laboratory Measurements* (Melinder 1998a). That material forms the basis for the charts and tables used in the IIR-publication *Thermo-physical properties of liquid secondary refrigerants* (Melinder, 1997). The present thesis reports on an update made since 1998, including re-

view work done on two additional fluids not covered in Melinder (1998a).

The thesis describes how the selection of property values results in tables and charts intended for the industry. Coefficients for polynomial equations are generated from these property values using a Matlab program and this material is intended as a useful tool for computer treatment. Aqueous solution of ethyl alcohol is used as example to see how this process is made. This choice of fluid can also be seen as a test of this method, as the basic thermo-physical properties of aqueous solutions of ethyl alcohol present more challenges than the other fluids examined.

A comparison is made of a few types of aqueous solutions used as secondary working fluids for two types of applications. The first example is bedrock heat pumps and the second is cooling cabinets in a supermarket. An effort is made to see how the additive concentration affects the thermal performance.

Most aqueous solutions used as single-phase secondary fluids can also be used as ice slurry, a fluid consisting of liquid and ice where small ice crystals are produced, usually with some type of ice generator. The ice crystals are then transported to the cooling object from which heat is removed when ice crystals melt. This results in less temperature change in the cooling object and makes it also possible to reduce the volume flow rate and to use smaller pipe dimensions in the system.

In order to choose a secondary fluid for ice slurry use and to make correct technical calculations of the ice slurry system there is a need to examine and evaluate thermo-physical properties and other aspects of ice and of the aqueous solution used. For dimensioning purposes it is of interest to estimate ice mass fraction and enthalpy values and enthalpy-phase diagrams can serve that purpose. This thesis presents enthalpy-phase diagrams made by author that besides isotherms contain lines with ice fraction and lines connecting enthalpies at freezing point and 1, 2, etc. to 10 K below the freezing point curve.

Key words: *Secondary working fluids, secondary refrigerants, indirect systems, aqueous solutions, ice slurry, thermo-physical properties, freezing point, density, viscosity, specific heat, thermal conductivity.*

Acknowledgments

Special thanks to Prof. Björn Palm who has been my supervisor for this thesis work, to Prof. Per Lundqvist who has assisted with fresh ideas and to Prof. Emeritus Eric Granryd, who for fifteen years was my supervisor in this research work and all the time has contributed to the development of the work!

Special thanks also to Inga Du Rietz who has provided kind and patient assistance in so many ways. Thanks to Anders Johansson and Fredrik Lagergren who encouraged me to write this thesis.

I also want to express my thanks and appreciation to all other colleagues at the Division of Applied Thermodynamics and Refrigeration, incl. Benny Andersson, Jaime Arias, Getachew Bekele, Erik Björk, Yang Chen, Joachim Claesson, Primal Fernando, Martin Forsén, Cecilia (Hägg) Fransson, Richard Furberg, Marino Grozdek, Peter Hill, Hans Jonsson, Nabil Kassem, Rahmat Khodabandeh, Claudi Martin Callizo, Jan-Erik Nowacki, Wahib Owhaib, Ali Rashid, Oxana Samoteeva, Samer Sawalha and Benny Sjöberg.

Prof. Torsten Fransson and others at the Dept. of Energy Technology have also been helpful in various ways! Tony Chapman and Birger Söderström have helped me with really good computer support.

Special thanks to Prof. Jesper Oppelstrup, NADA, KTH, who with kindness and great patience over the years has assisted me to establish equations and coefficients for computer treatment!

Warm thanks also to Ove Andersson and Åke Fransson at Umeå university whose laboratory tests have contributed to this work.

The Int. Institute of Refrigeration, IIR, and The Swedish Society of Refrigeration published the original material in 1997 in handbooks for the industry. The Swedish Energy Agency has provided some funds used for finishing this thesis work.

Last but not least my wife Ningsih and my children Daniel, Esther and Susanna have in a kind and patient way supported me and my work during this time!

List of publications

Journal paper

- Melinder, Å., Granryd, E., 2005, *Using property values of aqueous solutions and ice to estimate ice concentrations and enthalpies of ice slurries*, Int. Journal of Refrigeration, Volume 28, Number 1, p.13-19, Paris.

Conference papers

- Melinder, Å., 1998, *A Critical Review on Thermophysical Properties of Liquid Secondary Refrigerants*, Natural Working Fluids'98, IIR – Gustav Lorentzen Conf., Oslo.
- Melinder, Å., Granryd, E., 2001, *Using property values of water solutions and ice to estimate ice concentration and enthalpy values of ice slurries*, Third IIR Workshop on Ice Slurries, Lucern.
- Melinder, Å., 2001, *Enthalpy-phase diagrams as an aid in estimating the performance of ice slurries*, Fourth IIR Workshop on Ice Slurries, Osaka
- Melinder, Å., 2002a, *Enthalpy-phase diagrams of aqueous solutions for ice slurry applications*, Fifth IIR Workshop on Ice Slurries, Stockholm, May.
- Melinder, Å., 2002b, *Influence of temperature on latent heat benefits in ice slurry applications*, Zero leakage – Minimum Charge, IIF/IIR, Stockholm.
- Melinder, Å., 2002c, *Estimating the "transport capability" of ice slurries with the aid of enhanced enthalpy-phase diagrams*, Fifth Gustav Lorenzen Conference on Natural Fluids, IIF/IIR, Guangzhou.
- Melinder, Å., 2003a, *Comparing Properties of Aqueous Solutions for Liquid Only and Ice slurry Applications of Indirect Systems*, Proc. 21st Int. Congress of Refrigeration, IIR/IIF, Washington.
- Melinder, Å., 2003b, *Update on indirect systems and secondary fluids with focus on supermarket refrigeration*, Int. Conf. on Fluid and Thermal Conversion, FTEC2003, Bali, Indonesia.

- Melinder, Å., 2003c, *Comparing Thermophysical Properties of Liquid Only and Ice slurry Secondary Fluids*, Int. Conf. on Fluid and Thermal Conversion, FTEC2003, Bali, Indonesia
- Melinder, Å., 2004, *Secondary fluids for low operating temperatures*, Sixth Gustav Lorenzen Conference on Natural Working Fluids, IIF/IIR, Glasgow, UK.
- Melinder, Å., 2005, *Properties of various secondary fluids for supermarket applications*, Thermophysical Properties and Transfer Processes of Refrigerants, IIF/IIR, Vicenza, Italy
- Melinder, Å., Hägg, C., 2005, *Enthalpy-phase diagrams and the seesaw-principle – aids to describe various ice slurry conditions*, Thermophysical Properties and Transfer Processes of Refrigerants, IIF/IIR, Vicenza, Italy.
- Melinder, Å., 2006a, *Updating and extending secondary fluid data for various refrigeration and heat pump applications*, Seventh Gustav Lorenzen Conference on Natural Working Fluids, IIF/IIR, Trondheim
- Melinder, Å., 2006b, *Equilibrium properties of ice slurry based on aqueous solutions of ethyl alcohol*, Seventh Gustav Lorenzen Conference on Natural Working Fluids, IIF/IIR, Trondheim
- Melinder, Å., 2006c, *Equilibrium properties of ice slurry based on two chosen types of aqueous solutions*, Seventh IIR Conference on Phase Change Materials and Slurries for Refrigeration and Air Conditioning, Dinan, France
- Melinder, Å., 2006d, *Updating properties of secondary working fluids for various refrigeration systems*, Int. Conf. on Fluid and Thermal Conversion, FTEC2006, Jakarta

Handbooks, thesis works and reports

- Melinder, Å., 1985, *Köldbärare för värmepumptillämpningar. Studier av termodynamiska egenskaper och korrosion*; Report R114:1985 to the Swedish Council for Building Research
- Melinder, Å., Berendson, J., Granryd, E., Kyrk, B., 1989, *Köldbärare för värmepumptillämpningar. Termodynamiska egenskaper och korrosion för kaliumkarbonat samt jämförelse med andra köldbärare*, Report R18:1989 to the Swedish Council for Building Research.

- Melinder, Å. Granryd, E., 1992, Secondary Refrigerants for Heat Pumps and Low Temperature Refrigeration. A comparison of thermodynamic properties of aqueous solutions and non-aqueous liquids, Trita-REFR Report No 92/6, ISSN 1102-0245
- Melinder, Å., 1994, Thermodynamic Properties for Secondary Refrigerants - Tables and Charts, Handbook No. 12 of the Swedish Society of Refrigeration.
- Melinder, Å., 1997, Thermo-physical properties of liquid secondary refrigerants. Tables and Diagrams for the Refrigeration Industry, IIF/IIR, Paris
- Melinder, Å., 1997, Thermo-physical properties of liquid secondary refrigerants. Handbook Nr. 12, Swedish Society of Refrigeration
- Melinder, Å., 1998, *Thermophysical properties of liquid secondary refrigerants, A Critical Review on Literature References and Laboratory Measurements*, (Tekn. Lic. Thesis), KTH, Stockholm

TABLE OF CONTENTS

1 Secondary working fluids used in indirect systems	1
1.1 Indirect systems with secondary working fluids	1
1.1.1 Reasons for the move toward indirect systems	2
1.1.2 Temperature profiles of an indirect system	3
1.2 Advantages and drawbacks of indirect systems	4
1.2.1 Advantages of indirect systems compared to direct ones	4
1.2.2 Drawbacks of indirect systems in comparison with direct ones	5
1.3 Choice of secondary working fluid	5
1.3.1 What is involved in choosing a suitable secondary working fluid?	5
1.3.2 What requirements are to be fulfilled by an ideal secondary working fluid?	6
2 Basic thermophysical properties of aqueous solutions – description and comparison of fluids	7
2.1 Introduction	7
2.2 Freezing point, eutectic point and additive concentration	8
2.3 Density, ways of measuring density, mixing proportions and thermal volume expansion	10
2.3.1 Density	10
2.3.2 Ways of measuring density, index of refraction and mixing proportions	11
2.3.3 Thermal volume expansion	13
2.4 Specific heat and volumetric heat capacity	14
2.5 Thermal conductivity	16
2.6 Dynamic and kinematic viscosity	18
2.7 Boiling point	18
2.8 Surface tension	20
2.9 Conclusions	20
2.10 Nomenclature	22
3 Selection of property values of aqueous solutions	23
3.1 Summary	23

3.2 Summary of critical evaluation of thermo-physical property values given in Tech. Lic. Thesis	24
3.2.1 List of literature references for the evaluation of property values	25
3.2.2 Reference details and symbols used	30
3.3 Critical evaluation of some thermo-physical property values	31
3.3.1 Critical evaluation of specific heat of ethyl alcohol	31
3.3.2 Critical evaluation of viscosity of calcium chloride	33
3.3.3 Critical evaluation of property values of lithium chloride	37
3.3.4 Critical evaluation of property values of potassium formate	43
3.4 Measurements of some properties	50
3.4.1 Measurements of viscosity at KTH	50
3.4.2 Measurements of specific heat and thermal conductivity at Umeå university	52
4 Making thermophysical properties of aqueous solutions available for computer use	53
4.1 Introduction	53
4.2 Results of selection of property values of aqueous solutions using ethyl alcohol as example	53
4.3 Polynomial equations for computer treatment	56
4.4 Comparison of basic values from literature search with polynomial equation values for ethyl alcohol	58
4.5 Presenting aqueous solution values of ethyl alcohol.....	61
4.6 Presenting aqueous solution values of lithium chloride ...	67
4.5 Presenting aqueous solution values of potassium formate	72
5 Comparison of aqueous solutions as to heat transport, pressure drop and heat transfer	77
5.1 Introduction	77
5.2 Equations for heat transport, pressure drop and heat transfer	78
5.2.1 Heat transport and volumetric heat capacity	78
5.2.2 Friction coefficient, pressure drop and pumping power	78
5.2.3 Heat transfer and temperature differences	79
5.3 Comparison for a bedrock heat pump application	80

5.3.1 Example used for comparison of fluid types and concentrations	80
5.3.2 Heat transport ability and volumetric heat capacity	81
5.3.3 Viscosity and type of flow	81
5.3.4 Friction coefficient, pressure drop and pumping power	83
5.3.5 Heat transfer and temperature differences	85
5.4 Comparison for a supermarket cooling application	85
5.4.1 Example used to compare properties of aqueous solutions for cooling	85
5.4.2 Heat transport ability and volumetric heat capacity	87
5.4.3 Viscosity and type of flow	87
5.4.4 Friction coefficient, pressure drop and pumping power ...	87
5.4.5 Heat transfer and temperature differences	91
5.5 Discussion	91
5.6 Nomenclature	91
6 Basic thermophysical properties of aqueous solutions used as ice slurry	93
6.1 Introduction	93
6.2 Thermophysical properties of ice	94
6.3 Freezing point, ice fraction and density of ice slurries in equilibrium	97
6.3.1 Freezing point and ice mass fraction	97
6.3.2 Density and volumetric ice fraction	99
6.4 Heat transport ability and enthalpy change of aqueous solutions	101
6.4.1 Heat transport ability, enthalpy change and apparent specific heat	101
6.4.2 Heat transport ability of ice slurries and the volumetric enthalpy change.....	102
6.5 Enthalpy phase diagrams of aqueous solutions	103
6.5.1 Constructing enthalpy phase diagrams of aqueous solutions.....	103
6.5.2 Example of simple adiabatic ice slurry processes in equilibrium	105
6.5.3 Examples of simple adiabatic ice slurry mixing processes	109
6.6 Viscosity of ice slurries in equilibrium	111
6.7 Thermal conductivity of ice slurry	113
6.8 Nomenclature	115

7 Summary and conclusions.....	116
References	118

APPENDIX

Table 2.3. Conversion chart of additive concentrations	125
Table 2.4. Index of refraction of aqueous solutions as function of the additive concentration	125
Table 2.5. Thermal volume expansion of aqueous solutions with concentrations giving $t_F = -15^\circ\text{C}$ and water ...	126
Measuring results of viscosities	127
Calcium chloride - water	127
Lithium chloride - water	128
Potassium formate - water	129

APPENDIX PAPERS:

- PAPER 1:** Melinder, Å., 1998, *A Critical Review on Thermophysical Properties of Liquid Secondary Refrigerants*, Natural Working Fluids'98, IIR – Gustav Lorentzen Conf., Oslo
- PAPER 2:** Melinder, Å., 2002, *Enthalpy-phase diagrams of aqueous solutions for ice slurry applications*, Fifth IIR Workshop on Ice Slurries, Stockholm
- PAPER 3:** Melinder, Å., 2003, *Update on indirect systems and secondary fluids with focus on supermarket refrigeration*, Int. Conference on Fluid and Thermal Conversion, FTEC2003, Bali, Indonesia
- PAPER 4:** Melinder, Å., 2003, *Comparing Properties of Aqueous Solutions for Liquid Only and Ice slurry Applications of Indirect Systems*, Proc. 21st Int. Congress of Refrigeration, IIR/IIF, Washington
- PAPER 5:** Melinder, Å., 2004, *Secondary fluids for low operating temperatures*, Sixth Gustav Lorentzen Conference on Natural Working Fluids, IIF/IIR, Glasgow
- PAPER 6:** Melinder, Å., Granryd, E., 2005, *Using property values of aqueous solutions and ice to estimate ice concentrations and enthalpies of ice slurries*, Int. Journal of Refrigeration, Volume 28, Number 1, p.13-19, Paris

- PAPER 7:** Melinder, Å., 2005, *Properties of various secondary fluids for supermarket applications*, Thermophysical Properties and Transfer Processes of Refrigerants, IIF/IIR, Vicenza
- PAPER 8:** Melinder, Å., 2006, Updating and extending secondary fluid data for various refrigeration and heat pump applications, GLC 2006, Trondheim
- PAPER 9:** Melinder, Å., 2006, Equilibrium properties of ice slurry based on two chosen types of aqueous solutions, 7th Conference on Phase Change Materials and Slurries for Refrigeration and Air Conditioning, 2006, Dinan, France

1 Secondary working fluids used in indirect systems

1.1 Indirect systems with secondary working fluids

Secondary working fluids (secondary refrigerants, heat transfer fluids, antifreezes, brines) have long been used in various indirect refrigeration and heat pump systems. One secondary working fluid circuit transports heat from the cooling object or heat source to the evaporator and another may cool the condenser. Many terms are used for the fluids we here consider.

Many terms are used for the fluids we here consider. Why has *secondary working fluid* been chosen for this publication? The term is not restricted to just one function of the fluid, such as *heat carrier*, *heat transfer fluid* or *antifreeze*. It draws attention to an indirect or secondary system without using the often misunderstood term *secondary refrigerant*. The commonly used term *brine* should strictly only be used about aqueous salt solutions. The term *secondary working fluid* is in agreement with the Call for papers to the 22nd Int. IIR Congress of Refrigeration in Beijing, August 2007.

Aqueous solutions (water solutions) have long been used as single phase (liquid only) secondary working fluids for cooling in supermarkets, for ice rinks, heat recovery systems, heat pumps and other applications. However, aqueous solutions are increasingly used also for freezers in supermarkets and other applications in low temperature refrigeration. Aqueous solutions are among else also used in sorption based air conditioning equipment for dehumidification of air, usually operating at rather high temperatures (Arias 1999, Conde 2004, Kruse 2005, Lundqvist 2000, Mao and Hrnjak 1999, Pearson 1993).

Figure 1.1 shows the principles of a completely indirect cooling system, i.e. for a supermarket. In such an indirect system, the primary refrigerant can be confined to a compact chiller unit in the machine room. One secondary working fluid (1) cools the condenser and the dry cooler is usually placed outdoors. Another secondary working fluid (2) is cooled in the evaporator and used for cooling cabinets and cold storage.

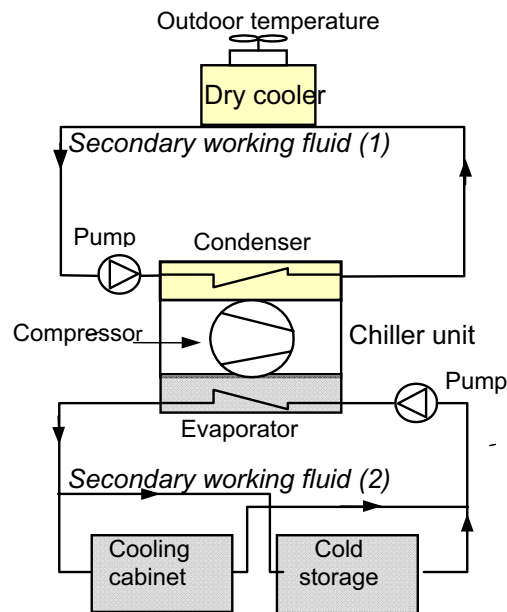


Figure 1.1 Completely indirect system

Research has the last decade been made with aqueous solutions as two-phase ice slurry for cooling in supermarkets, cooling of fish, shrimps and vegetables, transport cooling, energy storage as well as other applications. (Aittomäki 1999, Ben Lakhdar 2002, Egolf and Kauffeld 2005, Field and Kauffeld 2003, Fukusako 1999, Granryd 2002, Kauffeld, Egolf and Kawaji 2005, Malter 1996, Paul 1997, Snoek 1993, Wang 1999).

1.1.1 Reasons for the move toward more indirect systems

Indirect refrigeration plants and heat pumps with secondary working fluids are more and more used in in many countries. What are the reasons? The discovery that some refrigerants contribute to global environmental effects has led to the phasing out of CFC refrigerants and also an increased awareness that the use of any primary refrigerant should preferably be kept to a minimum. Large primary refrigerant leaks in supermarkets and large heat pumps have also focused attention on the use of indirect systems with secondary fluids.

With an indirect system it is possible to manufacture large quantities of factory built cooling units that can be used for a variety of applications. It is also possible to design the primary refrigeration unit as a compact chiller in a machine room with an extremely small refrigerant charge. All refrigerant piping work can be performed at the factory with minimal risk of leakage. Local construction of refrigerant piping can be avoided and installation work can be made in a more simple way (Lindborg 2003, Lundqvist 2000). See also [PAPER 3](#) (Melinder 2003b).

1.1.2 Temperature profiles of an indirect system

Before mentioning pro's and con's of an indirect refrigerating system, let's consider a schematic figure with temperature profiles of an indirect refrigerating system with temperature profiles of the evaporator side as given in [Figure 1.2](#).

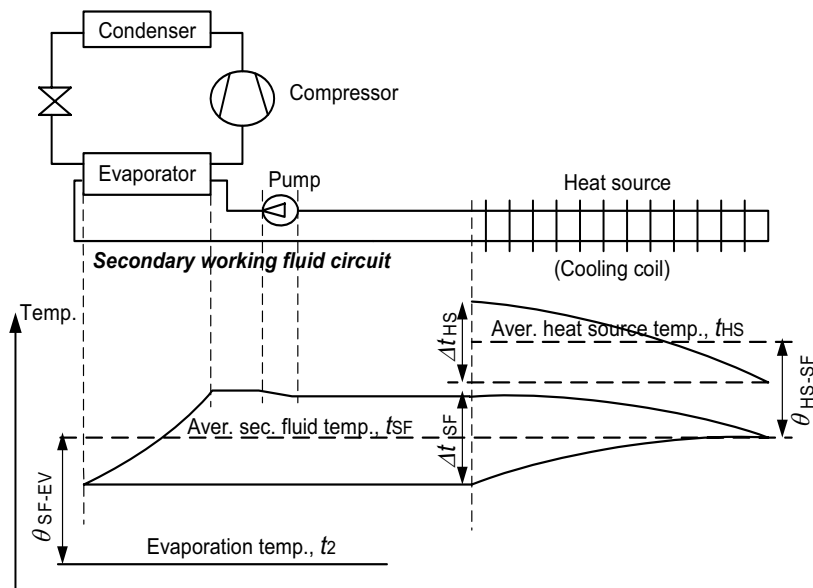


Figure 1.2. Scheme of indirect refrigerating system with temperature profiles of evaporator side

The following temperatures are indicated: evaporating temperature of refrigerant, t_2 , average temperature of heat source, t_{HS} , and average temperature of secondary working fluid, t_{SF} . The following temperature differences are indicated: average secondary working

fluid temperature change, Δt_{SF} , the difference between heat source temperature and average secondary fluid temperature, $\theta_{2,HS} = t_{HS} - t_{SF}$ and the difference between average secondary fluid temperature and evaporating temperature, $\theta_{2,EV} = t_{SF} - t_2$. (Compare Arias, Lundqvist 1999).

The additive concentration of the secondary working fluid is usually chosen to give a freezing point below the lowest temperature in the secondary fluid loop. This will be considered more in Chapter 5.3. (Compare Granryd and Melinder 2005, Melinder 1997).

Indirect systems with a secondary working fluid circuit may have a number of advantages but also some drawbacks in comparison with direct expansion (DX) systems depending on how the systems are built.

1.2 Advantages and drawbacks of indirect systems

1.2.1 Advantages of indirect systems compared to direct ones

- The primary refrigeration unit is usually designed as a factory built compact chiller in a machine room with an extremely small refrigerant charge. Such cooling units can be built in larger quantities as they can be used for a variety of purposes.
- Local construction of primary refrigerant piping is avoided and installation work is made in a more simple way. Such prefabricated chillers of high quality and with correct charge minimize refrigerant leakage. Equipment in contact with the refrigerant is confined to the machine room.
- High performing refrigerants, such as ammonia and hydrocarbons, may be used.
- Flooded evaporators in chillers for indirect systems can be used to 100 % while thermostatic expansion valves used in direct systems may limit the effective evaporator surface down to some 70 %.
- There are less technical limitations for the distance between cooled spaces and the machine room. Plastic piping may be used for secondary working fluid circuits.

- A more even temperature distribution gives more uniform formation of frost. Warm secondary fluid can be used for defrosting which shortens the time needed for each defrosting, gives longer intervals between defrosting periods and more even and favorable product temperatures.
- Indirect systems are usually designed and built as large systems and variations in load can be evened out and shared within different parts of a large system.
- The energy consumption of a well designed indirect system may over a year even be less than of a traditionally built direct system, partly also due to more even temperature distribution at the cooling object and fewer defrosts needed!

1.2.2 Drawbacks of indirect systems in comparison with direct ones

- The investment cost for an indirect system is usually higher than a direct (DX) system for the same application, as it introduces the added cost for pump, heat exchanger and secondary fluid, although the cost of primary refrigerant charge is much less.
- Added temperature difference is introduced, lowering the evaporation temperature as well as the coefficient of performance somewhat.
- Added pumping power, though it can be quite small compared to the cooling capacity.

In spite of these drawbacks of indirect systems in comparison with direct ones, the earlier mentioned advantages have been main reasons for the move toward more indirect systems. (Aittomäki 2003, Lindborg 2003, Lundqvist 2000, Mao and Hrnjak 1999, Melinder 2003b, Ure 2003, Yang 2001).

1.3 Choice of secondary working fluid

1.3.1 What is involved in choosing a suitable secondary working fluid?

On the warm side, for the condenser, water can usually be used as a secondary working fluid. Water is an excellent secondary working fluid for air conditioning and other applications where temperatures down to about +3°C are enough. The problem is to find flu-

ids for applications below 0°C, and the challenge is especially great for freezer applications.

There is a need for a good knowledge of thermophysical properties and other aspects of available secondary working fluids in order to make a good choice of fluid and to make reliable technical calculations of indirect refrigeration and heat pump systems. This thesis is meant to fill this need and to point out the requirements of such fluids.

1.3.2 What requirements are to be fulfilled by an ideal secondary working fluid?

The fluid should possess good thermophysical properties, making it possible to:

- Give sufficient freezing protection, yet not have higher concentration than necessary, in order to benefit more from the good properties of water.
- Transport large refrigeration capacity through pipes or tubes with a certain volume flow and temperature change of the fluid.
- Give good heat transfer leading to small temperature differences at heat transfer in evaporator and cooling object or heat source.
- Give small pressure drop for the system fluid flow in order to be able to use a simple pump with small pumping power.

The fluid should also be:

- Environmentally acceptable, non-toxic, biodegradable. Not flammable or dangerous to handle and be available at a reasonable price.
- Compatible with materials used in the system so that it does not give cause to problems in the plant due to corrosion.

No secondary working fluid fulfills all the mentioned requirements fully. We should find out which of the parameters that are especially important in the application we are considering and then try to choose the fluid that is best suited for that particular case (Granryd and Melinder 2005, Melinder 1997).

2 Basic thermophysical properties of aqueous solutions – description and comparison of fluids

2.1 Introduction

Of importance when comparing different secondary working fluids (secondary refrigerants, heat transfer fluids, brines antifreezes) for indirect systems are the following basic thermophysical properties: freezing point, density, viscosity, specific heat, and thermal conductivity. Reliable data of these properties are also needed to choose suitable fluid and to make technical calculations of an indirect refrigeration or heat pump system. Many handbooks over the decades have contained tables and/or charts with thermophysical properties of various secondary fluids (ASHRAE Fundamentals 2001, Bäckström 1965, Carrier's Handbook of Air Conditioning System Design 1965, CRC Handbook of Chemistry and Physics 1988-89, DOW The Glycols 1974, Hofmann 1959, Int. Critical Tables 1928-29, Plank 1988, Raznevic 1976, VDI Wärmeatlas 2002).

In order to mix the aqueous solutions to the right proportion this chapter also presents a conversion chart and table of additive concentrations in weight-% and volume-% (Figure 2.2 and Table 2.2). The chapter also gives some chemical information to determine the density, such as, molar concentration, mixing masses and refractive index of aqueous solutions. Thermal volume expansion is also given as an aid to determine the size of expansion vessels.

The types of aqueous solutions included and symbols used are given in Figure 2.1. A comparison of the fluid properties is here as an example given for additive concentrations of the solutions giving the freezing point $t_F = -15^\circ\text{C}$. This freezing point is often used in heat pumps with ground or rock as heat source and they may be used for many cooling applications.

A brief definition and description of each of the following basic thermophysical properties and some other properties is also given (CRC Handbook 1986-87, Geiringer 1962, Melinder 1997).

2.2 Freezing point, eutectic point and additive concentration

The **freezing point temperature**, t_F [°C], here means the temperature at which ice crystals begin to form in equilibrium if there is no sub-cooling. The freezing point curve is then the solubility curve of ice in the solution. It represents the temperatures at which ice and solution are in equilibrium for various additive concentrations. The freezing point should be below the lowest expected temperature and sufficiently ($\approx 10\text{K}$) below the normal operating temperature of the secondary working fluid so that the fluid without difficulty can be pumped through the system.

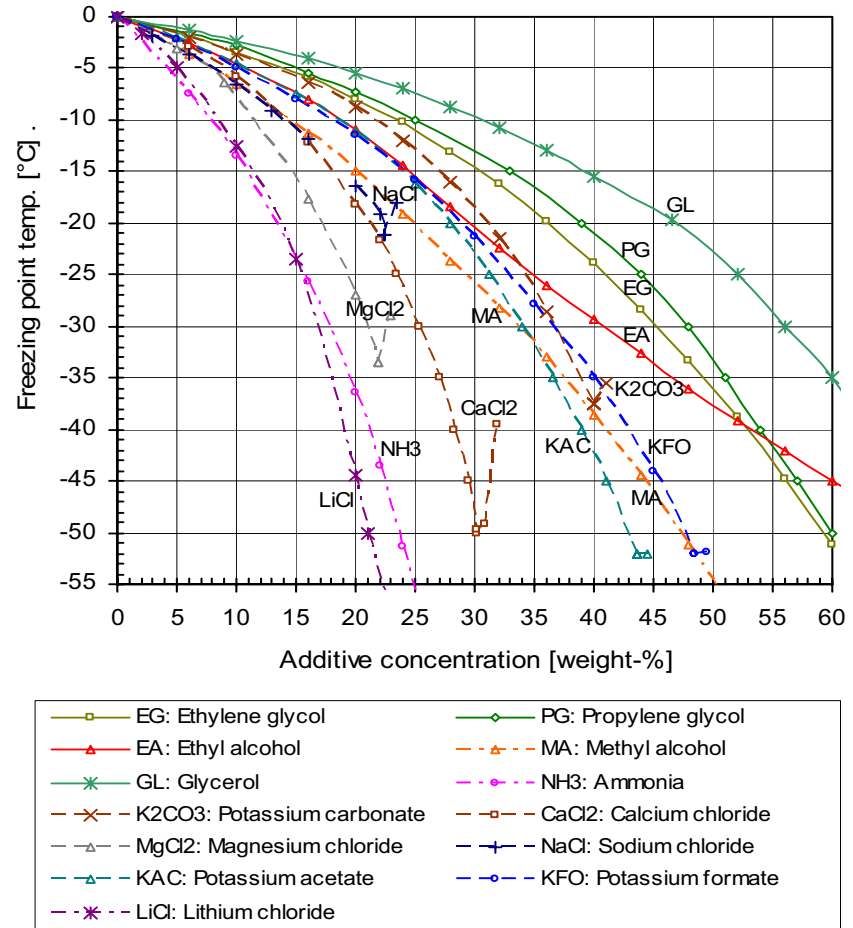


Figure 2.1. Freezing point temperature, t_F , of aqueous solutions as function of additive concentration by weight, c_A

Figure 2.1 gives freezing point curves of aqueous solutions as function of the concentration. Note the abbreviations or symbols used for the fluids in Figure 2.1. They are also used in several of the following figures and tables in chapters 2 and 5. Some fluids (NaCl, MgCl₂, CaCl₂, K₂CO₃) have well defined **eutectic temperature**, t_E [°C], as can be seen in the figure. The eutectic temperature is the highest temperature at which full solidification can be achieved (Fennema 1973).

The aqueous solutions do not usually freeze to solid ice when ice crystals begin to form and mix with the liquid. The risk for serious damage due to freezing up of a tube or heat exchanger is small with aqueous solutions. Total freezing takes place first when the temperature goes down to the eutectic point for the mixture. (Granryd and Melinder 2005). ASHRAE Fundamentals (2005) recommends at least 30% ethylene glycol to avoid damage of the system. However, if we know the lowest fluid temperature in the system, we may conclude that less additive concentration can be chosen. Compare discussion in chapter 5.3.

Note also in Figure 2.1 that the temperature often rises sharply at concentrations above the eutectic, as is the case with the chlorides where the over-eutectic curve is the solubility curve of additive in the solution.

Table 2.1. Chemical data, additive concentrations for $t_F = -15^\circ\text{C}$ and approximate eutectic points of aqueous solutions

Type of solution	Chem. symbol	Mol. weight	wt-% $t_F = -15^\circ\text{C}$	Appr. eutectic temp./ conc.
Ethylene glycol (EG)	C ₂ H ₆ O ₂	62.07	30.5	≈ -51°C/57%
Propylene glycol (PG)	C ₃ H ₈ O ₂	76.09	32.9	≈ -50°C/60%
Ethyl alcohol (EA)	C ₂ H ₆ O	46.07	24.5	≈ -118°C/93.5%
Methyl alcohol (MA)	CH ₄ O	32.03	20.0	≈ -116°C/87.7%
Glycerol (GL)	C ₃ H ₈ O ₃	92.09	39.2	≈ -47°C/67%
Ammonia	NH ₃	17.03	10.8	≈ -100°C/33%
Potassium carbonate	K ₂ CO ₃	138.20	27.0	≈ -37.5°C/40%
Calcium chloride	CaCl ₂	110.99	17.9	≈ -50°C/30.5%
Magnesium chloride	MgCl ₂	95.23	14.7	≈ -33°C/21.8%
Sodium chloride	NaCl	58.44	18.8	≈ -21°C/23.4%
Potassium acetate (KAC)	KC ₂ O ₂ H ₃	98.15	23.9	≈ -52°C/45%
Potassium formate (KFO)	KCO ₂ H	84.12	24.0	≈ -51°C/48%
Lithium chloride	LiCl	42.40	11.2	≈ -75°C/25%

Some of the organic aqueous solutions such as propylene glycol do not have a clear eutectic point but rather seem to enter a glassy state at very low temperatures. This seems to be the case also with some organic potassium salt mixtures.

Table 2.1 lists the aqueous solutions and gives the additive's chemical symbol, molecule weight and the approximate temperature and additive mass fraction or concentration of the eutectic solutions.

Note that Table 2.1 lists additive concentrations for the freezing point $t_F = -15^\circ\text{C}$ as the rest of this chapter makes a comparison of thermal properties for aqueous solutions with $t_F = -15^\circ\text{C}$.

In order to mix an aqueous solution to the right proportions it is often the concentration by volume rather than by weight, volume fraction rather than mass fraction of the additive, that is asked for or needed. This is especially true when the additive is in liquid form, as is the case with glycols and alcohols.

Figure 2.2 is based on ASHRAE Handbook - Fundamentals (2005) and Mellan (1977) and converts weight-% to volume-% for aqueous solutions of ethylene glycol, propylene glycol, ethyl alcohol and glycerol. For ethylene glycol and glycerol the concentration in volume-% is lower than in weight-%. For propylene glycol the values are about the same while for ethyl alcohol the concentration in volume-% is up to 8 % higher than in weight-%. For detailed values, see Appendix 1, Table 2.2.

2.3 Density, ways of measuring density, mixing proportions and thermal volume expansion

2.3.1 Density

Density of a homogenous material is the concentration of matter, measured by mass per unit volume. Correct **density**, ρ [kg/m^3], values of a secondary working fluid are important for several reasons. The concentration of a known solution is often measured with gas chromatography or refractive index. It may also be determined by checking the density with an aerometer (densimeter) or by weighing a known volume of the fluid or some other method. Most methods are today usually performed at 20°C .

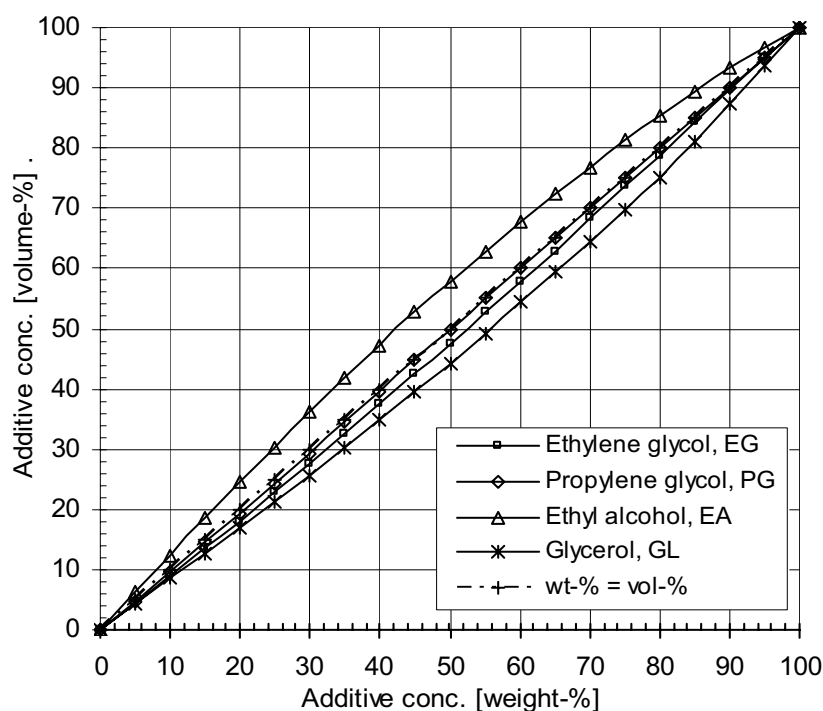


Figure 2.2. Conversion chart of additive concentration

Figure 2.3 gives density curves of aqueous solutions at 20°C as function of the additive concentration. The increase of density with additive concentration is rapid for the salts while the increase for the glycols is quite slow. Only the density of alcohols and ammonia is lower than that of water. The aerometer method to measure density is not very accurate with glycols as their density is near that of water and as inhibitors and other substances added to a commercial product usually changes the density. Glycol concentrations may be determined more accurately by index of refraction or gas chromatography. See also Table 2.3 in Appendix.

2.3.2 Ways of measuring density, index of refraction and mixing proportions

The refractive index, also called **index of refraction**, n , is defined as the ratio of the velocity of light in a vacuum to the velocity of light of a certain wave length in a given substance. The index of refraction of a fluid is then the measure of refraction experienced by a ray as it passes through a refracting interface, a surface separating two media of different densities.

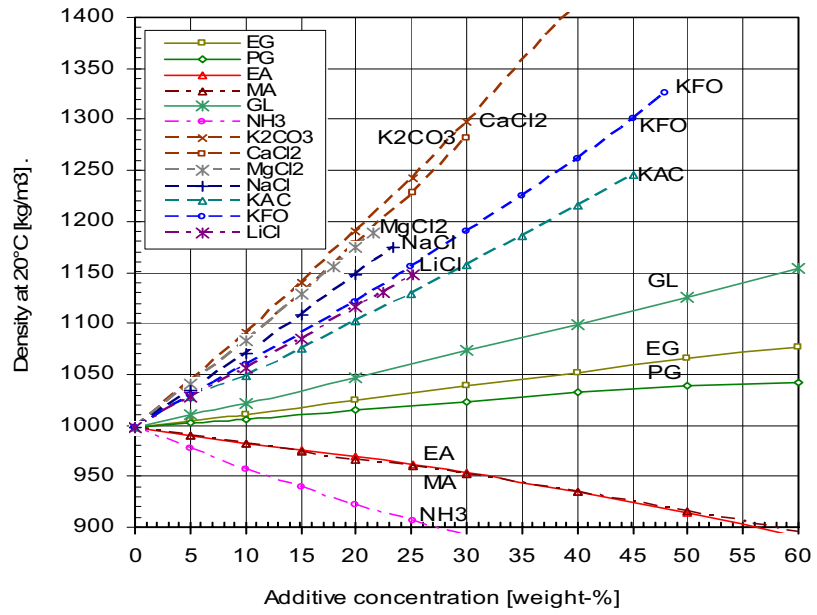


Figure 2.3. Density at 20°C, $\rho_{20^\circ\text{C}}$, as function of additive concentration by weight, c_A

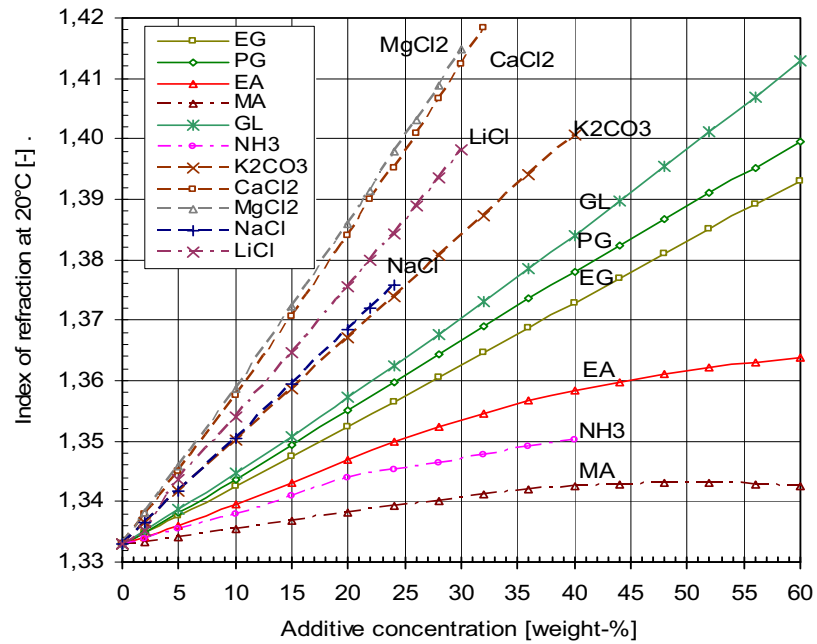


Figure 2.4. Index of refraction at 20°C, $n_{20^\circ\text{C}}$, as function of additive concentration by weight, c_A

Figure 2.4 gives values of the index of refraction of aqueous solutions at 20°C, $n_{20^\circ\text{C}}$, as function of the additive concentration. The increase with additive concentration is largest with salts and smallest with methyl alcohol and ammonia. As mentioned, refractive index is a common way to measure concentrations of glycols. For detailed values, see Table 2.4 in Appendix.

However, it should be possible to measure the additive concentration of any of these fluids with a “glycol meter” that has a scale for refractive index.

Table 2.2 is given as an aid to mix additive and water in the right proportions to get a certain freezing point. The table gives mixing proportions and values of refractive index, $n_{20^\circ\text{C}}$, of each of the aqueous solutions at 20°C, all as a function of the additive concentration that corresponds to the freezing point $t_F = -15^\circ\text{C}$. Table 2.2 also lists molar concentration and is mainly constructed by interpolation of values in CRC Handbook of Chemistry and Physics (1986-87). Values are not available for the organic potassium salts. For explanation of fluid symbols or abbreviations see Figure 2.1.

Table 2.2. Density, molar conc., mixing masses and refractive index of aqueous solutions with conc. giving $t_F = -15^\circ\text{C}$.

Type of additive	Additive Conc.	Density at 20°C	Molar Conc.	Additive Mass	Water Mass	Index of Refraction
	C_A wt-%	$\rho_{20^\circ\text{C}}$ kg/m ³	M kg-mol/l	C_s kg/m ³	C_w kg/m ³	$n_{20^\circ\text{C}}$ -
EG	30.46	1038.4	5.097	316.4	722.1	1.363
PG	32.90	1025.5	4.434	337.4	688.1	1.370
EA	24.55	962.4	5.128	236.3	726.2	1.350
MA	19.98	966.6	6.029	193.1	773.4	1.338
GL	39.17	1096.2	4.663	429.5	666.7	1.383
NH3	10.83	954.4	6.071	103.4	851.1	1.339
K2CO3	27.00	1264.7	2.473	341.8	922.9	1.379
CaCl2	17.92	1157.2	1.869	207.4	949.8	1.378
MgCl2	14.68	1125.2	1.736	165.3	959.9	1.371
NaCl	18.82	1138.3	3.665	214.2	924.10	1.366
LiCl	11.20	1062.9	2.809	119.1	943.8	1.357

2.3.3 Thermal volume expansion

The needed size of an expansion vessel can be determined by calculating the coefficient of thermal volume expansion. The expansion of a volume from V_0 to V_T over the temperature change ΔT

can with the coefficient of thermal volume expansion β [1/K] be expressed as:

$$V_T = V_0(1 + \beta \Delta T) \quad (2.1)$$

The value of β is a function of temperature. It is convenient for practical reasons to use the concept of **thermal volume expansion** of a fluid, TVE [%], estimated from the freezing point, t_F , to a chosen temperature, t . It can be calculated with the relation

$$TVE = 100 \cdot (\rho_F / \rho - 1) \quad (2.2)$$

where ρ_F is the density at the freezing point, t_F , and ρ is the density at a chosen temperature, t .

The largest expected volume change, ΔV [m³], from the freezing point, t_F , to a chosen temperature, t , is then for a volume V [m³]:

$$\Delta V = V \cdot (TVE/100) \quad (2.3)$$

Figure 2.5 gives as an example TVE curves for concentrations of the aqueous solutions giving $t_F = -15^\circ\text{C}$ and water (Compare Table 2.1). The figure shows that TVE from the freezing point $t_F = -15^\circ\text{C}$ to $t = 30^\circ\text{C}$ for all the aqueous solutions are between 1% and 2%. The TVE from the freezing point $t_F = -15^\circ\text{C}$ to $t = 100^\circ\text{C}$ is for glycols about 6%. The volume change between any two temperatures in the chart can be estimated by noting the values between these temperatures. For detailed values, see Table 2.5 in Appendix.

Thermal volume expansion is of interest for designers of heat pump and refrigeration systems as an aid to determine the size of expansion vessels.

2.4 Specific heat and volumetric heat capacity

The **specific heat** of a substance is simply the quantity of thermal energy (in J) required to raise the temperature of one unit mass (1 kg) of a substance one temperature unit (1 K). A high value of a fluid's specific heat, c_p [J/kgK], its thermal capacity, is favorable as it affects mass flow rate as well as the heat transfer coefficient.

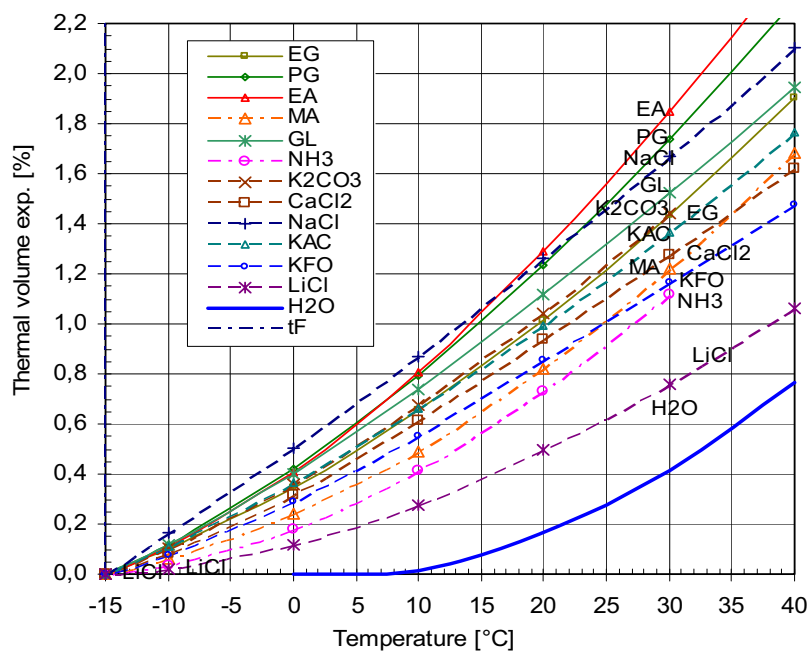


Figure 2.5. Thermal volume expansion, TVE , ($t_F = -15^\circ\text{C}$) as function of temperature

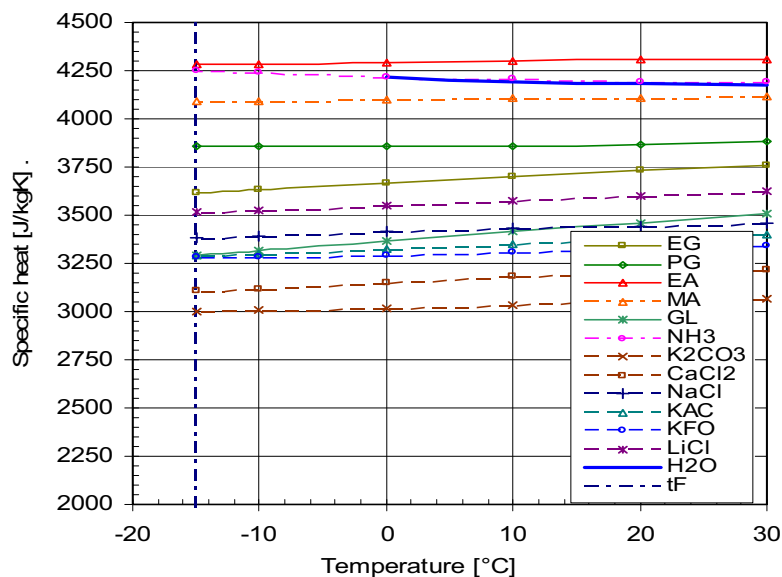


Figure 2.6. Specific heat of aqueous solutions ($t_F = -15^\circ\text{C}$) as function of temperature

Figure 2.6 gives specific heat for concentrations giving $t_F = -15^\circ\text{C}$. Note that the c_p -values for alcohols and ammonia are almost the same as for water while the c_p -values for salts are 20-30% lower.

An added reason for the importance of a high specific heat is that the **volumetric heat capacity**, ρc_p [kJ/m³K], has a direct influence on the volume flow rate, $\dot{V}$, needed for a given heating power, $\dot{Q}$, and temperature change, Δt . The simple relation can be written:

$$\dot{Q} = \dot{V} \rho c_p \Delta t \quad (2.4)$$

The volumetric heat capacity is then an index of the ability of a fluid to convey heat directly, by its motion, to a given destination.

Note in Figure 2.7 that the volumetric heat capacity, ρc_p of water and all aqueous solutions are within 15%, ethyl alcohol having highest ρc_p -value besides water. Glycerol, calcium chloride and the other salts have the lowest values.

2.5 Thermal conductivity

The thermal conductivity, k , is defined by Fourier's law for heat conduction (eqn. 2.5) as the negative of the rate of heat flux and the temperature gradient dT/dx over the thickness x and, the cross section area A . The relation is written

$$dQ/dt = -kAdT/dx \quad (2.5)$$

where the negative sign indicates that heat flows in the direction of decreasing temperatures.

A high value of **thermal conductivity**, k [W/mK], of a fluid is desirable as it contributes to good heat transfer and thereby decreases the temperature difference between fluid and tube wall in the evaporator and heat exchangers.

Figure 2.8 gives thermal conductivity curves for concentrations of the aqueous solutions giving $t_F = -15^\circ\text{C}$. The chloride salts and potassium formate have quite high values, close to water and about 30% higher values than propylene glycol, ethyl alcohol and glycerol.

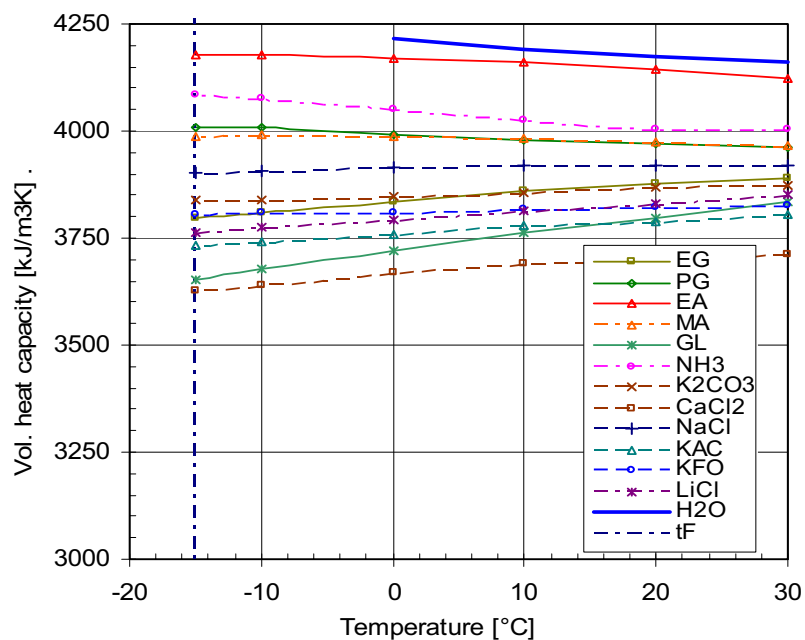


Figure 2.7. Volumetric heat capacity of aqueous solutions ($t_F = -15^\circ\text{C}$) as function of temperature

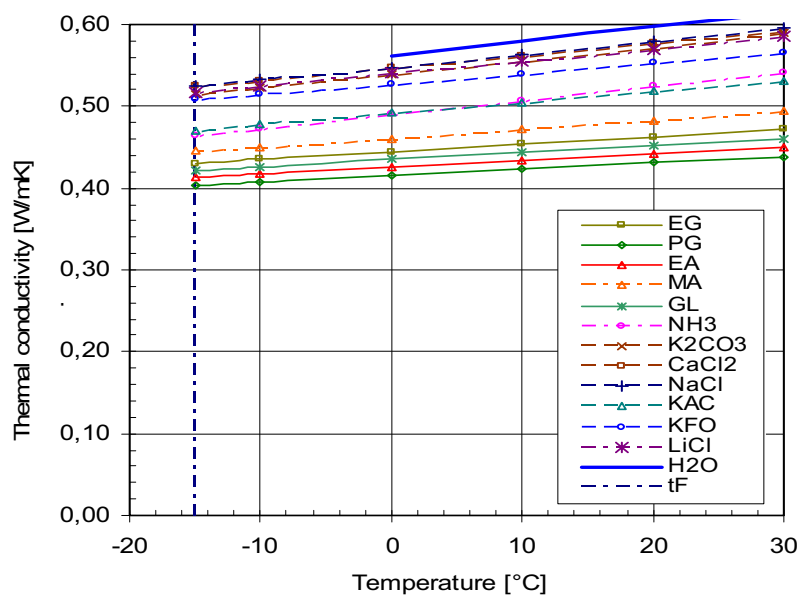


Figure 2.8. Thermal conductivity ($t_F = -15^\circ\text{C}$) as function of temperature

2.6 Dynamic and kinematic viscosity

It is important to determine the absolute or **dynamic viscosity**, μ [mNs/m²], and/or the **kinematic viscosity**, $\nu = \mu/\rho$ [mm²/s], at the operating temperature of a secondary fluid. The absolute viscosity μ or internal friction of a fluid is defined by the following relation

$$\tau = \mu \cdot du/dy \quad (2.6)$$

where τ is the shearing stress at the surface of contact and du/dy is the velocity gradient. The viscosity should be as low as possible as it directly influences the needed pumping power and the heat transfer.

which determines the type of flow that will prevail. Here w [m/s] is the fluid velocity and d [m] is the inner tube diameter.

The viscosity value is used when calculating the Reynolds number, Re ,

$$Re = wd/\nu = wd\rho/\mu \quad (2.7)$$

Figure 2.9 gives as an example kinematic viscosity for concentrations giving $t_F = -15^\circ\text{C}$. The values differ very much between the aqueous solutions and the viscosity increases sharply at low temperatures, especially with propylene glycol, ethyl alcohol and glycerol, even more so at higher additive concentrations and lower temperatures.

2.7 Boiling point

A low **boiling point temperature**, t_B [$^\circ\text{C}$] of the secondary fluid can create problems in a system, especially at higher temperatures. Evaporation of fluid can also take place at lower temperatures than the boiling point if the fluid is exposed to atmospheric air.

High concentrations of ammonia and alcohols give low boiling point temperature, t_B , and flash point, t_{FL} , which may make the fluid flammable. Ethyl alcohol solutions should in many applications not exceed 30 wt%, mainly because of low flash point. Figure 2.10 gives boiling point, t_B [$^\circ\text{C}$], as function of the additive concentration. Values mainly from Timmermans (1960), ASHRAE Fundamentals (2001) and Plank (1988). See further Melinder (1998a).

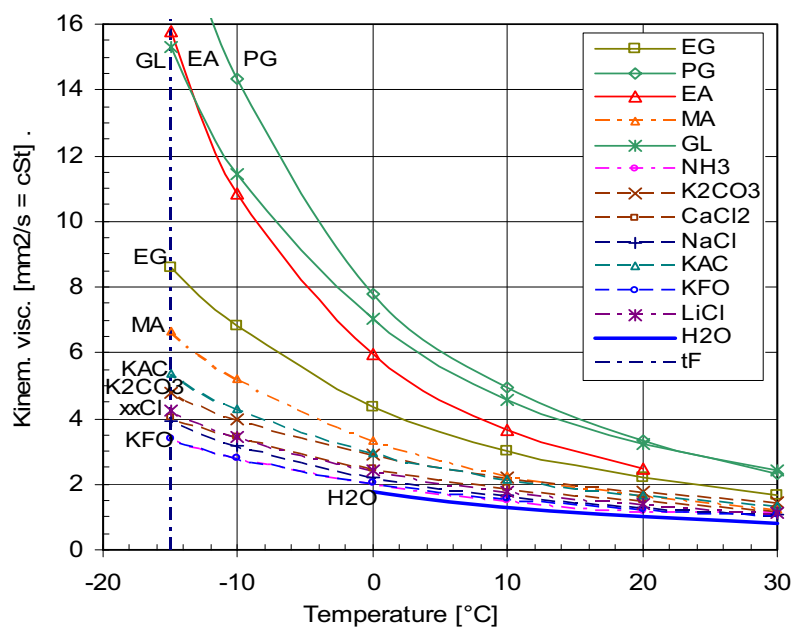


Figure 2.9. Kinematic viscosity ($t_F = -15^{\circ}\text{C}$) as function of temperature

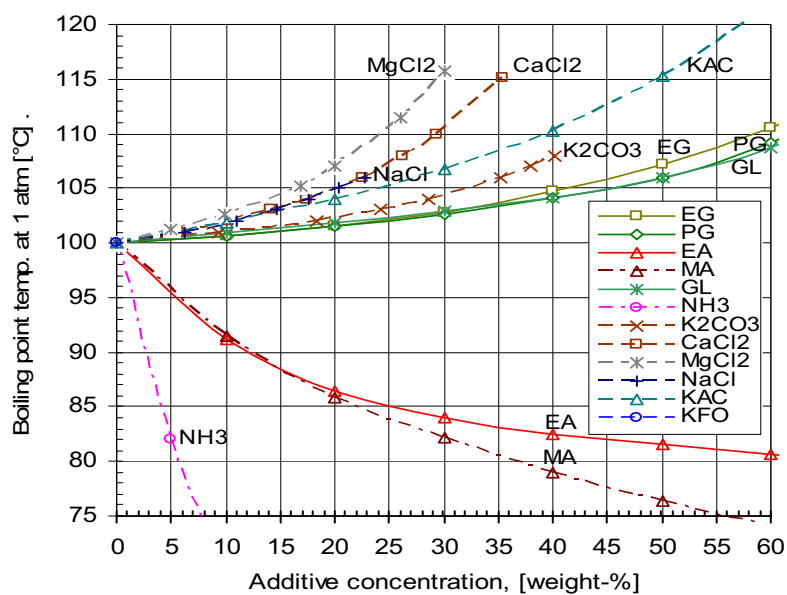


Figure 2.10. Boiling point temperature, t_B , of aqueous solutions as function of additive concentration by weight, c_A

In view of the flammability risk when using ethyl alcohol in indirect systems, safety regulations in Sweden require that the highest temperature in cooling systems has to be at least 5°C below the flash point for the current ethyl alcohol (ethanol) concentration. If the mass concentration is 30% or more the safety regulations are quite strict. [Figure 2.11](#) gives the flash point, t_{FL} [°C], as function of the additive concentration for ethyl alcohol solutions (Svensk Kylnorm, 1992).

2.8 Surface tension

The **surface tension**, γ [mN/m], of a fluid is the force per unit length that strives to keep the surface as small as possible. A low surface tension increases the tendency of the solution to leak out and increases the risk of foaming in the system and of cavitation in the pump. [Figure 2.12](#) gives surface tension as function of the additive concentration. Alcohols and glycols have rather low surface tension. Surface tension decreases with higher temperature as can be seen from values of water (at 0%). Values of surface tension are obtained among other sources from American Inst. of Physics (1927), CRC Handbook of Chemistry and Physics (1988-89), DOW - The Glycols (1974) and Int. Critical Tables (1928-29). See further Melinder (1998a).

2.9 Conclusions

A brief definition and description has in this chapter been given of each of the basic thermophysical properties and some other properties. A comparison of various fluids has been made for concentrations giving the freezing point, $t_F = -15^\circ\text{C}$.

It is evident that no secondary working fluid is ideal for all applications and we should look for the best one for each type of system and application. It is also essential to consider company information sheets of the commercial products.

In chapter 5 a comparison is made of different freezing points or concentrations of two of these aqueous solutions to see what impact the concentration has on the thermal behavior of the fluid.

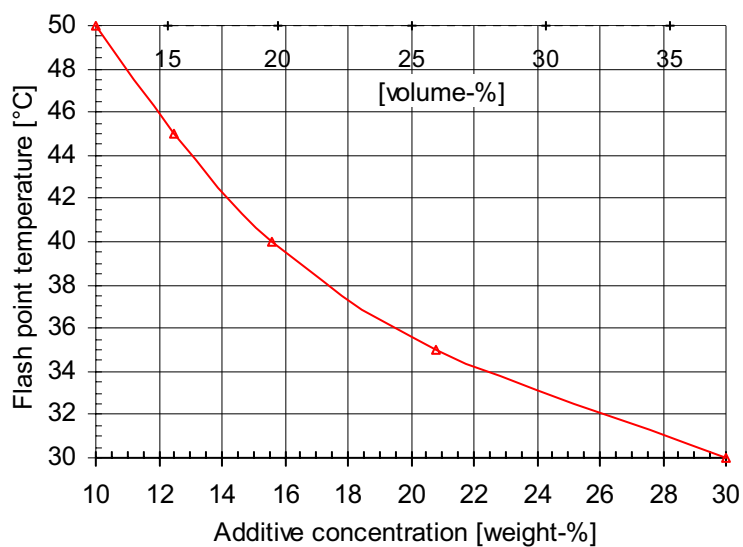


Figure 2.11. Flash point temperature, t_{FL} , of aqueous solutions of ethyl alcohol as function of additive concentration

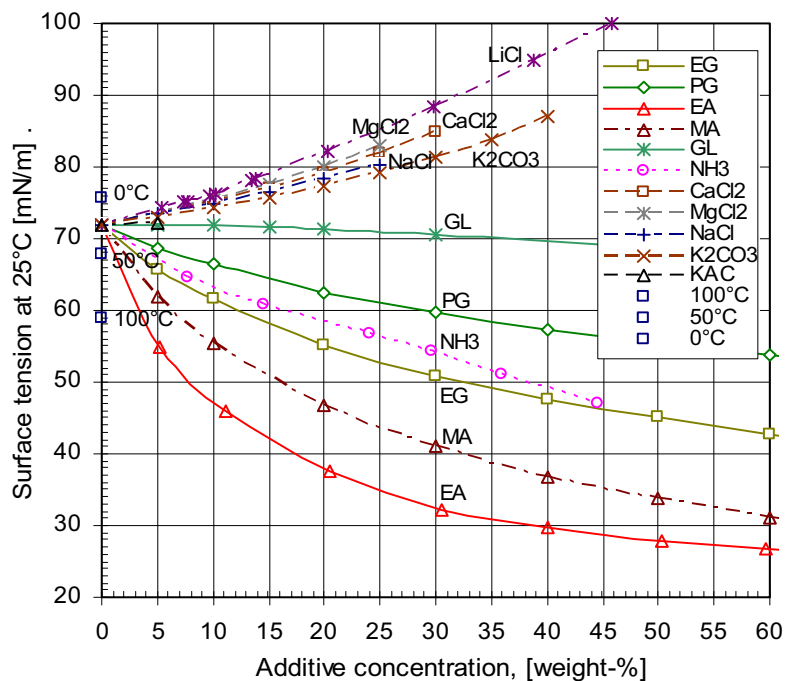


Figure 2.12. Surface tension, γ , of aqueous solutions as function of additive concentration by weight, c_A

2.10 Nomenclature

The following nomenclature is used in chapter 2.

A	cross section area [m ²]
c_A	additive concentration by weight [weight-%]
c_p	specific heat [J/kg K]
d	tube diameter [m]
dT/dx	temperature gradient [-]
du/dy	velocity gradient [-]
k	thermal conductivity [W/m K]
$\dot{Q}$	cooling capacity [W]
Re	Reynolds number; $Re = (w \cdot d)/\nu$ [-]
t, T	temperature [°C]
t_B	boiling point temperature [°C]
t_E	eutectic temperature [°C]
t_F	freezing point temperature [°C]
t_{FL}	flash point temperature [°C]
TVE	thermal volume expansion [%]
V	volume [m ³]
$\dot{V}$	volume flow [m ³ /s]
w	fluid flow velocity [m/s]
β	coefficient of thermal volume expansion [1/K]
$\Delta t, \Delta T$	temperature change [K]
ΔV	volume change [m ³]
γ	surface tension [mN/m]
μ	dynamic viscosity; $\mu = \nu \cdot \rho$ [Pa·s]
ν	kinematic viscosity [m ² /s]
ρ	density [kg/m ³]
ρc_p	volumetric heat capacity [kJ/m ³ K]
τ	shearing stress at surface of contact [Pa]

3 Selection of property values of aqueous solutions

3.1 Summary

One main purpose of this chapter is to summarize the review work reported on in the Engineering Licentiate Thesis: *Thermophysical properties of liquid secondary refrigerants, A Critical Review on Literature References and Laboratory Measurements* (Melinder 1998a). That material forms the basis for the charts and tables used in the IIR-publication *Thermophysical properties of liquid secondary refrigerants* (Melinder, 1997). An additional purpose of this chapter is to report on more recent review work done, including two additional fluids not covered in Melinder (1998a).

The main intention of this work has been to select thermophysical property data with good or acceptable technical accuracy of a number of aqueous solutions that can be used by the refrigeration and heat pump industry, rather than focusing on a limited number of property values or scientifically accurate measuring techniques.

Hence, a thorough literature search was made to find the most reliable property values of aqueous solutions for choice of secondary working fluid and technical calculations of refrigeration and heat pump systems. Detailed literature references are given for thermophysical properties of each aqueous solution. Some laboratory measurements were made of most of the fluids types when literature values were incomplete or deemed unreliable. Methods used are briefly described and results are given.

The aqueous solutions, without other additives, presented in Melinder (1998a) and Melinder (1997) are ethylene and propylene glycol, ethyl and methyl alcohol, glycerol, ammonia, potassium carbonate, calcium, magnesium and sodium chlorides as well as potassium acetate. The types of aqueous solutions that have been added in this thesis */ are potassium formate and lithium chloride.

*/ A commercial product based on betaine has been introduced on the market and in conference papers but property values from other sources are lacking (Jokinen 2003, Jokinen, Willems 2004).

Thermophysical property data given in Melinder (1997) are in the Melinder (1998a) compared with measured values and other litera-

ture reference data. A detailed list with reference data is given as well as charts with freezing point, density, viscosity, specific heat and thermal conductivity. For boiling point, and surface tension the information is limited to the data list. A brief discussion following each chart give main sources for the tables and charts in Melinder (1997) deviating literature values and other comments that may be of interest. A few of the charts show that some values in handbooks for the refrigeration industry deviate much from other basic literature sources.

Examples of this are viscosity of aqueous solutions of ethyl alcohol, specific heat of ammonia, thermal conductivity of sodium chloride, the glycols and alcohols. Viscosity measurements by author of potassium carbonate and calcium chloride (down to about -20°C) differ much from values in refrigeration handbooks at lower temperatures but are supported by laboratory loop tests at KTH and values from basic sources. See further Melinder (1998a) and Melinder (1998b).

The agreement in many of the charts is satisfactory from a technical standpoint. However, some of the charts indicate that more research and measurements would be of value to ensure that property values are sufficiently correct, especially at low temperatures where little data is found at temperatures below 0°C .

It is good to keep in mind that the commercial products on the market have corrosion inhibitors and or other added substances that will change property values to some extent.

3.2 Summary of critical evaluation of thermo-physical property values given in Tech. Lic. Thesis

A main purpose of this report is as mentioned to provide a critical review on the literature references and laboratory measurements that form the basis for charts and tables in Melinder (1997).

A detailed list of the sources of literature references and laboratory measurements for each thermophysical property is given for each aqueous solution. Many of these sources are introduced in charts from Melinder (1997) and observations and an evaluation is made.

3.2.1 List of literature references for the evaluation of property values

Literature sources and other references for thermophysical property values are in **a – b** listed with numbers and given in the same order as in Melinder 1997 and Melinder 1998a, references in **c**. only in Melinder 1998a. Literature sources and other references added in this Thesis are listed with numbers in **d**. In this chapter, these references are usually only marked with the number **xx** or [xx].

a. Main literature sources of aqueous solutions (Ref. 1 – 6)

1. CRC Handbook of Chemistry and Physics, 67th edition (1986-87)
2. Dow, Organic Chemicals / The Glycols (1974)
3. International Critical Tables of Numerical Data, Physics, Chemistry and Technology (1928-29)
4. Landolt-Börnstein; Eigenschaften der Materie in Ihren Aggregatzuständen (1960-1971)
5. Landolt-Börnstein; Numerical Data and Functional Relationships in Science and Technology; Vol. 1b (1977)
6. The Physico-chemical Constants of Binary Systems in Concentrated Solutions, Vol. 3 - 4; J. Timmermans (1960)

b. Additional literature sources and references (Ref. 7 – 42)

7. Akzo Chemie, "Hoesch Pa 9 rot" company data (1978)
8. American Institute of Physics Handbook, 3rd edition (1972)
9. ASHRAE Handbook, Fundamentals (1993)
10. Bayer, "Baysilone-Fluids" company data (1993)
11. Carrier, Handbook of Air Conditioning System Design, (1965)
12. Chemical Engineering, Vol. 1, J. M. Coulson, J F Richardson (1960)
13. Chevron Chemical Co. (Cryotech); "Chevron Heat Transfer Fluid" company data (1991)
14. *Development of Improved Secondary Refrigerants*; Int. Institute of Refrigeration (U K), S. F. Pearson (1993)
15. Die Kältemaschine, M. Hirsch (1932)
16. Dow Europe S.A., "Syltherm" company data
17. Handbook of Heat Transfer; W. M. Rohsenow, J. P. Hartnett (1973)

18. Handbook of Heat Transfer Media; P. L. Geiringer (1962)
19. Handbook of Refrigerating Engineering, Vol. 1, Fundamentals; W. R. Woolrich (1965)
20. Handbook of Thermodynamic Tables and Charts; K. Raznjevic (1976)
21. Handbuch der Kältetechnik, VI/B; R. Plank, (1988)
22. Heat transmission, W. H. Mc Adams (1954)
23. Hoechst, "Antifrogen KA" company data (1995)
24. Kältemaschinen Regeln (1958)
25. Kältetechnik ("Kylteknikern"), M. Bäckström, E. Emblik (1965)
26. Kältetechnik, Zeitschrift für das gesamte Gebiet der Kälteerzeugung und Kälteanwendung, Heft 4, 1950
27. Köldbärare för värmepumptillämpningar. Studier av termodynamiska egenskaper och korrosion; Å. Melinder; Report R114:1985 to the Swedish Council for Building Research (1985)
28. Köldbärare för värmepumptillämpningar. Termodynamiska egenskaper och korrosion för kaliumkarbonat samt jämförelse med andra köldbärare; Å. Melinder, J. Berendson, E. Granryd, B. Kyrk; Report R18:1989 to the Swedish Council for Building Research (1989)
29. Lange's Handbook of Chemistry, 13th edition (1985)
30. Macmillan's Chemical and Physical Data (1992)
31. Mellor's Comprehensive Treatise on Inorganic and Theoretical Chemistry, Vol. 2, Suppl. 3, Part 2 (1963)
32. Secondary Heat Transfer Systems and the Application of a New Hydrofluoroether; G. Sherwood, The 1995 International CFC and Halon Alternatives Conference (1995)
33. Secondary Refrigerants for Heat Pumps and Low Temperature Refrigeration. A comparison of thermodynamic properties of aqueous solutions and non-aqueous liquids; Å. Melinder, E. Granryd; Trita-REFR Report No 92/6, ISSN 1102-0245 (1992)
34. Secondary Refrigerants for Indirect Refrigeration and Heat Pump Systems; E. Granryd, Å. Melinder; Scanref No. 4, 1994
35. Thermodynamic Properties for Secondary Refrigerants - Tables and Charts; Å. Melinder, Handbook No. 12 of the Swedish Society of Refrigeration (1994)
36. Thermo-physical properties of liquid secondary refrigerants. Tables and Diagrams for the Refrigeration Industry, Melinder, Å., IIF/IIR, Paris (1997)
37. Tyforop; "Tyfoxit" company data (1995)

38. Verfahren der Kälteerzeugung und Grundlagen der Wärmeübertragung, E. Hofmann (1959)
39. VDI Heat Atlas (1993) (translation of VDI Wärmeatlas 6-th edition, 1991)
40. Wärmelehre und Wärmewirtschaft in Einzeldarstellungen; F. Bosnjakovic (1961)
41. Zeitschrift für Die Gesamte Kälte-Industrie (certain issues from 1918 - 1943)
42. 3M, "HFE-7100" company data (1996)

c. References of phase changing secondary fluids (Ref. 43 – 49)

43. Ammonia, carbon dioxide and water - the commercial refrigerants of the future?, M. Kauffeld, DTI Energy, Refrigeration and Heat Pump Technology (1995)
44. Binary Ice as a secondary refrigerant; J. Paul; 19th Int Congress of Refrigeration, Vol.4b (1995)
45. Ice slurry based district cooling systems; C. V. Snoek; D. Hampton; Proceedings, Int District Heating and Cooling Ass. 8th Ann. Cooling Conference, Tulsa, Oklahoma (1993)
46. Pressure drop characteristics of ice - water flows; C. V. Snoek, S. U. Joanis, R. V. Gupta; Int. Symp. on Two-Phase Flow Modelling and Experimentation, Rome (1995)
47. Properties of ice slurry, P. W. Egolf, et al; Applications for Natural Refrigerants, Aarhus (1996)
48. Recent advances in CO₂ refrigeration; J. Pettersen et al; 19th Int. Congress of Refrigeration; Vol.4b (1995)
49. Thermal study of an ice slurry used as refrigerant in a cooling loop; O. Bel et al; Applications for Natural Refrigerants, Aarhus (1996)

d. References added in Melinder 1998a (Ref. 50 – 63)

50. Chemical Engineers Handbook, J.H. Perry (1950)
51. Data Book of the Viscosity of Liquid, D. S. Viswanath; G. Natarajan (1989)
52. Handbuch der Kältetechnik, VII; R. Plank, (1959)
53. Heat Transfer and Pressure drop of Liquid Secondary refrigerants - Experiments and theory; Engineering Thesis work, Olivia Florian (1997)

54. Int. Journal of Thermophysics, Vol. 8, p.351 (1987) and Vol. 15, p.165 (1994)
55. Internal report of measurements at KTH for "Kylteknikern" [25]; (1957)
56. Internal report with measurements at Umeå university primarily of specific heat and thermal conductivity (1996)
57. MIDAS Database for Thermophysical Properties, Excerpt from R. Krauss, ITT, Stuttgart University, Germany (1995)
58. Refrigeration, Air Conditioning, and Cold Storage, R. C. Gunther (1957)
59. Tables of Physical and Chemical Constants, G. C. Kaye, T. H. Laby (1986)
60. Thermal Conductivity, Vol. 2, R. P. Tye (1969)
61. Thermophysical Properties of Matter, Volumes 3; Thermal Conductivity. Nonmetallic liquids and gases; Y. S. Touloukian et al (1970)
62. Thermophysical Properties of Matter, Volumes 6; Specific heat. Nonmetallic liquids and gases; Y. S. Touloukian et al (1970)
63. Wärme und Kältetechnik, Zeitschrift für Klimatechnik, Trockentechnik, Wärme- und Schallschutztechnik, February 1941

e. References added in this thesis (Ref. 64 – 88)

64. Aittiomäki, A., Lahti, A., 1997, Potassium formate as a second refrigerant, *Int. Journal of Refrigeration*, vol. 20, No 4, p.276-282.
65. Andersson, O., 1996, 2002, Internal reports of measurements of thermal conductivity and volumetric heat capacity with transient hot wire technique at Umeå univ.
66. Clariant, 1998, "Antifrogen KF", company data
67. Conde, M.R., 2004, *Properties of aqueous solutions of lithium and calcium chlorides: formulations for use in air conditioning equipment design*, Int. Journal of Thermal Sciences, Volume 43, p.367-383.
68. -
69. Evenmo, K., 1998, *A New Secondary Fluid*, Natural Working Fluids'98, IIR – Gustav Lorentzen Conf., Oslo.
70. Evenmo, K., 1998, Personal communication
71. Fransson, Å., 1996, 2002, Internal reports of measurements of specific heat with differential scanning calorimetry at Umeå univ.
72. Gmelins Handbuch der Anorganischen Chemie, Kalium, 22, 1937.
73. Gmelins Handbuch der Anorganischen Chemie, Lithium, 20, 1960.

74. Int. Journal of Thermophysics, Vol. 8, p.351, 1987
75. Int. Journal of Thermophysics, Vol. 15, p.165, 1994
76. Jokinen, J., *Betaine Based Heat Transfer Fluid as a Solution for Toxicity and Corrosion Problems in Heating and Cooling Systems*, 21st Int. Congress of Refrigeration, IIF/IIR, Washington D.C.
77. Kemira, 1998-99, "Freezium" company data
78. Melinder, Å., 1983-1999, Internal reports on measurements of viscosity at KTH with falling ball viscosimeter. (See Appendix).
79. Melinder, Å. 1998, *Thermophysical properties of liquid secondary refrigerants, A Critical Review on Literature References and Laboratory Measurements* (Tekn. Lic. Thesis), KTH, Stockholm.
80. Melinder, Å, 1998, *A Critical Review on Thermophysical Properties of Liquid Secondary Refrigerants*, Natural Working Fluids'98, IIR – Gustav Lorentzen Conf., Oslo.
81. Mellan, I., 1977, *Industrial Solvents Handbook*, New Jersey: Noyes Data Corporation, 133 p
82. Mellor's Comprehensive Treatise on Inorganic and Theoretical Chemistry, Vol. 2, Suppl. 3, Part 2 (1963)
83. Stakelbeck, H., Plank, R., 1929, *Ueber die Zähigkeit von Chlornatrium-, Chlorkalzium- und Chlormagnesium-lösungen in Abhängigkeit von Temperatur and Konzentration*, Zeitschrift für Die Gesamte Kälte-Industrie, p.105-112.
84. Söhnle, O., Novotny, P., 1985, Densities of solutions of inorganic substances, Physical Science Data 22.
85. Tyforop, 1997, "Tyfoxit F", company data
86. Vervoerd, M., Liem, N., van Gerwen, R.J.M., 1999, *Performance of a frozen food display cabinet with potassium formate as secondary refrigerant*, 20th Int. Congress of Refrigeration, IIF/IIR, Sydney.
87. Wimby, J.M., Berntsson, T.S., 1994, *Viscosity and density of aqueous solutions of LiBr, LiCl, ZnBr₂, CaCl₂ and LiNO₃*, 1. Single salt solutions, J. Chem. Engineering Data 39, p.68-72.
88. Yara formates - Coolants – "Hycool", 1997-2005, company data (from Hydro)

3.2.2 Reference details and symbols used

The IIR-publication (Melinder 1997) contains numerous charts and tables with basic thermophysical properties of secondary working fluids. A detailed list of references for each fluid and property is given in this chapter followed by a critical review with charts. Reference details and symbols used in the lists are here given.

The reference numbers from Melinder (1998a) are indicated with bold letters and refer to the references listed in 3.5. Some of these references indicate sources of measurements. These sources, or the number of sources are then given in brackets [...]. The numbers of main references and sources of measurements used to make up the tables and charts are underlined. The *estimated significance* of each source of the reference material is indicated by the following symbols given after the reference details:

- (*ℓ*,..) experimental values from one source (sometimes smoothed)
- (*x*₅,..) selected values from a number (*x*..) of sources
- (*t*,..) table giving property values (usually in handbook)
- (*c*,..) chart giving property values (usually in handbook)

- (*..,1±*) good accuracy, rather complete, incl. low temperatures
- (*..,1+*) good accuracy, rather complete, not incl. low temp.
- (*..,1÷*) good accuracy, not complete (scattered data or few temp.)
- (*..,2±*) less good accuracy, rather complete, incl. low temperatures
- (*..,2+*) less good accuracy, rather complete, not incl. low temp.
- (*..,2÷*) less good accuracy, not complete (scattered data or few temp.)

- (*..,3±*) poor accuracy, rather complete, incl. low temperatures
- (*..,3+*) poor accuracy, rather complete, not incl. low temp.
- (*..,3÷*) poor accuracy, not complete (scattered data or few temp.)
- (*..,4*) whole or vital part clearly inaccurate
- (*..,..°*) indicate that values are not given in same unit, conversion needed
- (*..,..**) indicate product with corrosion inhibitor or other substance added

3.3 Critical evaluation of some thermo-physical property values

Some of the charts and discussions in Melinder (1998a) are covered in Melinder (1998b). These include density of ethylene glycol, thermal conductivity of sodium chloride, freezing point and viscosity of calcium chloride.

Certain property values of some aqueous solutions have been changed especially at low temperatures since Melinder (1998a) was written. A discussion of some of these changed property values will be presented here, including specific heat of ethyl alcohol. The viscosity values of calcium chloride at low temperatures are reviewed. The reference lists of these aqueous solutions are also given.

Property values of aqueous solutions of lithium chloride and potassium formate have been added since the publication of Melinder (1998a) and are considered with a detailed reference list and charts with discussion. Charts give for these two aqueous solutions the basic thermophysical properties: freezing point; density; dynamic viscosity; specific heat and thermal conductivity.

3.3.1 Critical evaluation of ethyl alcohol

A detailed list of references for each property of ethyl alcohol is given while similar lists of glycols and other aqueous solutions can be seen in Melinder (1998a) that in chapter 3 is marked [79].

Freezing point: **1.** p.D-228 (*t,1*); **3.** Vol. IV, p.262 (*4 s,2°*); **4.** Vol. 2a, p.885 (*3 s,1°*); **6.** Vol 4, p.181-2 [Ross, 1954 (*e,2±*); Lalande, 1934 (*e,2±*); Benjamin, 1932 (*e,2±°*); Tarasenkova, 1928 (*e,2±*); Pushin and Glagoleva, 1923 (*e,2°±*); Pictet and Atschul, 1895 (*e,2±*); Pickering, 1893 (*e,2±*)]; **11.** p.4.26 (*t,2÷*); **18.** p.117 (*c,2*÷*); **19.** p.371 (*t,2°±*); **21.** p.670 (*t,2÷*); **25.** p.52 (*c,3±*); **29.** p.10-75,76 (*t,2°±*); **39.** p.Dd15-16 (*t,2±*).

Density: **1.** p.D-228 (*t,2*); **3.** Vol. III, p.116,117 (*19 s,1+*); **5.** p.129 (*25 s,1+*); **6.** Vol 4, p.183-7 [Harms, 1938 (*e,1°÷*); Spills, 1936 (*e,1÷*); Wiley and Harder, 1935(*e,1°÷*); Brun, 1931 (*e,1÷*); Kreitling, 1892 (*e,1+*); Mendelejeff, 1869 (*e,1+*)]; **11.** p.4.41 (*c,2°±*); **18.** p.115 (*t,2°÷*); **20.** p.72 (*t,2÷*); **21.** p.670 (*t,2÷*); **25.** p.52 (*c,3±*); **39.** p.Dd15-16 (*t,2±*).

Dynamic viscosity: 1. p.D-228 (*t,2*); 3. Vol. V, p.22 (2*s,1+*); 4. Vol. 5a, p.367 (4*s,1+*); 6. Vol 4, p.190-3 [Spells, 1936 (*e,2÷*); Whightman et al, 1914 (*e,2+*); Bingham et al, 1913 (*e2+*); Dunstan et al, 1909 (*e,2+*); Noack, 1886 (*e,2+*)]; 11. p.4.40 (*c,4°±*); 18. p.115 (*t,3°÷*); 20. p.85 (*t,2÷*); 21. p.670 (*t,2÷*); 25. p.52 (*c,3±*); 39. p.Dd15-16 (*t,2±*); 79 and App.1 [Melinder, 1992 (*e,2±*)].

Specific heat: 3. Vol. V, p.116 (7*s,2°+*); 5. p.310-11 (3*s,c,2+*); 6. Vol 4, p.205 [Blacet et., 1931 (*e,2÷*); Bose, 1907 (*e,2+*)]; 11. p.4.42 (*c,2°±*); 20. p.65 (*t,2÷*); 21. p.670? (*t,2÷*); 25. p.52 (*c,3±*); 39. p.Dd15-16 (*t,2±*); 71. [Fransson, 1996 (*e,2÷*)].

Thermal conductivity: 4. Vol V/b, p.94 and Vol. T. 4b, p.567 (*t,1±*); 11. p.4.43 (*c,2°±*); 18. p.115 (*c,2°*); 20. p.96 (*t,2÷*); 21. p.670 (*t,2÷*); 25. p.52 (*c,3±*); 39. p.Dd15-16 (*t,2±*).

Boiling point: 6. Vol 4 p. 171-80 [Bosnakovic and Grumbt, 1931 (*e,2+*), Brun, 1931 (*e,2+*); Öman and Gunnelius, 1925 (*e,2+*); 21. p.668-70 (*t,2÷*).

Surface tension: 1. p.F-32 (*t,1°+*); 3. Vol. IV. p.467 (4*s,1+*); 6. Vol 4, p.194-7 [Teitelbaum et al, 1951 (*t,1°+*); Valentine and Holes, 1938 (*t,2°+*); Ernst et al, 1936 (*t,2÷*)]; 8.p.2-209 (*t,2÷*); 17. p.2.61 (*t,1÷*); 30. p.30 (*t,1°+*).

Specific heat of ethyl alcohol – water (See Fig 3.1)

A discussion of changed specific heat values of ethyl alcohol are presented here followed by some comments about specific heat values of ethylene and propylene glycols. Similar discussions are made of the other basic properties of ethyl alcohol in Melinder (1998a) on p. 27 - 31.

Discussion: Curves for high concentrations and low temperatures in Fig 3.1 have been adjusted to fit measured values by Bose [3] and [6], measured values of 47% and 50% by Fransson [71] (with DSC, see also 3.3.2) that all show a steep slope below room temperature. Values in [11] and [21] are in rather good agreement, though curves for high concentrations and low temperatures in [11] are less steep. Note also measured values of 10% from Umeå univ. are in good agreement with measured values by Bose. Note the curve for water, 0%, that is of right level, though a little steep.

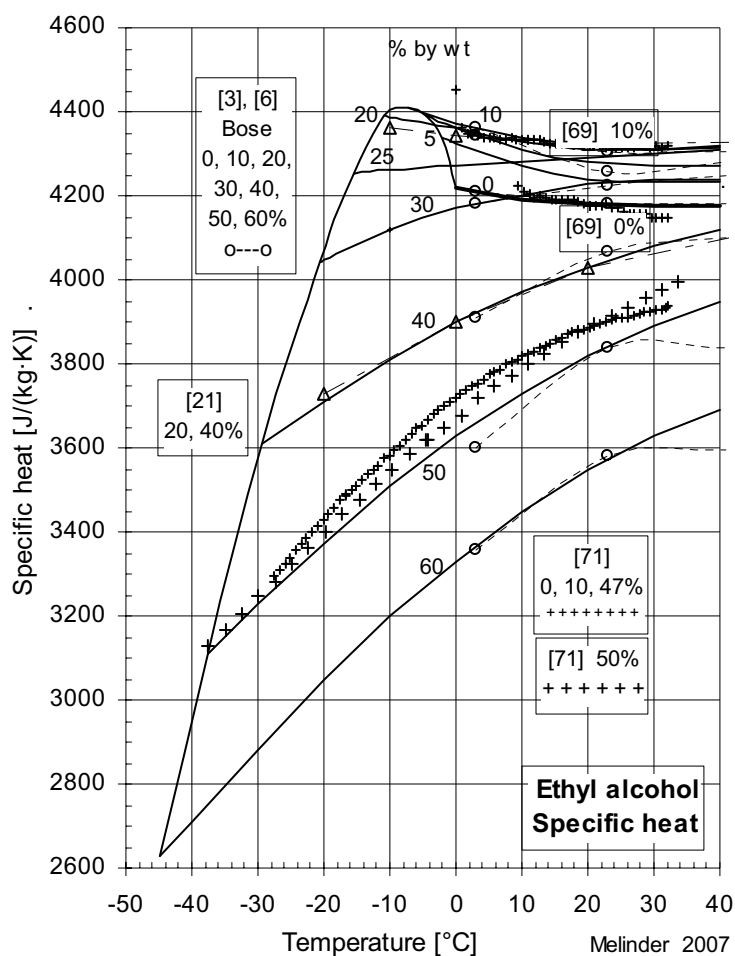


Figure 3.1. Specific heat of ethyl alcohol - water

Specific heat of ethylene and propylene glycols

Values of specific heat for ethylene glycol were already adjusted for high concentrations and low temperatures. Values for propylene glycol have now been adjusted in about the same way as with ethylene glycol and ethyl alcohol. See Melinder (1998a).

3.3.2 Critical evaluation of viscosity of calcium chloride

A discussion of viscosity values of calcium chloride are presented here to draw attention to the fact that less accurate values for low

temperatures still are used in main sources of literature values such as ASHRAE Fundamentals, 2005 (Compare Melinder, 1998a).

Calcium chloride - water:

Dynamic viscosity: **1.** p.D-225 (*t,2*); **3.** Vol. V, p.14 (*5_s,2°÷*); **4.** Vol. 5a, p.342 (*3_s,1°÷*); **6.** Vol 3, p.762-63 [Lyons and Riley, 1954 (*e,2÷*); Stackelbeck and Plank, 1929 (*e,2-4±*); Hechler, 1904 (*e,2°±*); **9.** p.18.2 (*c2-4±*); **11.** p.4.32 (*c,2-4°±*); **15.** p.531 (*c,2-4°±*); **21.** p.618 (*t,2-3÷*); **24.** p.86 (*t,2-4±*); **25.** p.47 (*c,3±*); **38.** p.454-5 (*t,2-4°±*); **39.** p.Dd10 (*t,2-4±*); **41.** 1927, p.144 (*t,2-3°+*); 1929, p.110 (*t,2-4°±*); **79** and **App.1** [Melinder, 1983-1996 (*e,2±*)].

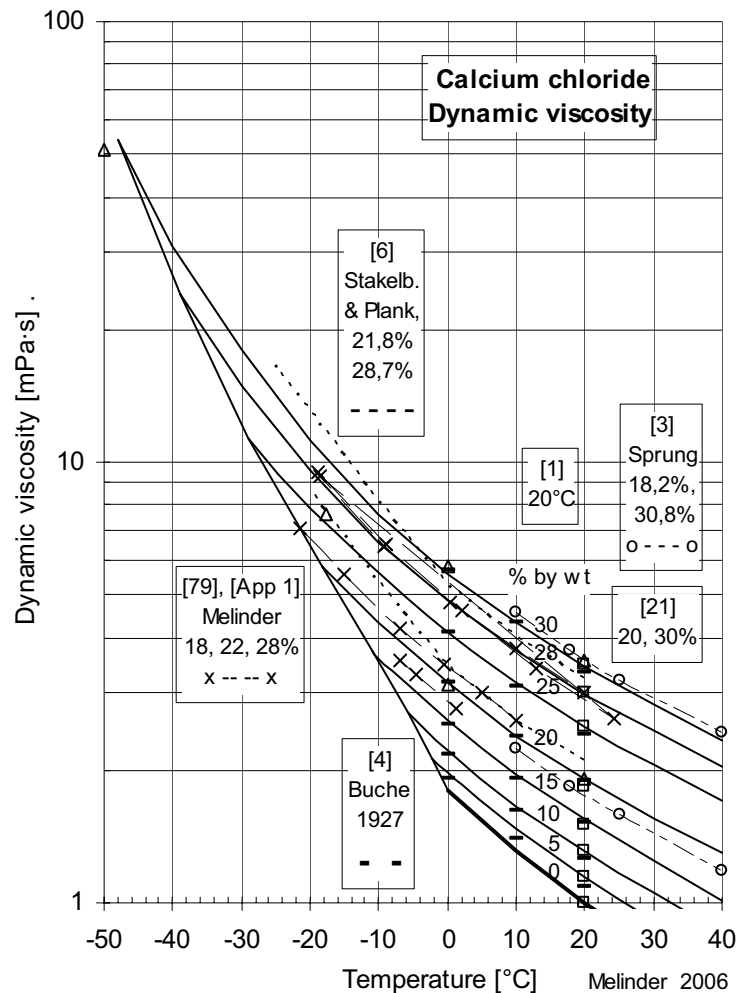


Figure 3.2. Dynamic viscosity of calcium chloride – water

Discussion of dynamic viscosity of calcium chloride - water

Curves from Chart C8b in [36] are based on [1] and measurements made by Sprung (1876) [3], Buche (1927) [41] and author (see Appendix). Values from Stakelbeck & Plank [41] (1929, p.105-112) used by ASHRAE Fundamentals, 2005 and most other publications are about 30% higher at -20°C than values from measurements made by author [App.1].

The article in [41] explains that a constant used for the calculations had to be extrapolated between -5.5°C and -25°C . Note that the curves from [41] are very straight for temperatures below 0°C in a linear-logarithmic representation. The low temperature values by author are supported by tests made by Björn Kyrk in a laboratory loop at KTH as described in [28] Melinder (1989), on p.61-63 in [79] (Melinder 1998a), in Melinder (1998b) and on the following pages.

The laboratory test rig by Kyrk was designed to measure heat transfer and pressure drop of liquids flowing in a tube and the liquid was circulated with a pump in a closed tube loop. The pressure drop, Δp , over this section, that had a sharp inlet device, to ensure transition between turbulent and laminar flow at Reynolds number, $Re \approx 2300$, was measured with a mercury manometer and the velocity w with a flow meter. Density, ρ , and dynamic viscosity, μ , for the liquid were determined at the mean temperature of the liquid [28].

Pressure drop results with calcium chloride - water

Pressure drop results from tests with a concentration of calcium chloride - water, giving the freezing point -25°C , were represented in charts where the friction coefficient $f_1 (= f/2)$ was plotted as a function of the Reynolds number, Re . (Compare equations 5.4 and 5.5 in chapter 5.2). See Figures 3.3 and 3.4 and discussion below.

Note that in Figure 3.3 with values from [41] that transition between laminar and turbulent flow takes place at lower Re -values for lower temperatures ($t = -10^{\circ}\text{C}$ and -20°C). In Figure 3.4 with values from author [79], transition takes place at approximately the same Re -values for all temperatures.

Similar tests and observations were made with heat transfer measurements and these are recorded on p.62-63 in Melinder (1998a).

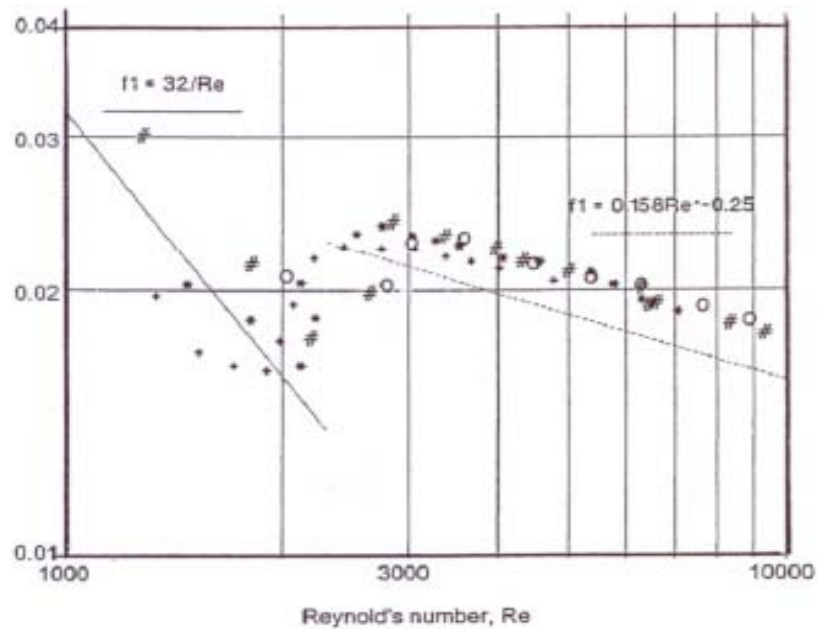


Figure 3.3. Results of pressure drop tests of calcium chloride - water with property values from Stakelbeck [41]

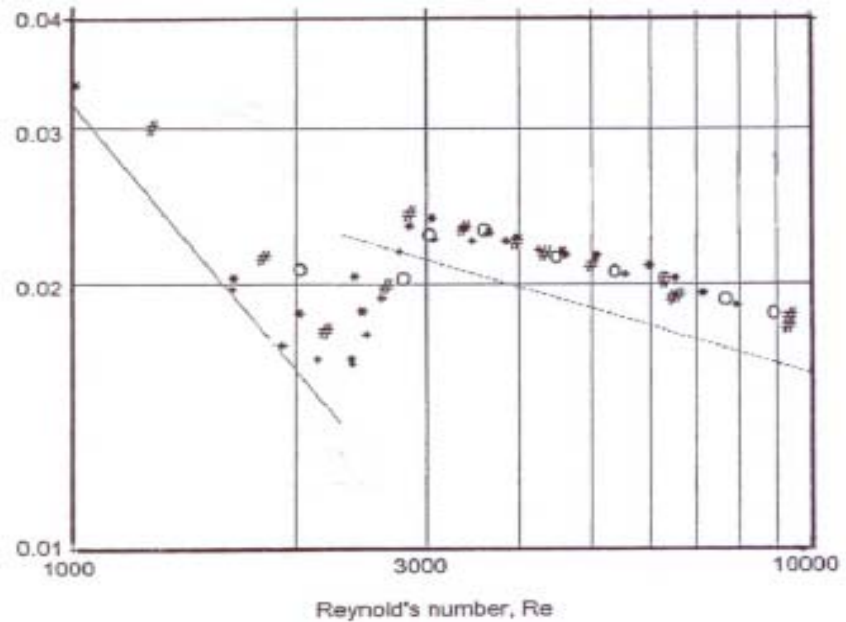


Figure 3.4. Results of pressure drop tests of calcium chloride - water with property values from author [79]

These results show that the viscosity values presented by author are basically correct and that values from [41] are too high at these lower temperatures. See also Melinder (1998a), p. 59–65.

3.3.3 Critical evaluation of property values of lithium chloride

Recently aqueous solutions of lithium chloride have been considered for use in refrigerating and airconditioning systems (Conde 2004, Melinder 2004, Wimby et al 1994). A list and detailed discussion of property values of aqueous solutions of lithium chloride that may be used at very low temperatures is here for the first time presented in detail by author.

Lithium chloride - water

Freezing point: **1.** p.D-235 (*t,1÷*); **6.** Vol 3, p.614-615 [Maran, 1956 (*e,3±*); Voskresenskaja & Yanatieva, 1937 (*e,2±*); Rodebush, 1918 (*e,2÷*)]; **65.** Andersson, 1996 (*e,2÷*); **73.** p.337 (*t,2±*) p.354-55 (*4 s,2°±*).

Density: **1.** p. D-235 (*t,1÷*); **3.** Vol. III, p.77 (*51 s,1+*); **5.** p.59 (*29 s,1+*); **6.** Vol 3, p.619 [Fontell, 1927 (*e,1÷*); **73.** p.344 (*31 s,1+*); **84.** p.135 (*t,2+*).

Dynamic viscosity: **1.** p. D-235 (*t,1÷*); **4.** Vol. 5a, p.314-315 [Langyel & Tamaz, 1964 (*s,1°÷*); **6.** Vol 3, p.621-623 [Jacopetti, 1942 (*e,2÷*); Alfimoff, 1917 (*e,2°÷*); Green, 1908 (*e,2°÷*); **73.** p.349-51 (*t,2°÷*); **App.1.** [Melinder, 1996 (*e,2±*)].

Specific heat: **5.** p.263-264 (*6 s,2°÷*); **6.** Vol 3, p.627 [Jauch, 1921 (*e,2°÷*); Tucker, 1920 (*e,2°÷*); Green, 1908 (*e,2°÷*); **71.** [Fransson, 1996 (*e,1÷*)];

Thermal conductivity: **4.** Vol V/b, p.92 [Riedel, 1948 (*t,2÷*)]; **6.** Vol 3, p.628 [Kapustinski & Ruzavin, 1955 (*t,2÷*)]; **65.** [Andersson, 1996 (*e,1÷*)]; **67.** Conde, 1967 [Uemura, (*c,2÷*)]; **73.** p.355 (*e,2°÷*).

Boiling point: **6.** Vol 3, p.614 [Johnston, 1906 (*e,2+*); Gerlach, 1886 (*e,2+*)].

Surface tension: **6.** Vol 3, p.622 [De Block, 1925 (*e,2+*); Lineberger, 1899 (*e,2+*)]; **73.** p.349 (*t,2°÷*).

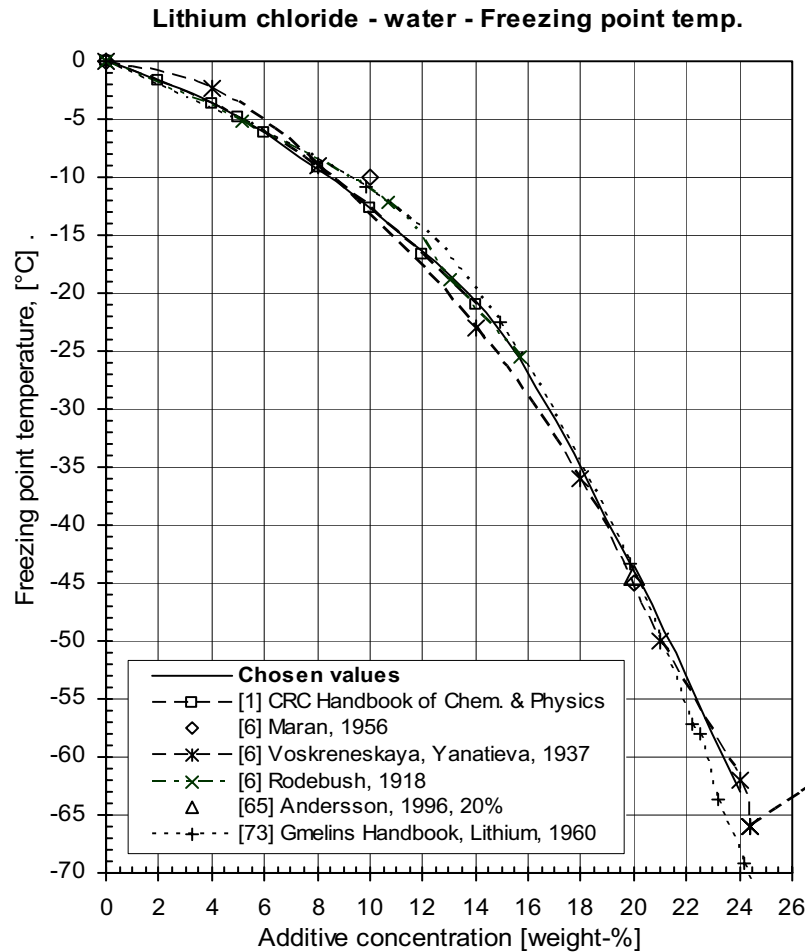


Figure 3.5. Freezing point of lithium chloride – water

Discussion: The curve for the chosen values is in good agreement with [1] (values given to 14%), Rodebush [6] (values given to 18%) and fit measured values by Voskreneskaya and Yanatieva [6] rather well. Note also the good agreement of the value for 20% obtained from measurements of thermal conductivity at Umeå univ. [65] (Compare Fig 3.9). Values by Maran [6] and in [73], p.336-37 are for the “ice curve” and indicate a lower eutecticum ($t_E = -75^\circ\text{C}$) than Voskreneskaya and Yanatieva [6] ($t_E = -66^\circ\text{C}$).

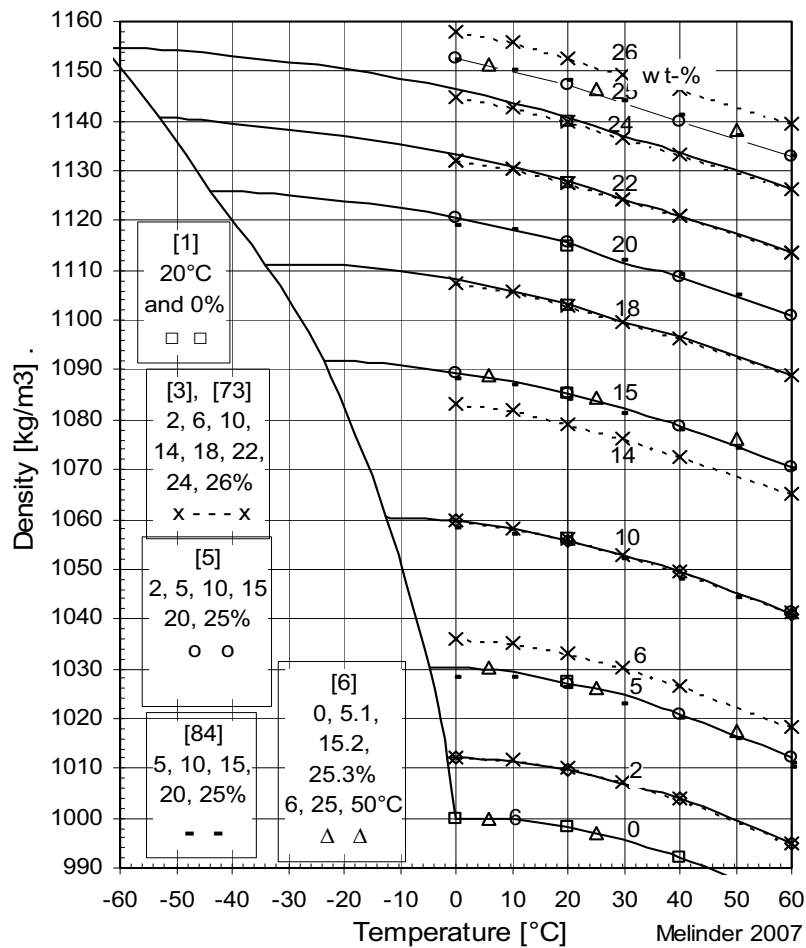


Figure 3.6. Density of lithium chloride – water

Discussion: The curves for the chosen values for density are in good agreement with values from many sources, some with values down to 0°C in [1], [3], [5], [6], [73] and [84]. No values are given for temperatures below 0°C. The curves follow the pattern of other chlorides, such as calcium chloride, below 0°C (Melinder, 1998a).

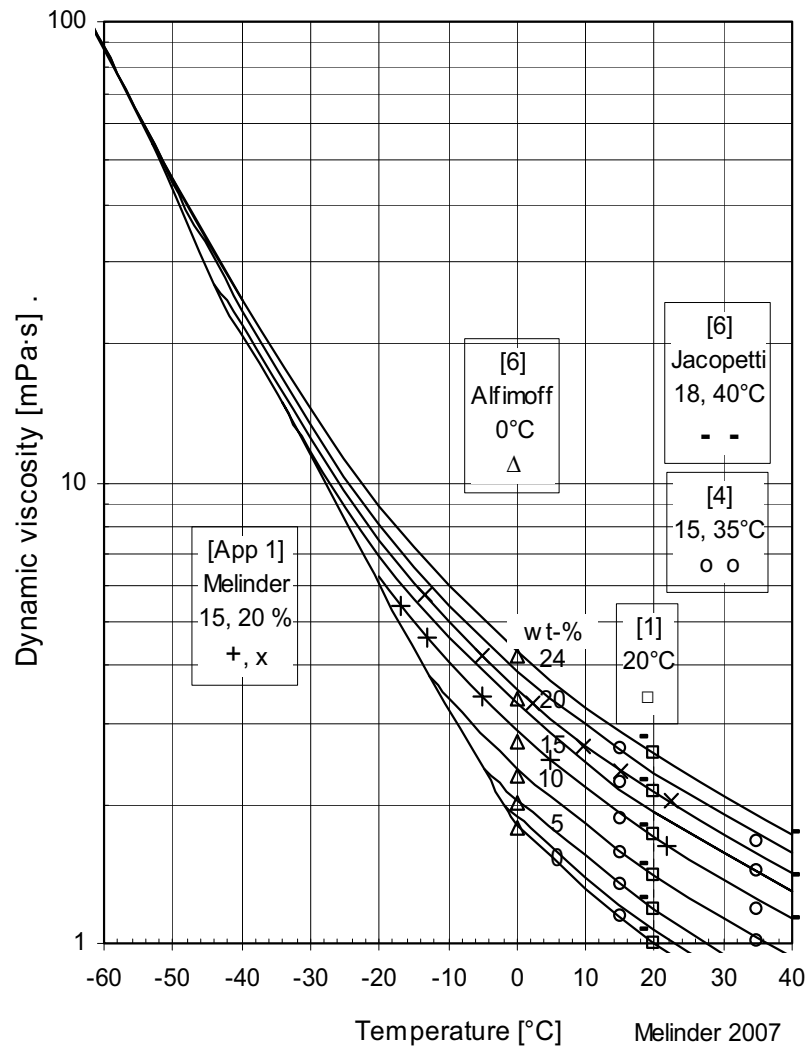


Figure 3.7. Dynamic viscosity of lithium chloride – water

Discussion: The curves for the chosen values for viscosity are in good agreement with values from [1] as well as measured values by Langyel & Tamaz [4], Alfimoff and Jacopetti [6] down to 0°C. However, only measured values by author [App.1] are given below 0°C, down to -15 to -20°C. The curves below -20°C are extended much based on the freezing point values.

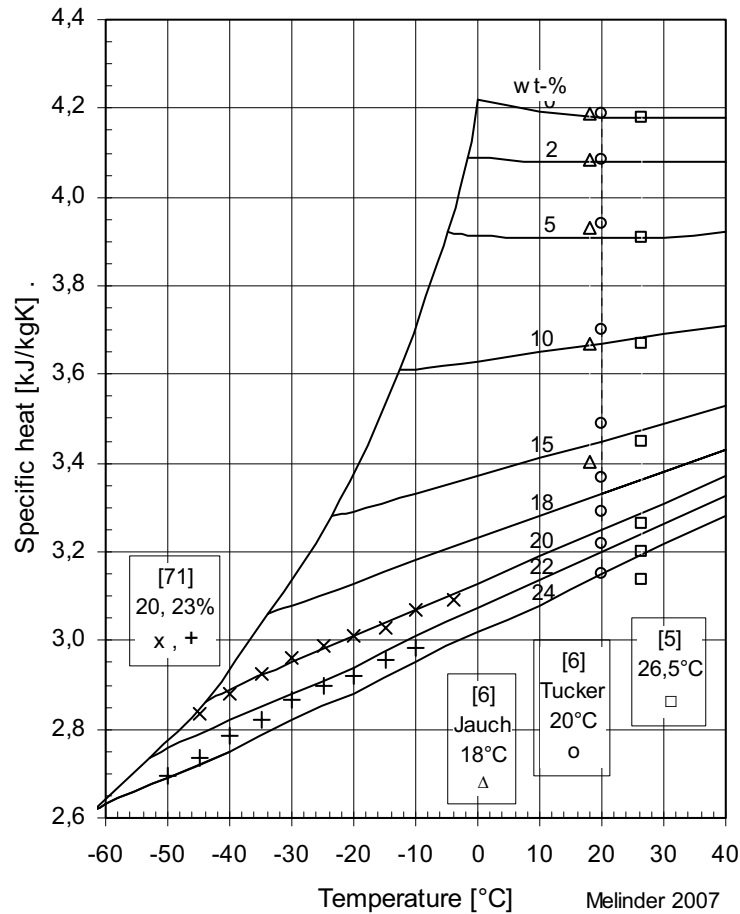


Figure 3.8. Specific heat of lithium chloride – water

Discussion: The curves for the chosen values for specific heat are in good agreement with values from 5 sources in [5] as well as values by Jauch and Tucker [6] all for room temperatures. However, measured values by Fransson [71] for 20% and 23% are given down to -45 and -50°C. The slope of the freezing point is estimated and curves for lower additive concentrations can be drawn to their freezing point values that are established in Figure 3.5.

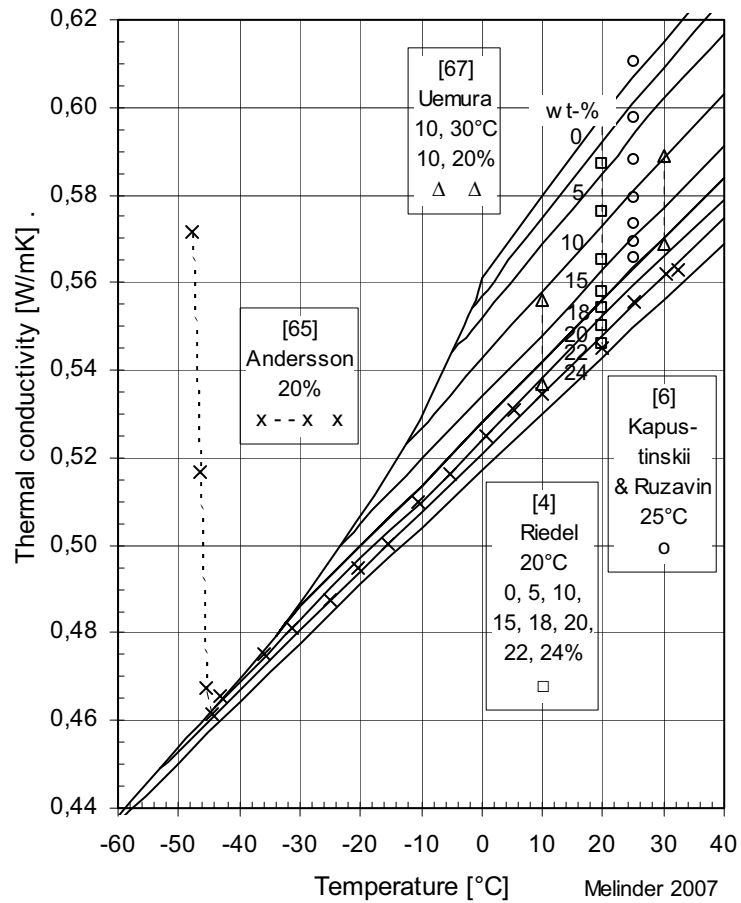


Figure 3.9. Thermal conductivity of lithium chloride – water

Discussion: The curves for the chosen values for specific heat are in good agreement with values from Riedel [4] as well as values by Kapustinski & Ruzavin [6] and Uemura [67]. However, measured values by Andersson [65] for 20% are given down and passed the freezing point at about -45°C. The slope of the freezing point is estimated and curves for lower additive concentrations can as in Figure 3.8 be drawn to their freezing point values that are established in Figure 3.5.

3.3.4 Critical evaluation of property values of potassium formate

Over a decade ago aqueous solutions of potassium formate were considered for use in refrigerating systems as a secondary fluid with very low viscosity at low temperatures (Aittiomäki, Lahti 1997, Evenmo 1998, Vervoerd et al 1999). A list and detailed discussion of property values of potassium formate is here presented in detail by author.

Very little data material was found in the general literature search. However, when doing this research there were four commercial products providing data that seemed rather independent from each other. Data for two of these products, Antifrofen KF and Freezium, were presented as function of additive concentration, the other two, Hycool and Tyfoxit F, were given only as function of cooling limit. How could these data properly be compared and combined? My approach was first to establish a freezing point curve as there were some literature values to compare with. The next step was to determine the additive concentration of the products from their density value at 20°C as that is the property value that a company would find of high importance and could easily establish.

Potassium formate - water (not yet done)

Freezing point: 6. Vol 3, p.588 [Sidgwich and Gentle, 1922 (*e,2±*)]; 66. (*c,3*±*); 69. (*e,2±*); 70. (*e,3÷*); 77. (*t,2±*).

Density (and TVE): 6. Vol 3, p.588 [de Garcia 1920, (*t,2÷*)]; 84. p.199 (*t,2÷*).

Dynamic viscosity: App.1 [Melinder, 1996 (*e,2*±*)].

Specific heat: Only company data.

Thermal conductivity: 65. (*c,2±*).

Boiling point and Surface tension: Only company data.

In view of a strong interest from the industry for property values of potassium formate I tried to find an acceptable way to make use of information from the mentioned four commercial products. See Figures 3.10-15 and the discussions.

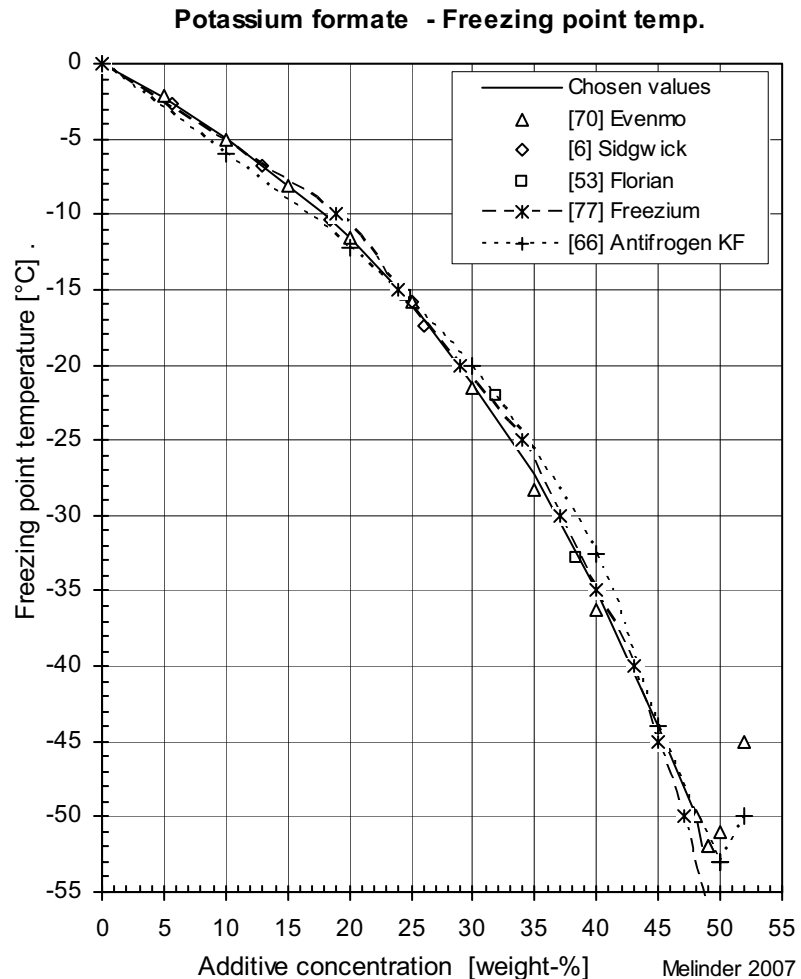


Figure 3.10. Freezing point of potassium formate – water

Discussion: The curve for the chosen values is in good agreement with measured values by Sidgwick and Gentle [6] as well as values given by Evenmo [70] and Florian [53]. Note also the values by two commercial products, Antifrogen KF [66] and Freezium [77]. The agreement with the other sources is not as good but acceptable. The low freezing point indicated near 50% by [77] is not supported by [66] nor by measurement results given by Evenmo [69]. Below -52°C we may get a metastable phase that could turn glassy state as may also be the case with propylene glycol at high concentrations (See Melinder, 1998a, p.21).

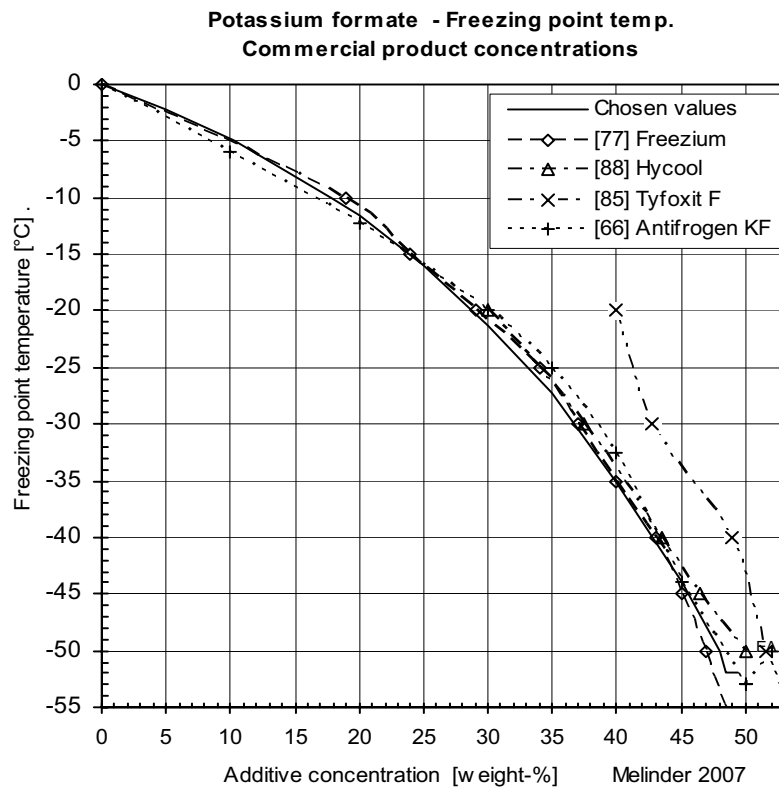
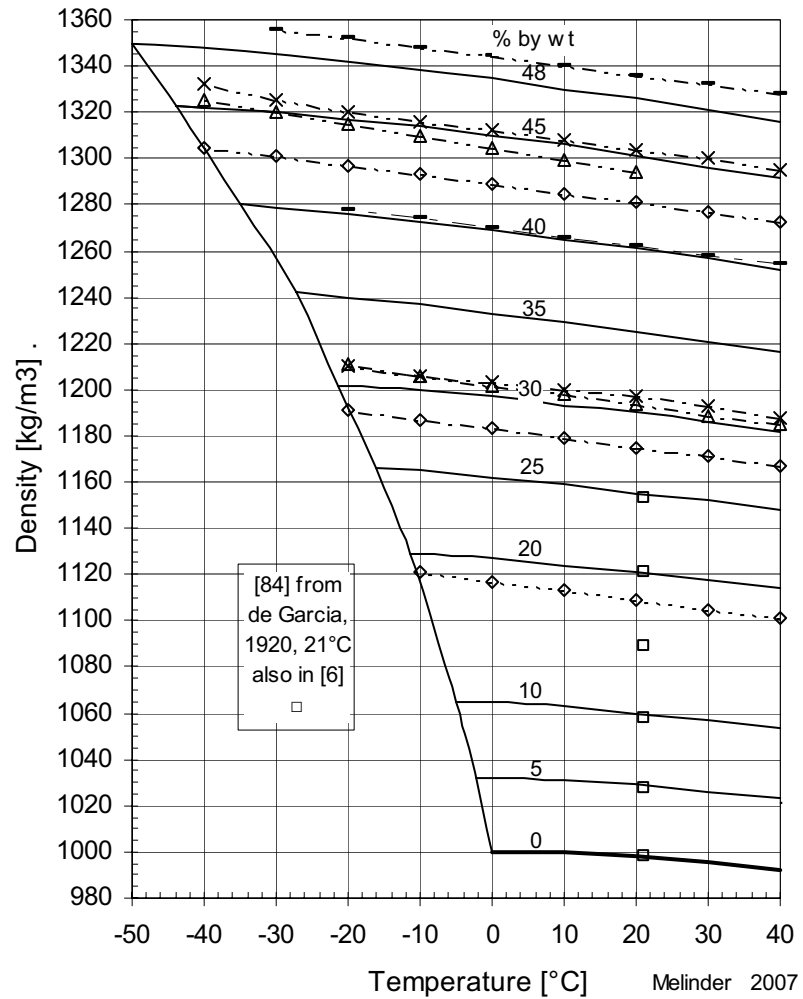


Figure 3.11. Freezing point of some commercial products based on potassium formate – water

Discussion: The curve for the chosen values (from Figure 3.10) is in Figure 3.11 plotted with four commercial products, besides Antifrogen KF [66] and Freezium [77], also Hycol [88] and Tyfoxit F [85] that both give cooling limit rather than freezing point. From the figure it is obvious that the additive concentrations chosen for Tyfoxit F are unnecessarily high leading to poorer performance of the fluid.

In each of Figures 3.12-15 the following sources appear and are commented on in the discussions.

- - x - - [66] Antifrogen KF -40; 43%	- - ◇ - - [77] Freezium -40; 43%
- - △ - - [88] Hycol -40; 43,5%	- - — - - [85] Tyfoxit F -40; 49%
- - × - - [66] Antifrogen KF -20; 30%	- - ◇ - - [77] Freezium -20; 29%
- - △ - - [88] Hycol -20; 30%	- - — - - [85] Tyfoxit F -20; 40%
- - ◇ - - [77] Freezium -10; 19%	



- · × · - [66] Antifrogen KF -40; 43%	- · ◇ · - [77] Freezium -40; 43%
- · △ · - [88] Hycool -40; 44%	- · — · - [85] Tyfoxit F -40; 49%
- · × · - [66] Antifrogen KF -20; 30%	- · ◇ · - [77] Freezium -20; 29%
- · △ · - [88] Hycool -20; 30%	- · — · - [85] Tyfoxit F -20; 40%
- · ◇ · - [77] Freezium -10; 19%	

Figure 3.12. Density of potassium formate – water

Discussion: The only literature data found beside the commercial product data are from de Garcia [6], [84]. Density values for the commercial products put in the same figure and from this their additive concentrations are estimated and used also in Figures 3.13 to 3.15.

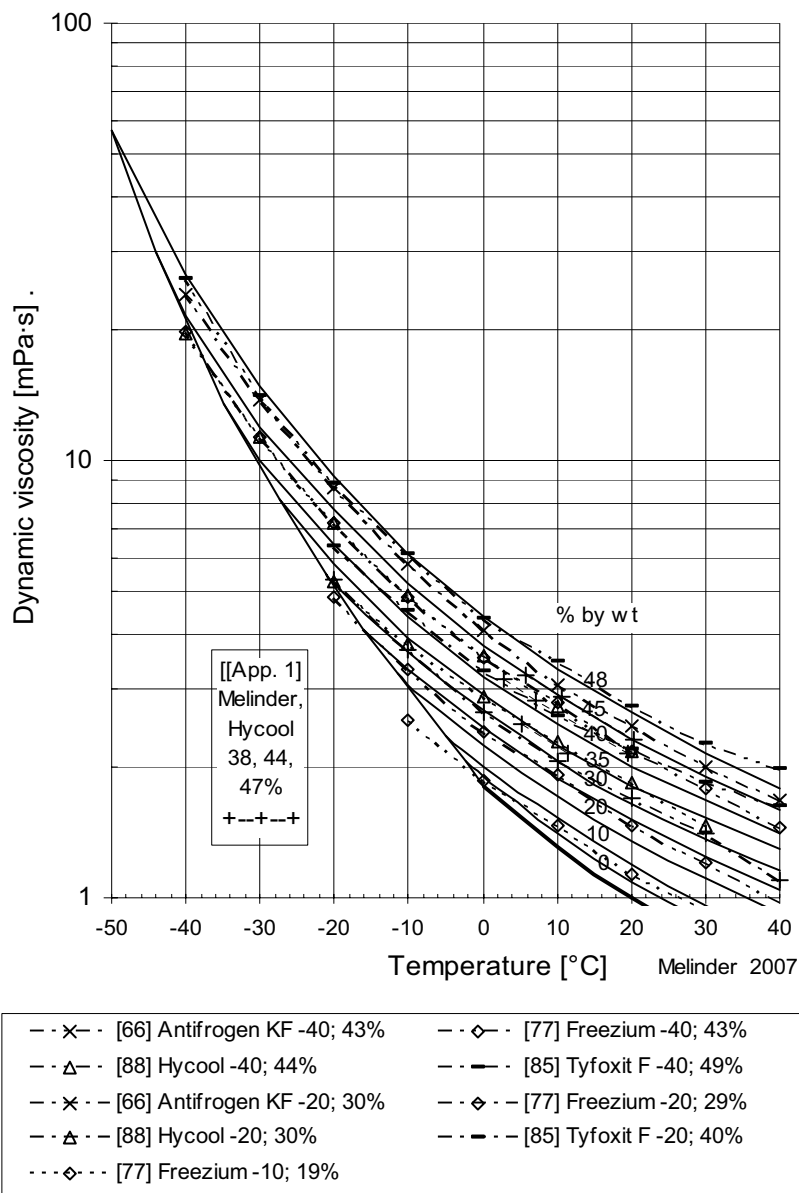


Figure 3.13. Dynamic viscosity of potassium formate – water

Discussion: The only literature data found beside the commercial product data are from measurements of three additive concentrations of Hycool by author [Appendix 1]. The agreement is rather good though Freezium -10 give too low values.

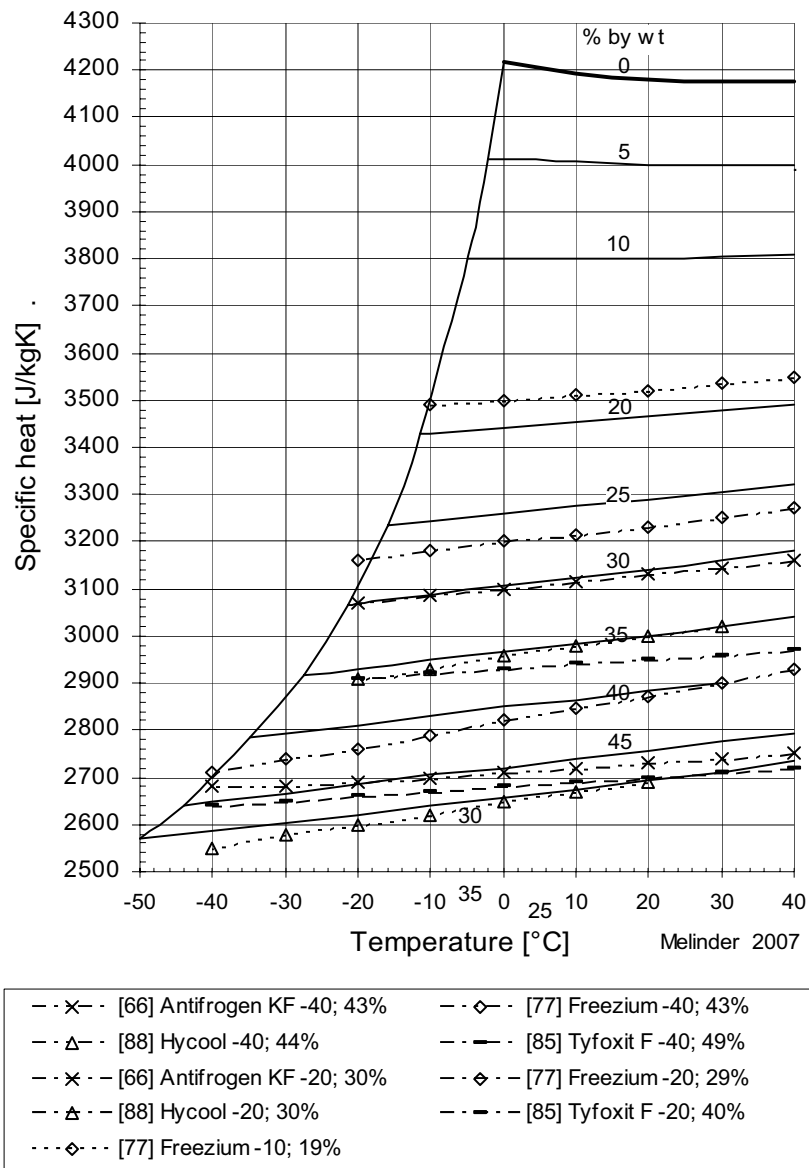
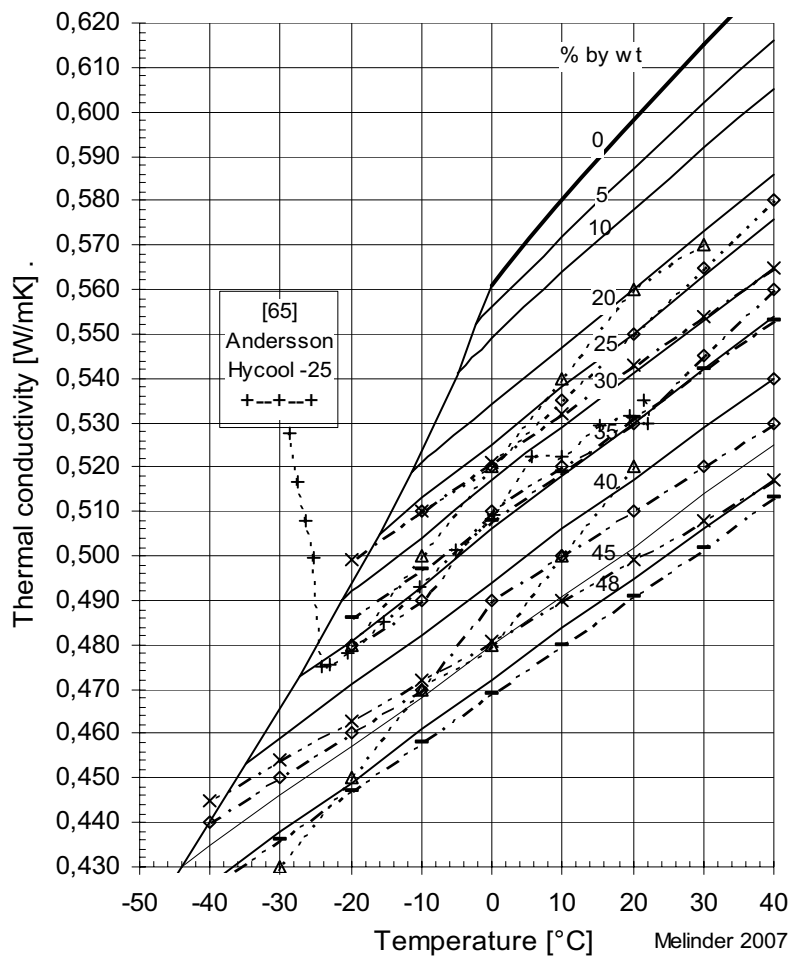


Figure 3.14. Specific heat of potassium formate – water

Discussion: The only data found are commercial product data. The values of Tyfoxit F have a different slope and are higher than what can be expected of the high additive concentrations. One reason for this deviation could be that the values are based on volumetric heat capacity measurements with the transient hot-wire technique (See 3.4.2 and also Melinder, 1998a, p.18 and 24).



--x-- [66] Antifrogen KF -40; 43%	--◇-- [77] Freezium -40; 43%
--△-- [88] Hycool -40; 44%	--■-- [85] Tyfoxit F -40; 49%
--x-- [66] Antifrogen KF -20; 30%	--◇-- [77] Freezium -20; 29%
--△-- [88] Hycool -20; 30%	--■-- [85] Tyfoxit F -20; 40%
---◇--- [77] Freezium -10; 19%	

Figure 3.15. Thermal conductivity of potassium formate – water

Discussion: The only data found beside the commercial product data are measured values by Andersson [65] for Hycool -25 given down and passed the freezing point at about -25°C. The higher values below the freezing point are no doubt due to ice crystals. The values of the commercial products have different slopes and the chosen values take this into consideration.

3.4 Measurements of some properties

Some laboratory measurements of basic thermophysical properties have as a background to this work been made at KTH since 1983 and also at Umeå university since 1996. These measurements were mainly made when literature values were incomplete or seemed unreliable. The different thermophysical properties are already briefly described (Chapter 2). A brief description of the methods and equipment used for these measurements is given here.

3.4.1 Measurements of viscosity at KTH

Measurements of viscosity were carried out with a Höppler Falling ball Viscosimeter (Figure 3.16) attached to a refrigerating unit (See Figure 3.17). The falling tube of glass of the viscosimeter was filled with the sample of the liquid to be tested and a falling ball of glass. The falling time of the ball between the two marking lines was measured manually with a chronometer. A falling ball was chosen that gives a falling time of 30 - 300 seconds. The time was measured manually with a chronometer.

The calculation of the dynamic viscosity is done with the following formula:

$$\mu = \tau(\rho_1 - \rho_2) \cdot K \quad (3.1)$$

where μ = dynamic viscosity (Pa·s); τ = average falling time of ball (sec.); ρ_1 = density of ball (kg/m³); ρ_2 = density of liquid (kg/m³); K = ball constant.

The set up for measurements before 2002 is described in Melinder (1998a). The set up for measurements since 2002 is as follows. Stable low temperatures were obtained in the following way: The glass enclosed container around the falling tube was connected with a circuit consisting of the heat exchanger of a refrigerating unit, a small pump, an electric heater and a thermostat that could be set to desired temperature. This whole circuit was filled with an alcohol - water solution that was pumped around and deaired. Stable temperatures were obtained with a thermostat. The equipment put a limit to about -20°C for the lowest temperature. The temperature was measured with thermocouples connected to a recorder. In Figure 3.17 the viscosimeter is upside down ready for start of a measurement.

Measurements and calculations were carried out with almost all aqueous solutions. The number of measurements, mean temperature, maximum temperature variation, density of liquid, ball constant, ball density and average falling time are also given and the viscosity result is recorded both as dynamic (absolute) and kinematic viscosity. Results for aqueous solutions of ethyl alcohol, lithium chloride and potassium formate are given in Appendix 1. Results for other solutions are given in 4.2.3 in Melinder (1998a).



Figure 3.16. Höppler Falling ball Viscosimeter.



Fig 3.17. Höppler viscosimeter and measurement equipment

3.4.2 Measurements of specific heat and thermal conductivity at Umeå university

Some laboratory measurements were made by personnel at Umeå university when literature values were incomplete or were deemed unreliable. This univ. has advanced technology and equipment to make accurate measurements.

Measurements of specific heat were carried out with a Perkin-Elmer differential scanning calorimeter (DSC-2). The principle for the measurements of specific heat, c_p , is among else described in Int. Journal of Thermophysics (1994) [75]. Temperature calibration was made with standard materials at heating. The rate of cooling was 5 K/minute. The specific heat data obtained for samples were determined during cooling.

The transient hot-wire technique used to determine the thermal conductivity and volumetric heat capacity was developed at the Department of Experimental Physics of Umeå University and is among else described in Int. Journal of Thermophysics (1987) [74]. Measurements of the thermal conductivity, k (or λ), and the volumetric heat capacity, ρc_p , of samples of aqueous solutions were made at atmospheric pressure in the temperature range of 200-320 K. During measurements the temperature was decreased at a constant rate of approximately 0.01 K/s.

Common tap water was used for the test experiments. Samples were heated up to 320 K and degassed before experiments. The largest discrepancy between experimental and reference data for k (or λ) is less than 2%, for ρc_p , about 5% and for c_p about 3%. These results confirm that convection effects are reasonably small in the methods used. Both experimental and smoothed data of thermal conductivity and specific heat outlined can be used for different kinds of engineering calculations, but the ρc_p data obtained should be used only as qualitative results.

Some measuring results with certain aqueous solutions of ethylene glycol, glycerol, calcium chloride and potassium acetate are in Melinder (1998a) given as charts of specific heat and thermal conductivity. Approximate crystallization or freezing point temperatures are indicated graphically as a drastic change in the value of thermal conductivity (See Figure 3.9).

4 Making thermophysical properties of aqueous solutions available for computer use

4.1 Introduction

As mentioned in earlier chapters, aqueous solutions are used in various types of indirect refrigeration and heat pump systems. Reliable basic thermo-physical data is needed for choice of suitable fluid and for technical calculations of the system. This chapter describes how the selection of property values results in tables and charts intended for the industry. The chapter also describes the process to transform these property values to a material suitable for computer treatment (Melinder 2006).

4.2 Results of selection of property values of aqueous solutions using ethyl alcohol as example

Based on the work described in the previous chapter, property values have been selected, generally for even concentrations or mass fractions of the aqueous solutions examined. With the help of a Matlab program, coefficients are generated from these property values and together with polynomial equations this material can be a useful tool for computer treatment. This is described in 4.3.

To see how this process is made, aqueous solutions of ethyl alcohol, one of the most versatile secondary working fluids, is used as example. This choice of fluid can also be seen as a test of this method, as the basic thermophysical properties of aqueous solutions of ethyl alcohol seem to present more challenges than most other fluids examined. This is true about the freezing point curve, density, specific heat and viscosity. Hence, the choice of ethyl alcohol is a good test case for the equations used for computer treatment.

The results for aqueous solutions of lithium chloride and potassium formate, salts that may be used at quite low temperatures, are also given as these types have been added since Melinder (1997) and Melinder (1998a) were compiled.

Based on the work described in chapter 3, Table 4.1 gives basic thermophysical properties of aqueous solutions of ethyl alcohol.

The following symbols are used:

- t_F freezing point [$^{\circ}\text{C}$]
- c_A additive concentration [wt-%]
- t fluid temperature [$^{\circ}\text{C}$]
- ρ density [kg/m^3]
- c_p specific heat [$\text{J}/\text{kg K}$]
- k thermal conductivity [$\text{W}/\text{m K}$]
- μ dynamic viscosity [mPa s]

Table 4.1. Basic thermophysical properties of ethyl alcohol

Chemical formula: $\text{C}_2\text{H}_5\text{OH}$; $M = 46.07$

t_F [$^{\circ}\text{C}$]	c_A [wt-%]	t [$^{\circ}\text{C}$]	ρ [kg/m^3]	c_p [J/kgK]	k [W/mK]	μ [mPa s]
0	0	40	992.2	4177	0.631	0.653
		30	995.7	4177	0.615	0.798
		20	998.2	4181	0.598	1.002
		10	999.7	4192	0.580	1.306
		0	999.9	4218	0.561	1.792
-2.09	5	40	983.1	4235	0.588	0.77
		30	986.6	4235	0.576	0.96
		20	989.5	4240	0.562	1.24
		10	990.8	4270	0.548	1.70
		0	991.4	4325	0.532	2.45
		-2.09	991.2	4340	0.528	2.70
-4.47	10	40	974.9	4270	0.554	0.90
		30	978.7	4270	0.542	1.16
		20	982.0	4280	0.529	1.53
		10	983.8	4310	0.516	2.17
		0	984.9	4360	0.503	3.25
		-4.47	985.1	4390	0.498	4.20
-7.36	15	40	966.8	4320	0.518	1.03
		30	971.2	4310	0.507	1.35
		20	975.2	4320	0.496	1.84
		10	978.0	4340	0.485	2.70
		0	979.9	4370	0.474	4.20
		-7.36	980.5	4410	0.466	6.5

(cont.)

Table 4.1. Basic thermophysical properties ... (cont.)

t_F	c_A	t	ρ	c_p	k	μ
-10.92	20	40	958.8	4315	0.484	1.16
		30	963.8	4310	0.475	1.55
		20	968.8	4310	0.466	2.16
		10	972.5	4330	0.457	3.22
		0	975.7	4360	0.448	5.15
		-10	977.8	4385	0.439	9.7
		-10.9	978.0	4390	0.438	10.3
-15.45	25	40	950.5	4310	0.454	1.27
		30	957.0	4300	0.446	1.73
		20	962.0	4290	0.438	2.45
		10	967.0	4280	0.430	3.71
		0	971.0	4270	0.422	6.10
		-10	974.5	4260	0.414	11
		-15.45	975.5	4254	0.410	17
-20.47	30	40	940.8	4240	0.423	1.36
		30	948.0	4240	0.417	1.86
		20	954.0	4230	0.411	2.68
		10	960.0	4200	0.405	4.08
		0	965.4	4170	0.398	6.80
		-10	969.5	4120	0.391	12.2
		-20	972.9	4050	0.385	25
		-20.47	973.0	4045	0.385	26
-29.3	40	40	920.1	4120	0.369	1.47
		30	927.7	4080	0.365	2.01
		20	935.1	4030	0.361	2.90
		10	942.5	3970	0.357	4.35
		0	949.4	3900	0.352	7.05
		-10	956.0	3810	0.348	12.5
		-20	962.3	3710	0.344	24.5
		-29.3	967.5	3610	0.340	50
-37.7	50	40	897.6	3950	0.322	1.48
		30	905.8	3890	0.320	2.00
		20	914.0	3820	0.317	2.84
		10	921.3	3730	0.315	4.20
		0	929.4	3630	0.312	6.60
		-10	937.0	3510	0.309	11
		-20	945.0	3370	0.307	20
		-30	953.0	3230	0.304	41
-44.9	60	-37.7	959.0	3110	0.302	75
		40	874.3	3690	0.280	1.44
		30	882.7	3630	0.279	1.92
		20	891.3	3550	0.278	2.65
		10	899.3	3450	0.277	3.78
		0	907.3	3330	0.276	5.75
		-10	915.5	3200	0.275	8.85
		-20	923.5	3050	0.274	14.7
		-30	931.7	2880	0.273	26.5
		-40	940.0	2710	0.272	55
		-44.9	943.7	2630	0.272	85

4.3 Polynomial equations for computer treatment

The purpose of this part of the chapter is to make basic thermophysical property values available for computer treatment. To make technical calculations of refrigeration and heat pump systems we need, for the chosen freezing point or concentration and the fluid temperature, to know values for basic thermophysical properties such as density, specific heat, thermal conductivity, and dynamic viscosity. Such properties are in 4.2 presented for aqueous solutions of ethyl alcohol. Other property values can then easily be calculated with a method or procedure developed with the kind help of Prof. Jesper Oppelstrup and M. Sc. Thomas Oppelstrup, NADA, KTH. A mathematically suitable polynomial equation form was chosen. A Matlab program was made to generate coefficients to the equations from a set of basic property values.

The basic polynomial equation form chosen for most of these thermophysical properties is

$$f = \sum C_{ij} \cdot (x-x_m)^i \cdot (y-y_m)^j \quad (4.1)$$

where $i+j \leq 5$ and where C_{ij} is the coefficient for each term.

However, the logarithmic equation form of the same polynomial is used for dynamic viscosity:

$$\log f = \sum C_{ij} \cdot (x-x_m)^i \cdot (y-y_m)^j \quad (4.2)$$

also with $i+j \leq 5$.

The freezing point temperature, t_F , or concentration, c_A , can be chosen as input parameter x and the fluid temperature, t , as input parameter y .

Mean values of all x and y values in a whole table, x_m and y_m , have been calculated. In an effort to minimize errors $(x-x_m)$ and $(y-y_m)$ appear in the polynomials given above.

Max values for i , j and $i+j$ have for each table been chosen to minimize the max deviation from the values in the table. Choosing $i \leq 5$, $j \leq 3$ and $i+j \leq 5$ we can write the basic equation so that the coefficients appear in the same order as in the tables:

$$\begin{aligned}
f = & C_{00} + C_{01} \cdot (y - y_m) + C_{02} \cdot (y - y_m)^2 + C_{03} \cdot (y - y_m)^3 + \\
& C_{10} \cdot (x - x_m) + C_{11} \cdot (x - x_m) (y - y_m) + C_{12} \cdot (x - x_m) \cdot (y - y_m)^2 + \\
& C_{13} \cdot (x - x_m) \cdot (y - y_m)^3 + C_{20} \cdot (x - x_m)^2 + C_{21} \cdot (x - x_m)^2 (y - y_m) + \\
& C_{22} \cdot (x - x_m)^2 \cdot (y - y_m)^2 + C_{23} \cdot (x - x_m)^2 \cdot (y - y_m)^3 + C_{30} \cdot (x - x_m)^3 + \\
& C_{31} \cdot (x - x_m)^3 \cdot (y - y_m) + C_{32} \cdot (x - x_m)^3 \cdot (y - y_m)^2 + C_{40} \cdot (x - x_m)^4 + \\
& C_{41} \cdot (x - x_m)^4 \cdot (y - y_m) + C_{50} \cdot (x - x_m)^5
\end{aligned} \quad (4.3)$$

Table 4.2 gives coefficients of aqueous solutions of ethyl alcohol obtained from the Matlab program. The freezing point, t_F , is chosen as x and the temperature, t , as y . Symbols from Table 4.1.

Table 4.2. Coefficients for properties of ethyl alcohol

i	j	c_A	ρ	c_p	k	μ
0	0	3.00E+01	9.61E+02	4.19E+03	4.03E-01	1.48E+00
0	1	1.12E-04	-5.37E-01	2.83E+00	6.54E-04	-4.74E-02
0	2	-1.55E-05	-3.21E-03	-3.58E-02	3.48E-07	4.33E-04
0	3	6.60E-07	1.49E-05	5.22E-04	-2.11E-08	-3.03E-06
1	0	-1.04E+00	1.58E+00	2.30E+01	5.16E-03	-1.38E-02
1	1	-2.81E-05	2.17E-02	-5.87E-01	2.22E-05	-4.57E-05
1	2	-4.74E-07	-2.15E-04	3.67E-03	3.72E-08	5.97E-06
1	3	-2.45E-08	7.94E-07	5.66E-05	-2.01E-10	-1.03E-07
2	0	2.00E-03	-3.93E-02	-2.39E-01	1.39E-05	-8.21E-04
2	1	-5.01E-06	3.26E-04	-6.32E-03	3.48E-07	1.83E-05
2	2	5.43E-08	3.94E-07	3.37E-05	-5.85E-09	-1.63E-07
2	3	-2.33E-09	-6.60E-09	8.83E-07	2.16E-11	-3.68E-09
3	0	-5.19E-04	3.65E-04	-1.49E-02	2.64E-06	-2.74E-05
3	1	2.04E-07	-7.91E-06	5.73E-04	4.22E-08	7.24E-07
3	2	2.04E-09	1.81E-07	-1.20E-06	-2.79E-10	-2.64E-09
4	0	-2.42E-05	7.41E-05	-1.08E-03	1.80E-07	-1.54E-06
4	1	1.49E-08	-1.46E-07	1.77E-05	1.14E-09	2.37E-08
5	0	-4.65E-07	2.13E-06	-1.45E-05	3.80E-09	-2.58E-08
xm =		-20.2024	-20.2024	-20.2024	-20.2024	-20.2024
ym =		8.1578	8.1578	8.1578	8.1578	8.1578

Max relative errors, absolute where tabulated data is zero */

	c_A	ρ	c_p	k	μ
Error	0,1690 */	0,0006	0,0137	0,0044	0,0284
Row nr	5 */	25	22	41	67

The error value marked */ for c_A is an absolute error where for $t_F = 0^\circ\text{C}$ and $t = 0^\circ\text{C}$, $c_A = -0.17$, but should be 0. This and the other rel. error values may be acceptable for technical calculations.

4.4 Comparison of basic values from literature search with polynomial equation values for ethyl alcohol

The basic polynomial equation and the coefficients obtained are put into an Excel sheet that generates values of the basic properties and other properties as well. In order to see how well the Matlab program predicts values for an aqueous solution, the series of property values in Table 4.1 and values with even additive concentrations generated from the Matlab program and Excel sheet are plotted in the same figures. Series from Table 4.1 are given with full lines while series generated from the Matlab program are given with bold dotted lines. This will give a good indication of how useful this tool can be to calculate thermophysical properties of aqueous solutions used as secondary working fluids. Figures 4.1 to 4.4 give density, specific heat, thermal conductivity and dynamic viscosity of aqueous solution of ethyl alcohol.

Discussion of Figures 4.1-4: The figures give basic properties of ethyl alcohol – water and show a good agreement between the basic values (full lines) and the values based on the Matlab program and the Excel sheet (dotted lines).

Deviations in Figure 4.1, giving density, such as between high and low temperatures near curves from 10 to 25% concentration ($t_F = -5^\circ\text{C}$ to -15°C) are small ($\leq 0.2\%$ of the density values).

Deviations in Figure 4.2, giving specific heat, such as between high and low temperatures between curves for 40% concentration and $t_F = -30^\circ\text{C}$ are rather small ($\leq 1\%$ of the specific heat values). The peak between 10 and 15% also is handled rather well by the computer program.

Deviations in Figure 4.3, giving thermal conductivity, such as between high and low temperatures between curves for 50% concentration and $t_F = -40^\circ\text{C}$, are rather small ($\leq 2\%$ of the thermal conductivity values).

Deviations in Figure 4.4, giving dynamic viscosity, are quite uniform and generally relatively small ($\leq 2\text{-}3\%$ of the viscosity values). Note that the peak between 30 and 40% is handled rather well by the computer program.

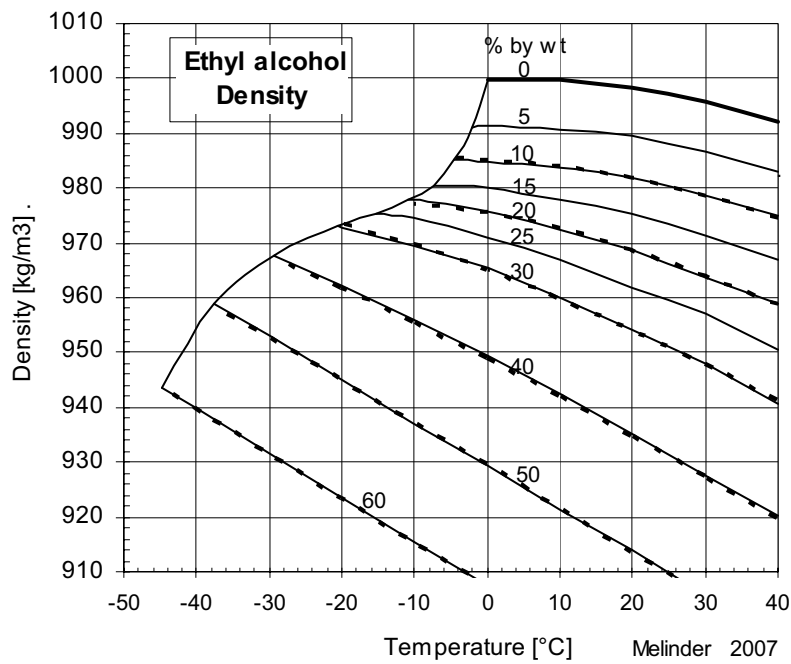


Figure 4.1. Density of ethyl alcohol – water

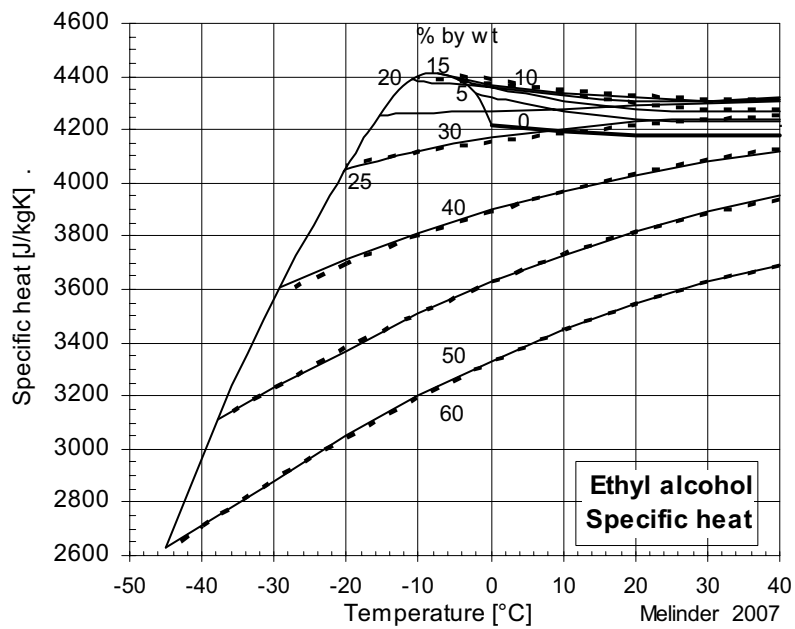


Figure 4.2. Specific heat of ethyl alcohol - water

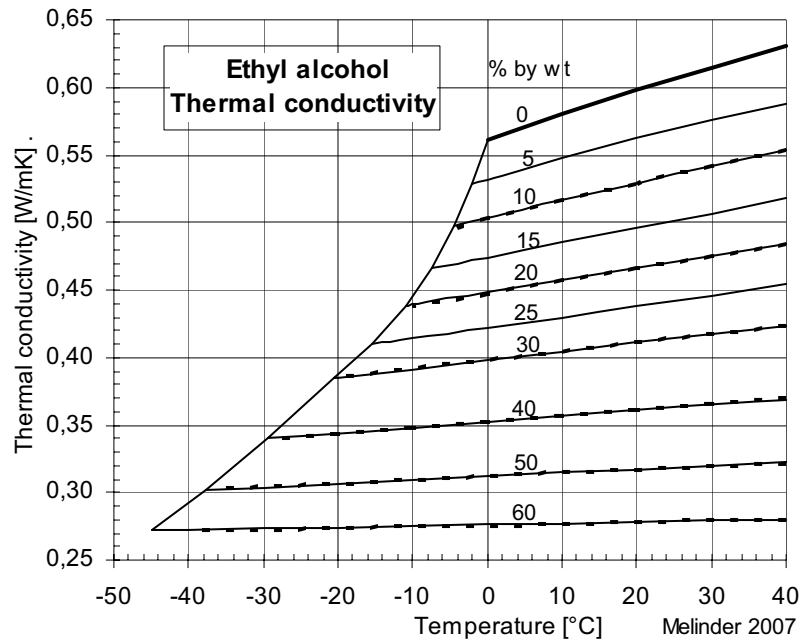


Figure 4.3. Thermal conductivity of ethyl alcohol - water

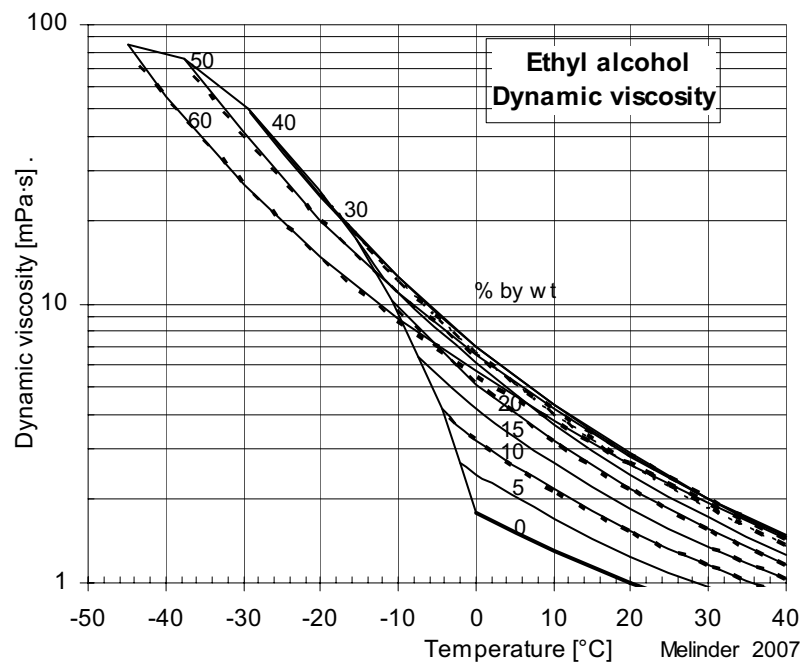


Figure 4.4. Dynamic viscosity of ethyl alcohol - water

4.5 Presenting aqueous solution values of ethyl alcohol

This part describes how selected property values, polynomial equations, Matlab program already described and an Excel program results in tables and charts intended for the industry. Using freezing point and temperature as input values, the other property values can be generated as seen in Table 4.3.

Table 4.3. Example with basic thermophysical properties of aqueous solutions (here for $t_F = -20^\circ\text{C}$ and $t = -10^\circ\text{C}$)

	t_F	t	c_A	ρ	c_p	k	μ
Water solution:	[°C]	[°C]	[wt-%]	[kg/m³]	[J/kgK]	[W/mK]	[mPas]
Validity limits	: 50<x<0	x<y<100					
Ethylene glycol	-20	-10	36,19	1058	3494	0,415	7,99
Validity limits	: 50<x<0	x<y<100					
Propylene glycol	-20	-10	39,41	1045	3620	0,384	21,67
Validity limits	-45<x<0	x<y<40					
Ethyl alcohol	-20	-10	29,73	970	4127	0,393	12,22
Validity limits	-73<x<0	x<y<40					
Methyl alcohol	-20	-10	24,87	969	3914	0,431	5,42
Validity limits	-40<x<0	x<y<40					
Glycerol	-20	-10	46,28	1129	3155	0,407	19,18
Validity limits	-84<x<0	x<y<20					
Ammonia	-20	-10	13,46	953	4239	0,460	2,94
Validity limits	37,5<x<0	x<y<40					
Potassium carbonate	-20	-10	30,95	1322	2877	0,519	6,25
Validity limits	-50<x<0	x<y<40					
Calcium chloride	-20	-10	20,82	1196	2992	0,530	4,48
Validity limits	33,5<x<0	x<y<40					
Magnesium chloride	-20	-10	16,59	1151	3192	0,505	5,58
Validity limits	-21<x<0	x<y<40					
Sodium chloride	-20	-10	22,62	1182	3305	0,530	4,22
Validity limits	-50<x<0	x<y<40					
Potassium acetate	-20	-10	27,91	1155	3232	0,467	5,67
Validity limits	-50<x<0	x<y<40					
Potassium formate	-20	-10	28,94	1192	3121	0,506	3,55
Validity limits	-75<x<0	x<y<40					
Lithium chloride	-20	-10	13,66	1083	3405	0,522	3,97

The results obtained can generate values for any freezing point. Table 4.4 gives property values of ethyl alcohol for $t_F = -5^\circ\text{C}$, -10°C , -15°C , -20°C , -30°C and -40°C . For symbols see Table 4.1.

Table 4.4. Basic thermophysical properties of ethyl alcohol

t_F [°C]	c_A [wt-%]	t [°C]	ρ [kg/m³]	c_p [J/kgK]	k [W/mK]	μ [mPas]
-5	11.1	40	972.8	4287	0.546	0.92
		30	976.9	4285	0.534	1.20
		20	980.3	4310	0.522	1.60
		10	982.7	4350	0.509	2.25
		0	984.0	4396	0.496	3.48
		-5	984.2	4419	0.490	4.51
-10	18.7	40	960.8	4313	0.493	1.12
		30	966.0	4315	0.484	1.50
		20	970.5	4333	0.474	2.09
		10	974.1	4361	0.464	3.09
		0	976.6	4390	0.455	5.00
		-10	978.1	4415	0.445	9.10
-15	24.5	40	951.5	4298	0.456	1.26
		30	957.5	4291	0.448	1.71
		20	963.0	4292	0.440	2.42
		10	967.7	4295	0.433	3.67
		0	971.6	4297	0.425	6.05
		-10	974.5	4291	0.417	11.16
-20	29.7	-15	975.6	4284	0.413	15.89
		40	941.8	4259	0.425	1.37
		30	948.5	4239	0.419	1.86
		20	954.8	4219	0.412	2.65
		10	960.5	4196	0.406	4.03
		0	965.6	4166	0.399	6.67
-30	40.7	-10	970.0	4127	0.393	12.22
		-20	973.6	4074	0.387	25.24
		40	918.4	4119	0.366	1.47
		30	926.0	4078	0.362	2.02
		20	933.5	4024	0.358	2.88
		10	940.8	3957	0.354	4.35
-40	53.1	0	947.8	3877	0.349	7.03
		-10	954.6	3783	0.345	12.4
		-20	961.1	3675	0.341	24.0
		-30	967.1	3553	0.337	52.2
		40	890.8	3862	0.308	1.45
		30	899.0	3812	0.306	1.99
		20	907.1	3740	0.305	2.81
		10	915.2	3650	0.303	4.11
		0	923.2	3542	0.301	6.33
		-10	931.1	3419	0.299	10.4
		-20	939.1	3280	0.297	18.6
		-30	947.1	3129	0.295	36.5
		-40	955.1	2967	0.293	79.9

Based on these basic thermophysical properties, other properties can easily be obtained with simple relations. See Table 4.5.

Table 4.5. Example with thermophysical properties (for $t_F = -20^\circ\text{C}$ and $t = -10^\circ\text{C}$) based on the basic properties in Table 4.4.

	t_F [°C]	t [°C]	c_A [wt-%]	ν [mm ² /s]	Pr [-]	ρc_p [kJ/m ³ K]
Water solution:						
Validity limits	-50<x<0	x<y<100				
Ethylene glycol	-20	-10	36,19	7,55	67,2	3697
Validity limits	-50<x<0	x<y<100				
Propylene glycol	-20	-10	39,41	20,73	204,1	3785
Validity limits	-45<x<0	x<y<40				
Ethyl alcohol	-20	-10	29,73	12,60	128,4	4003
Validity limits	-73<x<0	x<y<40				
Methyl alcohol	-20	-10	24,87	5,59	49,2	3795
Validity limits	-40<x<0	x<y<40				
Glycerol	-20	-10	46,28	16,99	148,7	3562
Validity limits	-84<x<0	x<y<20				
Ammonia	-20	-10	13,46	3,08	27,1	4041
Validity limits	37,5<x<0	x<y<40				
Potassium carbonate	-20	-10	30,95	4,73	34,6	3803
Validity limits	-50<x<0	x<y<40				
Calcium chloride	-20	-10	20,82	3,75	25,3	3579
Validity limits	33,5<x<0	x<y<40				
Magnesium chloride	-20	-10	16,59	4,85	35,3	3674
Validity limits	-21<x<0	x<y<40				
Sodium chloride	-20	-10	22,62	3,57	26,3	3908
Validity limits	-50<x<0	x<y<40				
Potassium acetate	-20	-10	27,91	4,91	39,2	3734
Validity limits	-50<x<0	x<y<40				
Potassium formate	-20	-10	28,94	2,98	21,9	3719
Validity limits	-75<x<0	x<y<40				
Lithium chloride	-20	-10	13,66	3,66	25,9	3688
Input values:	-20	-10				

These tables are in the Excel program extended to make simple calculations for heat transport, pressure drop and heat transfer. This is described further in chapter 5. Similar tables can be generated for each of the aqueous solutions.

In the updated handbook the property values in Table 4.1 will be presented in the same table with the values in Table 4.4. Values from both these tables will also be plotted in the same diagrams.

The good agreement seen in Figures 4.1 to 4.4 can also be observed between the basic series of values from Table 4.1 and the series of values for even freezing points from Tables 4.4 obtained from the Matlab program and the Excel program.

Discussion of Figures 4.5-8:

Here these series of values are plotted in the same figures, Figure 4.5 to 4.8 for aqueous solutions of ethyl alcohol. Series with even concentrations from Table 4.1 are given with full lines. Series with even freezing points from Table 4.4 are given with dashed lines. The freezing points presented are: $t_F = -5^\circ\text{C}$, -10°C , -15°C , -20°C , -30°C and -40°C .

Deviations are generally quite small: In Figure 4.5 $\leq 0.2\%$ of the density values; in Figure 4.6 $\leq 1\%$ of the specific heat values; in Figure 4.7 $\leq 1\%$ of the thermal conductivity values; in Figure 4.8 deviations are quite uniform and $\leq 2\text{-}3\%$ of the viscosity values.

This rather good agreement between the basic values (full lines) and the values for even freezing points based on the Matlab and Excel programs (dashed lines) for ethyl alcohol is a good indication that we can expect similar or even better agreement for the other aqueous solutions. As earlier mentioned, the basic thermo-physical properties of aqueous solutions of ethyl alcohol present more challenges than most other fluids examined.

The moderate deviations are also a good indication that we can expect any computer program based on the Matlab and Excel programs to represent the basic values well.

In 4.6 and 4.7, tables and figures with basic properties of aqueous solutions of lithium chloride and potassium formate are presented in a similar way.

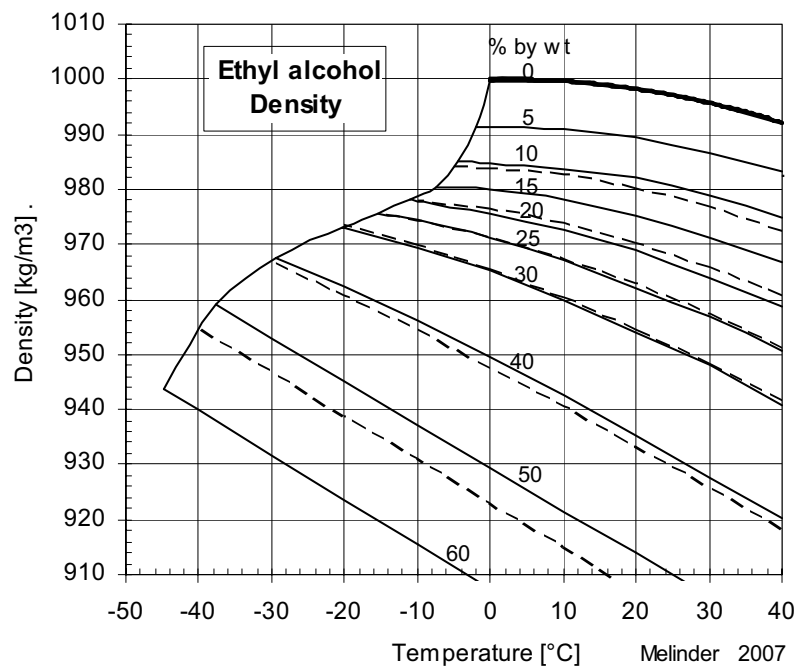


Figure 4.5. Density of ethyl alcohol - water

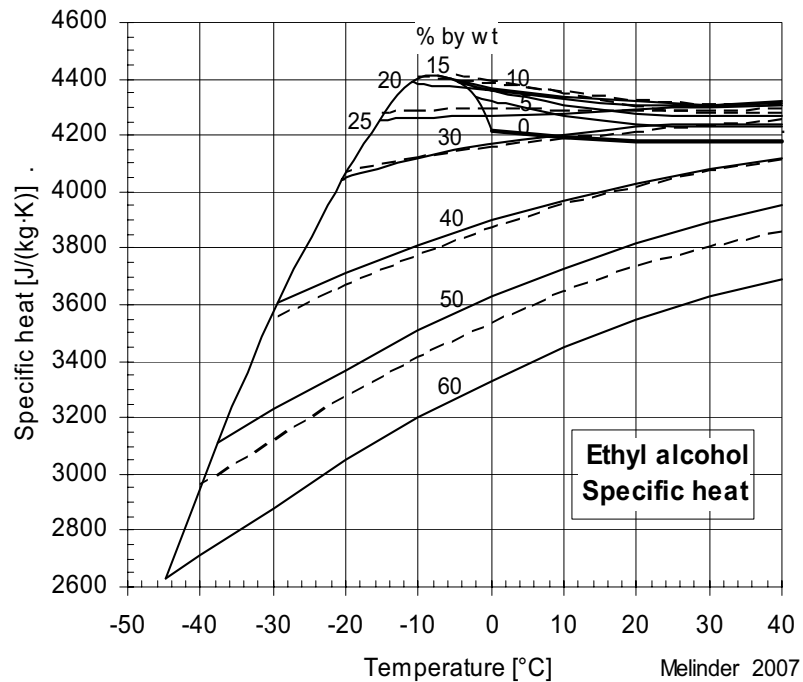


Figure 4.6. Specific heat of ethyl alcohol - water

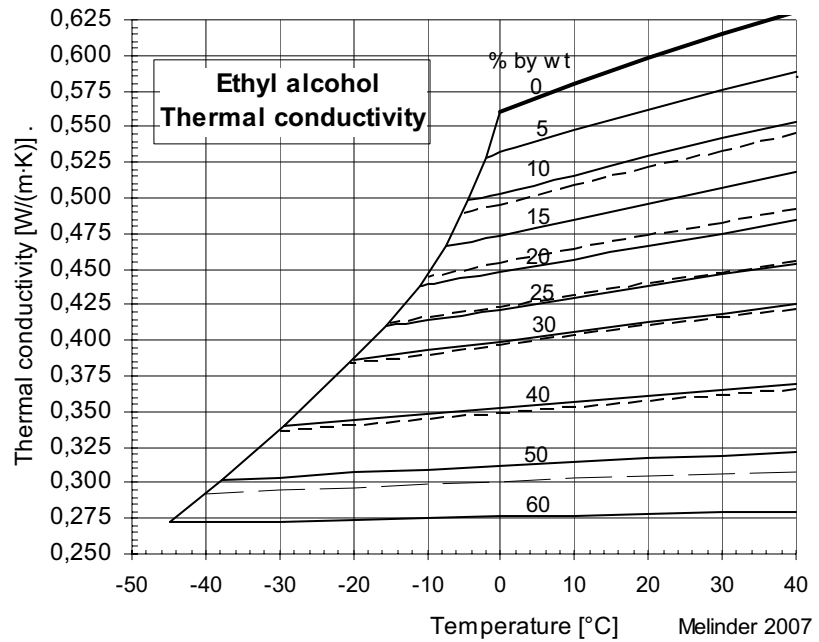


Figure 4.7. Thermal conductivity of ethyl alcohol - water

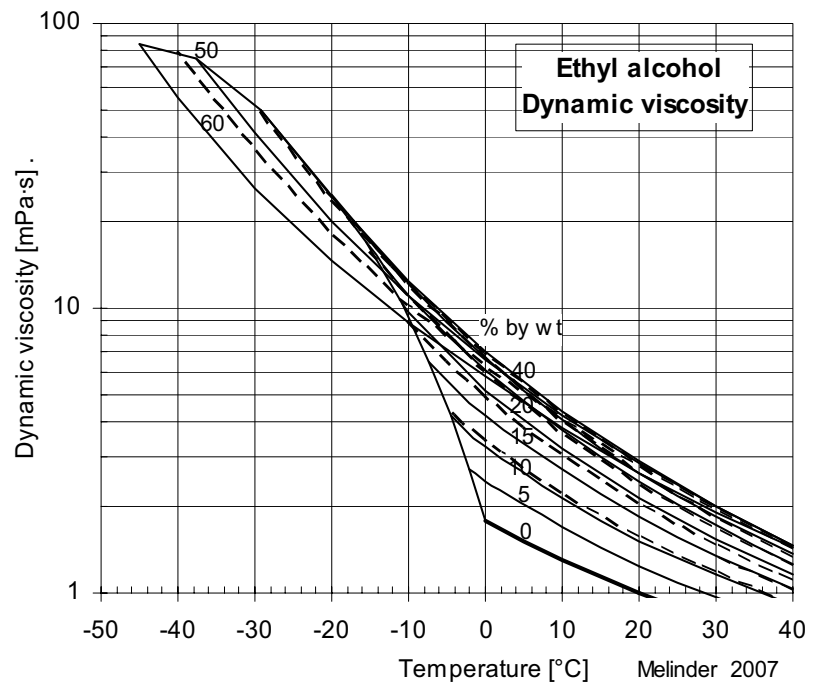


Figure 4.8. Dynamic viscosity of ethyl alcohol - water

4.6 Presenting aqueous solution values of lithium chloride

The critical evaluation of property values of aqueous solutions of lithium chloride presented in 3.3.3 has resulted in the following property values. Table 4.6 gives basic property values of aqueous solutions of lithium chloride. In Figures 4.9 to 4.12, series with even concentrations are plotted together with series with even freezing points.

Table 4.6. Basic thermophysical properties of lithium chloride

Freezing Point	Concentration	Temperature	Density	Specific heat	Thermal conduct.	Dynamic viscosity
t_F °C	Conc. wt-%	t °C	ρ kg/m ³	c_p J/kgK	k W/mK	μ mPa·s
-1.72	2	40	1004	4080	0.625	0.69
		30	1007	4080	0.609	0.86
		20	1010	4080	0.592	1.07
		10	1011	4080	0.575	1.39
		0	1012	4090	0.557	1.90
		-1.72	1012	4090	0.554	2.01
-4.86	5	40	1021	3920	0.617	0.78
		30	1024	3910	0.602	0.94
		20	1027	3910	0.585	1.18
		10	1029	3910	0.569	1.55
		0	1030	3915	0.552	2.06
		-4.86	1030	3920	0.544	2.40
-10	8.6	40	1042	3766	0.607	0.89
		30	1045	3748	0.592	1.08
		20	1048	3731	0.577	1.34
		10	1050	3715	0.561	1.72
		0	1052	3702	0.545	2.32
		-10	1052	3694	0.529	3.33
-12.6	10	40	1050	3710	0.603	0.93
		30	1053	3690	0.589	1.13
		20	1056	3670	0.573	1.41
		10	1058	3650	0.558	1.84
		0	1060	3630	0.543	2.45
		-10	1060	3610	0.527	3.45
-15	11.5	-12.6	1061	3610	0.523	3.80
		40	1059	3652	0.599	0.99
		30	1062	3626	0.585	1.20
		20	1065	3599	0.570	1.48
		10	1067	3574	0.555	1.90
		0	1069	3549	0.540	2.56
		-10	1070	3528	0.525	3.67
		-15	1070	3518	0.517	4.51

Table 4.6. Basic thermophysical properties ... (cont.)

t_F	Conc.	T	ρ	c_p	K	μ
-20	13.7	40	1071	3572	0.594	1.08
		30	1075	3539	0.580	1.30
		20	1078	3505	0.566	1.61
		10	1080	3471	0.551	2.06
		0	1082	3437	0.536	2.77
		-10	1083	3405	0.522	3.97
		-20	1084	3374	0.507	6.10
-23.5	15	40	1079	3530	0.591	1.12
		30	1082	3490	0.577	1.37
		20	1085	3450	0.563	1.70
		10	1088	3410	0.548	2.19
		0	1089	3370	0.534	2.91
		-10	1091	3330	0.520	4.1
		-20	1092	3290	0.505	6.3
-30	16.8	-23.5	1092	3280	0.500	7.8
		40	1090	3472	0.587	1.21
		30	1093	3426	0.573	1.47
		20	1096	3379	0.559	1.82
		10	1099	3332	0.545	2.34
		0	1101	3284	0.531	3.14
		-10	1102	3236	0.516	4.5
-40	19.1	-20	1103	3189	0.502	6.8
		-30	1104	3142	0.488	11.1
		40	1104	3400	0.582	1.33
		30	1107	3343	0.568	1.62
		20	1110	3287	0.554	2.01
		10	1113	3230	0.540	2.59
		0	1115	3173	0.526	3.47
-44.5	20	-10	1117	3116	0.512	4.89
		-20	1118	3058	0.498	7.29
		-30	1119	3000	0.484	11.64
		-40	1120	2941	0.471	20.03
		40	1109	3370	0.579	1.40
		30	1112	3310	0.566	1.71
		20	1116	3250	0.553	2.12
		10	1118	3190	0.538	2.68
		0	1121	3130	0.524	3.57
		-10	1123	3070	0.510	5.02
		-20	1124	3010	0.497	7.55
		-30	1125	2950	0.483	12
		-40	1126	2890	0.469	20
		-44.5	1126	2860	0.463	26

Cont.

Table 4.6. Basic thermophysical properties ... (cont.)

t_F	Conc.	t	ρ	c_p	K	μ
-50	21.2	40	1116	3333	0.576	1.47
		30	1120	3269	0.563	1.79
		20	1123	3205	0.549	2.24
		10	1126	3142	0.535	2.88
		0	1128	3078	0.522	3.85
		-10	1130	3014	0.508	5.38
		-20	1132	2950	0.494	7.91
		-30	1134	2886	0.481	12.3
		-40	1135	2821	0.467	20.3
		-50	1135	2755	0.454	35.9
-56	22.5	40	1123	3295	0.572	1.55
		30	1127	3230	0.559	1.94
		20	1130	3160	0.546	2.41
		10	1133	3095	0.532	3.1
		0	1136	3030	0.519	4.1
		-10	1139	2960	0.505	5.7
		-20	1140	2890	0.492	8.4
		-30	1142	2830	0.479	13
		-40	1143	2760	0.465	21
		-50	1144	2690	0.452	36
-60	23.3	-56	1145	2650	0.445	50
		40	1128	3273	0.569	1.66
		30	1132	3204	0.557	2.01
		20	1135	3135	0.544	2.51
		10	1138	3066	0.530	3.23
		0	1141	2998	0.517	4.33
		-10	1143	2930	0.504	6.02
		-20	1146	2862	0.491	8.75
		-30	1147	2793	0.477	13.3
		-40	1149	2725	0.464	21.3
		-50	1150	2657	0.451	36.0
		-60	1150	2589	0.438	64.4

*/ Chemical formula: LiCl; M = 42,40

Discussion of Figures 4.9-12:

Figure 4.9-12 show good agreement between the basic values (full lines) and the values for even freezing points based on the Matlab and Excel programs (dotted lines). This can be seen by the fact that the curves are parallel to each other and The freezing points presented are: $t_F = -5^\circ\text{C}$, -10°C , -15°C , -20°C , -30°C , -40°C -50°C and -60°C . Deviations are generally quite small: In Figure 4.9 $\leq 0.2\%$ of the density values; in Figure 4.10 $\leq 1\%$ of the specific heat values; in Figure 4.11 $\leq 1\%$ of the thermal conductivity values; in Figure 4.12 deviations are quite uniform and $\leq 2\text{-}3\%$ of the viscosity values.

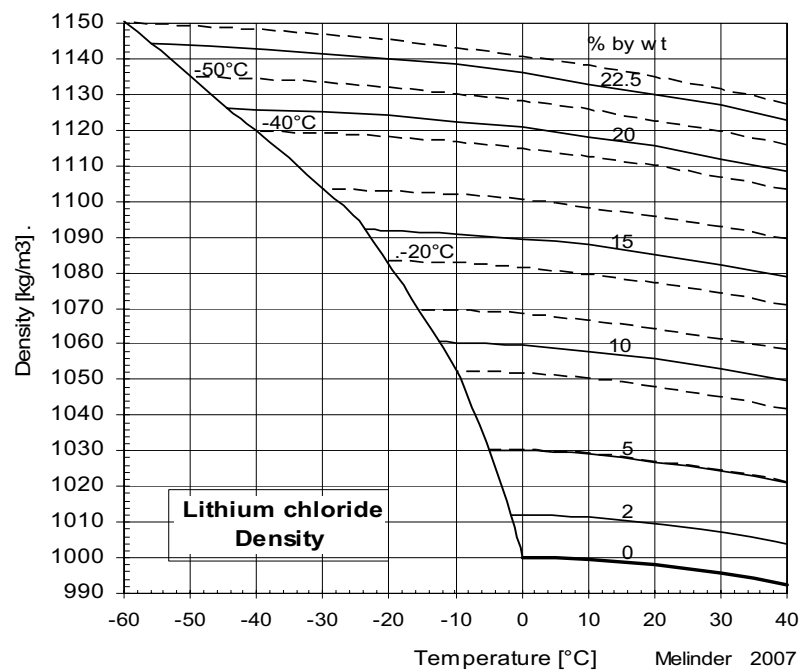


Figure 4.9. Density of lithium chloride - water

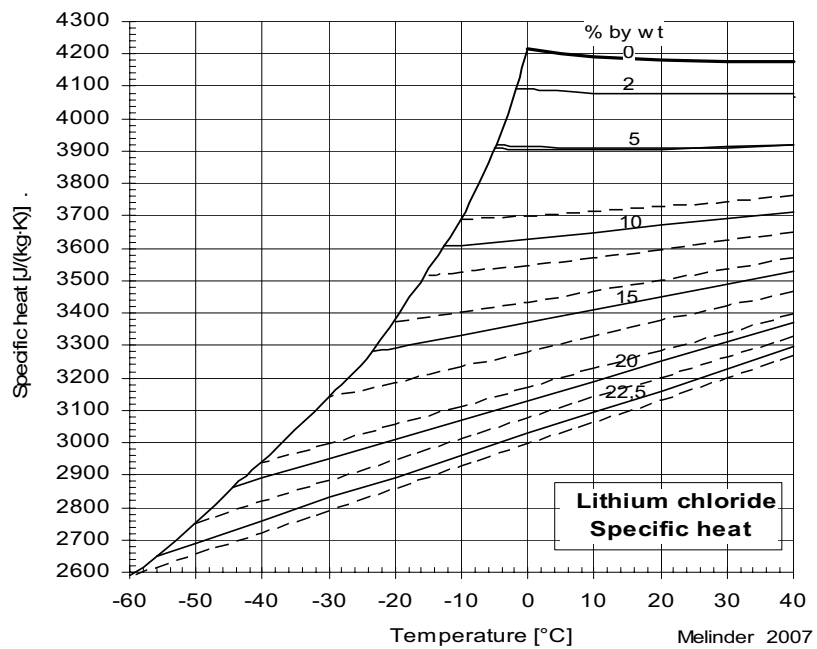


Figure 4.10. Specific heat of lithium chloride - water

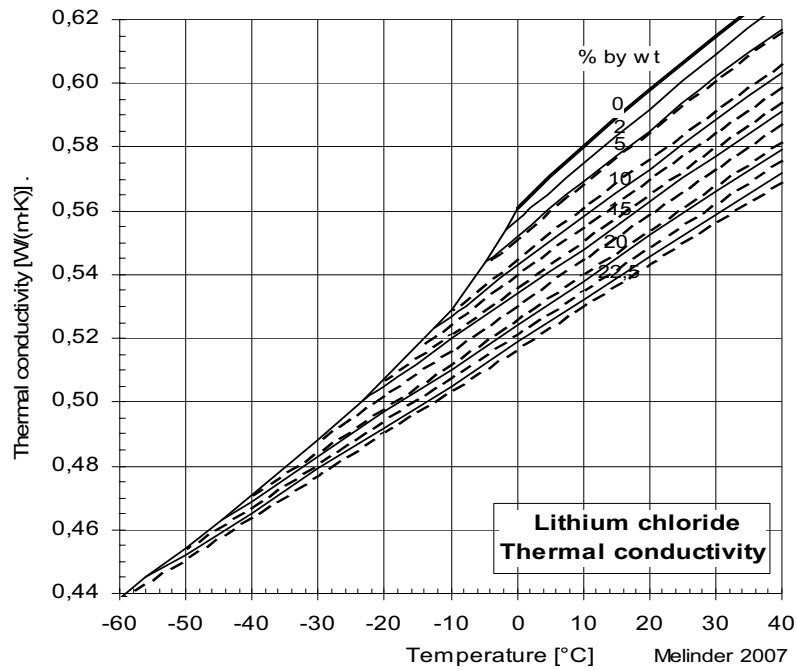


Figure 4.11. Thermal conductivity of lithium chloride - water

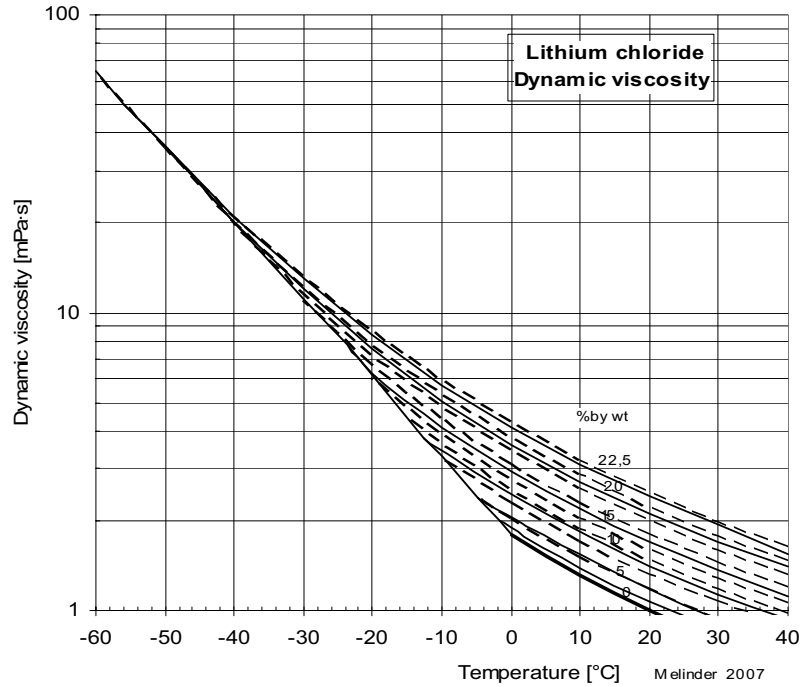


Figure 4.12. Dynamic viscosity of lithium chloride - water

4.7 Presenting aqueous solution values of potassium formate

The critical evaluation of property values of aqueous solutions of potassium formate presented in 3.3.4 has resulted in the following property values. Table 4.7 gives basic property values of aqueous solutions of potassium formate. In Figures 4.13-4.16, series with even concentrations are plotted together with series with even freezing points. Deviations are smaller than with ethyl alcohol.

Table 4.7. Basic thermophysical properties of potassium formate

Freezing Point	Concentration	Temperature	Density	Specific heat	Thermal conduct.	Dynamic Viscosity
t_F °C	Conc. wt-%	t °C	ρ kg/m ³	c_p J/kgK	k W/mK	μ mPa·s
-2.2	5	40	1023	4000	0.616	0.72
		30	1026	4000	0.602	0.88
		20	1029	4000	0.587	1.09
		10	1031	4005	0.572	1.41
		0	1032	4010	0.556	1.9
		-2.2	1032	4010	0.552	2.05
-5	10.2	40	1054	3815	0.605	0.78
		30	1057	3808	0.592	0.95
		20	1060	3803	0.578	1.18
		10	1062	3801	0.563	1.50
		0	1064	3801	0.548	2.00
		-5.0	1065	3803	0.541	2.35
-10	17.9	40	1102	3552	0.589	0.90
		30	1105	3539	0.576	1.08
		20	1108	3527	0.564	1.33
		10	1111	3517	0.551	1.68
		0	1114	3507	0.537	2.20
		-10	1115	3500	0.523	3.02
-11.5	20	40	1114	3490	0.586	0.92
		30	1118	3480	0.573	1.11
		20	1121	3465	0.560	1.35
		10	1124	3455	0.547	1.70
		0	1127	3440	0.534	2.23
		-10	1129	3430	0.521	3.08
-15	24.0	-11.5	1129	3428	0.519	3.25
		40	1141	3356	0.577	1.01
		30	1144	3340	0.565	1.21
		20	1148	3325	0.552	1.47
		10	1151	3309	0.540	1.84
		0	1154	3294	0.527	2.40
		-10	1157	3280	0.515	3.28
		-15	1158	3273	0.508	3.91

Table 4.7. Basic thermophysical properties ... (cont.)

t_F °C	Conc. wt-%	t °C	ρ kg/m ³	c_p J/kgK	k W/mK	μ mPa·s
-16,0	25	40	1148	3320	0.576	1.04
		30	1152	3305	0.563	1.24
		20	1155	3290	0.550	1.51
		10	1159	3275	0.538	1.88
		0	1162	3260	0.525	2.44
		-10	1165	3245	0.513	3.32
		-16,0	1166	3235	0.505	4.1
-20	28,9	40	1174	3207	0.568	1.12
		30	1178	3190	0.555	1.33
		20	1181	3173	0.543	1.61
		10	1185	3155	0.531	2.01
		0	1189	3138	0.518	2.61
		-10	1192	3121	0.506	3.55
		-20	1194	3103	0.494	5.14
-21,3	30	40	1182	3180	0,565	1,15
		30	1186	3160	0,553	1,36
		20	1190	3140	0,541	1,64
		10	1193	3125	0,529	2,06
		0	1197	3105	0,517	2,69
		-10	1200	3085	0,504	3,66
		-20	1202	3070	0,492	5,25
		-21,3	1202	3065	0,490	5,50
-30	36,8	40	1229	2995	0,550	1,32
		30	1233	2977	0,538	1,57
		20	1237	2959	0,526	1,89
		10	1242	2941	0,514	2,34
		0	1246	2923	0,502	3,01
		-10	1249	2905	0,490	4,11
		-20	1252	2886	0,478	6,0
		-30	1255	2866	0,466	9,5
-35,0	40	40	1252	2920	0,540	1,40
		30	1257	2900	0,529	1,67
		20	1261	2885	0,517	2,00
		10	1265	2865	0,506	2,50
		0	1269	2850	0,494	3,20
		-10	1273	2830	0,482	4,40
		-20	1276	2810	0,471	6,40
		-30	1279	2795	0,459	10,0
		-35	1280	2785	0,453	13,5

Table 4.7. Basic thermophysical properties ... (cont.)

t_F °C	Conc. wt-%	t °C	ρ kg/m ³	c_p J/kgK	k W/mK	μ mPa·s
-40	42,9	40	1274	2846	0,531	1,50
		30	1279	2827	0,520	1,79
		20	1284	2808	0,509	2,16
		10	1288	2790	0,497	2,68
		0	1292	2772	0,486	3,46
		-10	1296	2755	0,474	4,74
		-20	1299	2737	0,463	6,98
		-30	1302	2719	0,451	11,2
		-40	1304	2700	0,440	20,0
-50,0	48	40	1316	2735	0,517	1,78
		30	1321	2710	0,506	2,15
		20	1326	2695	0,495	2,65
		10	1330	2675	0,484	3,32
		0	1335	2655	0,472	4,4
		-10	1338	2640	0,461	6,1
		-20	1342	2620	0,449	9,2
		-30	1345	2605	0,438	14,8
		-40	1348	2585	0,426	26,5
		-50	1350	2570	0,415	57

*/ Chemical formula: KHCO₂; M = 84.12

Discussion of Figures 4.13-16:

Figure 4.13-16 show good agreement between the basic values (full lines) and the values for even freezing points based on the Matlab and Excel programs (dotted lines). The freezing points presented are: $t_F = -5^\circ\text{C}$, -10°C , -15°C , -20°C , -30°C , and -40°C .

As with lithium chloride, deviations for potassium formate are rather small: In Figure 4.9 $\leq 0.2\%$ of the density values; in Figure 4.10 $\leq 1\%$ of the specific heat values; in Figure 4.11 $\leq 1\%$ of the thermal conductivity values; in Figure 4.12 deviations are quite uniform and generally relatively small ($\leq 2\text{-}3\%$ of the viscosity values).

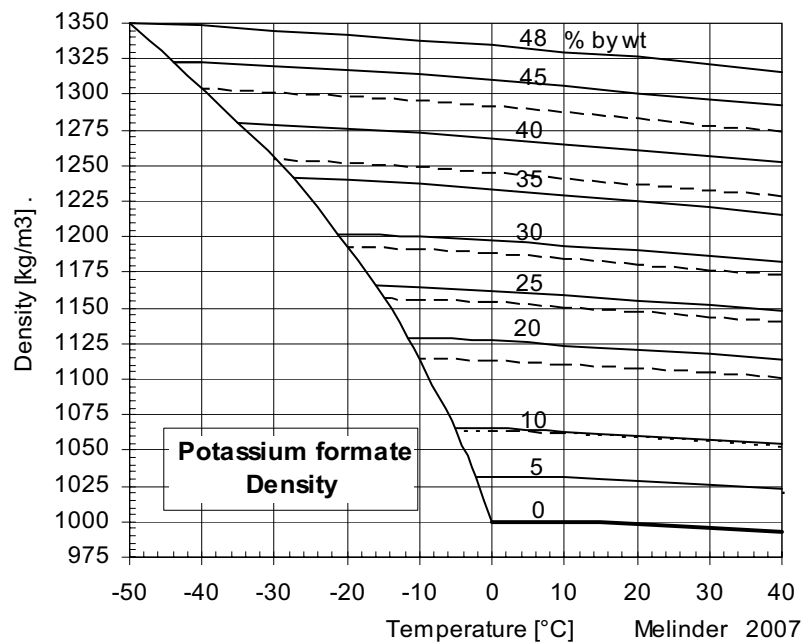


Figure 4.13. Density of potassium formate - water

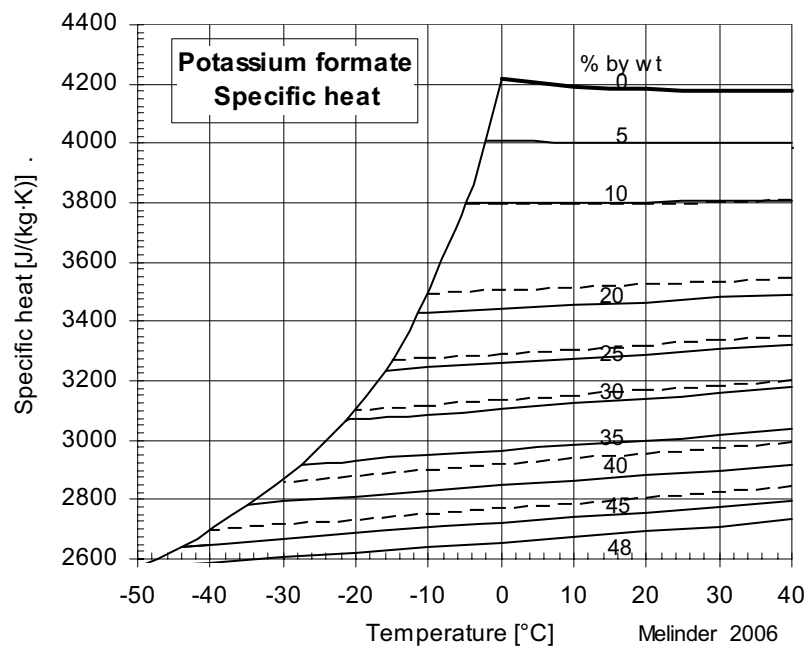


Figure 4.14. Specific heat of potassium formate – water

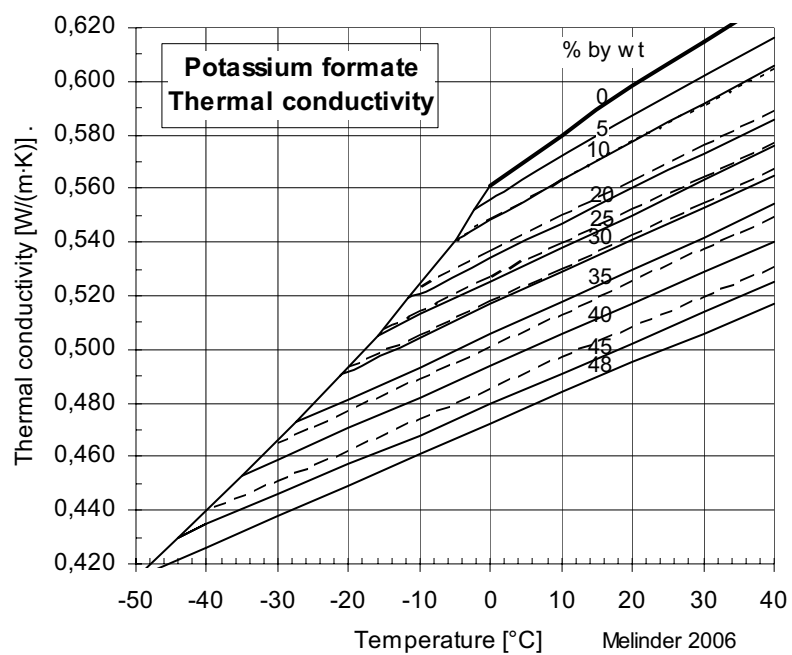


Figure 4.15. Thermal conductivity of potassium formate – water

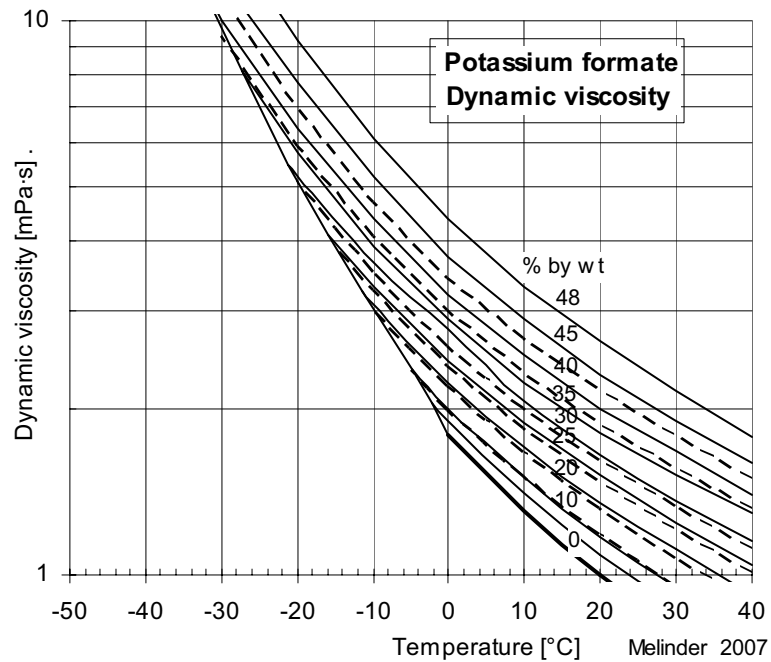


Figure 4.16. Dynamic viscosity of potassium formate – water

5. Comparison of aqueous solutions as to heat transport, pressure drop and heat transfer

5.1. Introduction

Indirect systems with secondary working fluids are used extensively in heat pumps and in supermarkets in Sweden. Using such indirect systems make it possible to minimize the primary refrigerant charge and leakage which is important from environmental aspects. A secondary working fluid should have enough freezing security, good transport and heat transfer abilities and the viscosity should not be too high at the operating temperatures, as it determines the type of flow occurring and the pressure drop.

This chapter compares a few types of aqueous solutions used as secondary working fluids for two types of applications. The first is bedrock heat pumps and the second is cooling cabinets in supermarkets. Low temperature applications are considered in [PAPER 5](#) (Melinder 2004) and [PAPER 7](#) (Melinder 2005).

The additive mass fraction or concentration of the aqueous solutions is often higher than necessary for the application. How does the additive concentration affect the thermal performance? As an aid to find out, the aqueous solutions used in the examples are compared with different concentrations, giving the freezing points $t_F = -10^\circ\text{C}$, -15°C and -20°C for heat pumps and $t_F = -15^\circ\text{C}$ and -20°C for cooling cabinets.

Typical examples are used to compare the fluids as to volumetric heat capacity, viscosity, Reynolds number, needed pumping power as well as heat transfer that affect temperature differences in heat exchangers. The result is presented in figures that also give indications how the type of flow influences these properties.

What equations are used in the comparisons chosen for heat transport, heat transfer and pressure drop?

5.2 Equations for heat transport, heat transfer and pressure drop

5.2.1 Heat transport and volumetric heat capacity

The cooling capacity, $\dot{Q}$ [W] that can be transported through pipes or tubes with a given secondary fluid flow rate and temperature change can for a fluid be obtained with the following relations:

$$\dot{Q} = m' c_p \Delta t = \dot{V} \rho c_p \Delta t \quad (5.1)$$

In these equations, m' [kg/s] is the mass flow rate, $\dot{V}$ [m³/s] the volume flow rate, Δt [K] the temperature change between fluid exiting and entering the cooling object, ρ [kg/m³] the fluid density and c_p [J/kgK] its specific heat. The cooling capacity, $\dot{Q}$, is then proportional to the volumetric heat capacity, ρc_p .

5.2.2 Friction coefficient, pressure drop and pumping power

In order to use a simple pump with moderate pumping power the fluid should have low viscosity giving small pressure drop for the system fluid flow especially with laminar flow. The pumping power is

$$\dot{E}_p = \Delta p_f \cdot \dot{V} / \eta_p \quad (5.2)$$

In this equation, η_p [–] is the efficiency of the pump (incl electric motor) and Δp_f [Pa] is the pressure drop due to friction for flow in a tube. This pressure drop due to friction, Δp_f , can be estimated with the following equation, where f is the friction coefficient, w is the fluid velocity, L is the total tube length and d is the inner tube diameter.

$$\Delta p_f = (f/2) \cdot \rho \cdot w^2 \cdot L / d \quad (5.3)$$

The friction coefficient for laminar flow ($Re \leq 2300$) is:

$$f = 64 / Re \quad (5.4)$$

where

$$Re = wd/\nu = wd\rho/\mu \quad (5.5)$$

In eqn 5.5 the μ [Ns/m²] = $\rho\nu$ is the dynamic viscosity and ν [m²/s] is the kinematic viscosity.

A commonly used friction coefficient equation for turbulent flow (valid for $3000 < Re < 10000$) is Blasius relation:

$$f = 0.316/Re^{0.25} \quad (5.6)$$

A friction coefficient given in Gnielinski (1976) is valid for turbulent flow as well as the transitional region ($Re \geq 2300$):

$$f = (0.79 \cdot \ln Re - 1.64)^{-2} \quad (5.7)$$

In our comparison we will use equations 5.4 and 5.7 that together cover all types of flow that are of interest here.

The basic thermo-physical property that has most influence the friction coefficient is the viscosity. The influence of viscosity is stronger with laminar flow than with turbulent flow as is indicated by the equations above.

5.2.3 Heat transfer and temperature differences

Good heat transfer is desired in order to get small temperature differences between the liquid and inner tube wall in the collector tube and also the evaporator. This temperature difference, $\mathcal{G}_m$, can for a circular tube (with $A = \pi d \cdot L$) be estimated with

$$\dot{Q} = \alpha \cdot A \cdot \mathcal{G}_m \quad (5.8)$$

The heat transfer coefficient, α [W/m²K], of a fluid with laminar flow is here estimated with the equation by Sieder and Tate (1956) that is valid for $Re \leq 2300$ and where the geometry has a large influence:

$$Nu = 1.86 \cdot (Re \cdot Pr)^{1/3} \cdot (d/L_{str})^{1/3} (\mu/\mu_w)^{0.14} \quad (5.9)$$

Here the Nusselt number is defined $Nu = \alpha d/k$, the Reynolds number $Re = wd/\nu$ and the Prandtl number $Pr = \mu c_p/k$ where $\mu = \nu\rho$ is the fluid viscosity at bulk temperature and μ_w at wall temperature.

One equation valid for heat transfer at turbulent flow in circular tubes is:

$$Nu = 0.023 \cdot Re^{0.8} \cdot Pr^{1/3} \quad (5.10)$$

This equation is valid for flow in relatively long smooth tubes in turbulent flow with $10^4 < Re < 2 \cdot 10^5$ but it can with some uncertainty be extrapolated for use down to Re about $3 \cdot 10^3$.

The heat transfer coefficient of a fluid with non-laminar flow is here calculated with the following equation by Gnielinski, that with the friction coefficient in equation 5.7 can be used for the transition region as well as for turbulent flow, as it is valid for $2300 < Re < 10^6$ and $Pr > 0.5$ (Gnielinski, 1976):

$$Nu = \frac{(f/8)(Re-1000)Pr}{1 + 12.7(f/8)^{1/2}(Pr^{2/3} - 1)} \quad (5.11)$$

Tests at our department of the influence of Reynolds number on heat transfer indicate the following: For $Re < 3000 - 4000$ the heat transfer coefficient decreases rapidly due to transition to laminar flow in harmony with what equation 5.11 gives (See Melinder 1989, Melinder 1998a,b, Granryd and Melinder 2005)

5.3 Comparison for a bedrock heat pump application

5.3.1 Example used for comparison of fluid types and concentrations

In order to make a comparison of the fluids for a domestic heat pump application with rock as heat source the following temperatures and other parameters are chosen. Typical operating temperatures for the fluids differ at different places and are here set between $+4^\circ\text{C}$ to -4°C as indicated in Figures 5.1-5. In order to see how the additive concentration and freezing point of the aqueous solutions influence performance, aqueous solutions of propylene glycol, ethyl alcohol and glycerol are compared with concentrations giving the freezing points of $t_F = -10^\circ\text{C}$, -15°C and -20°C .

The reason for the comparison of the fluids at three different concentrations is the following. The freezing point temperature should be below the operating temperature of the secondary fluid. Aque-

ous solutions do not freeze to solid ice even if the temperature is somewhat lower than the freezing point except near eutectic concentration (See chapter 2.1). The fluid used should have enough but not more than needed freezing security (5-10 K) as a higher concentration of the freezing point depressant additive will mean poorer thermophysical properties as we will see in this example.

A domestic heat pump with rock as heat source use circular collector tubes with the length $L = 200$ m and an inner diameter of $d_i = 0.035$ m. Let us assume that a heat flow of $\dot{Q} = 5000$ W is drawn from the ground and transported with the fluid having the velocity $w = 0.6$ m/s. The pumping efficiency is set to $\eta_p = 0.4$.

In a domestic heat pump with rock as heat source the tube parts that are near the ground surface can be well insulated. The coldest part would most likely be at the evaporator and the freezing point of the fluid could be chosen near the evaporation temperature.

5.3.2 Heat transport ability and volumetric heat capacity

The refrigerating capacity $\dot{Q}$ [W] that can be transported with a given volume flow and temperature change is proportional to the volumetric heat capacity, ρc_p [kJ/m³K].

How do the various concentrations or freezing points of the three aqueous solutions compare as to volumetric heat capacity that determines the heat transport ability of the fluid?

Figure 5.1 shows that ethyl alcohol gives 5-20 % higher ρc_p values than propylene glycol and glycerol. The high values of ethyl alcohol at $t_F = -10^\circ\text{C}$ and also -15°C are consistent with Figures 3.1, 4.2 and 4.6 showing a higher value of c_p than water. The volumetric heat capacity decreases with higher additive concentrations.

5.3.3 Viscosity and type of flow

The kinematic viscosity, $\nu = \mu/\rho$, is a crucial property as it determines the type of flow we get, whether laminar or not. The Reynolds number, $Re = wd/\nu = wdp/\mu$, is inversely proportional to the viscosity. A high viscosity makes it impossible to keep the flow turbulent with a reasonable pumping power in a collector tube and

also in a heat exchanger with circular tubes. How do the chosen fluids and concentrations compare?

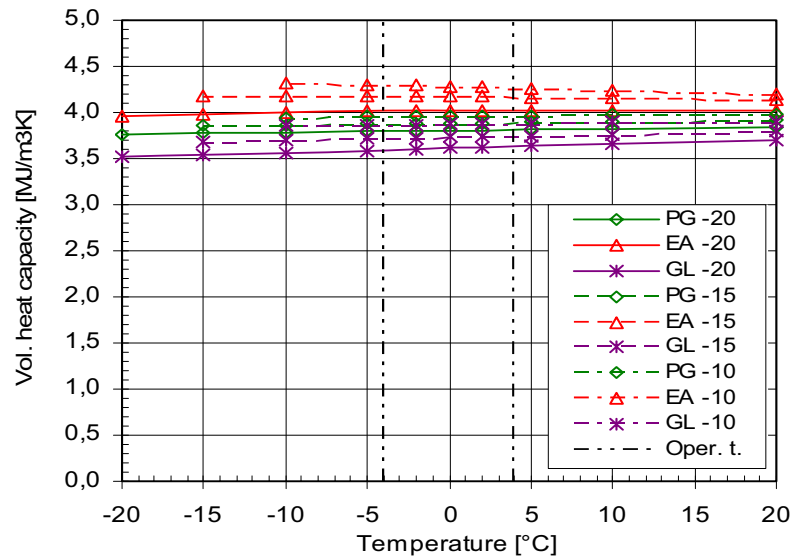


Figure 5.1. Volumetric heat capacity as function of fluid temp.

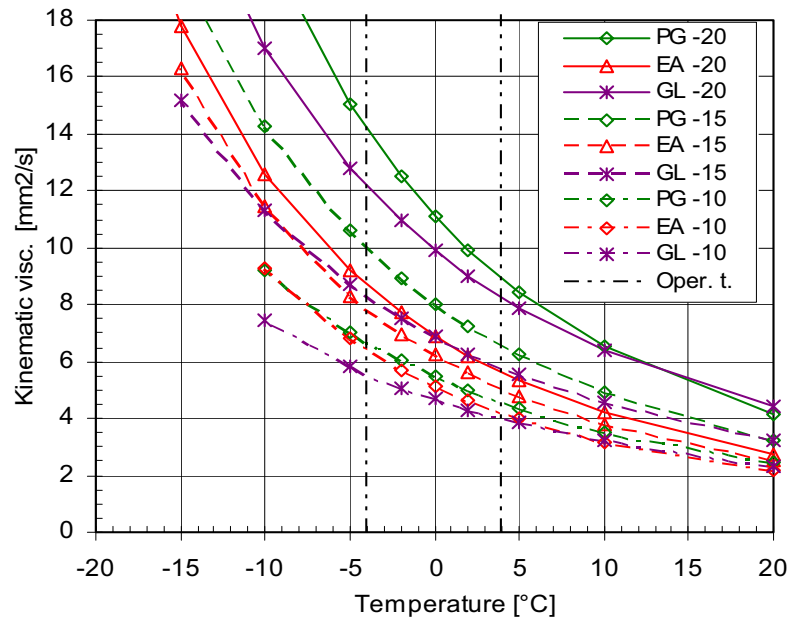


Figure 5.2. Kinematic viscosity as function of fluid temperature

Figure 5.2 shows that the viscosity of ethyl alcohol solutions is much less affected by the additive concentration or freezing point than propylene glycol and glycerol solutions. This is consistent with the viscosity peak at $\approx 40\%$ seen in Figures 4.4 and 4.8. However, at concentrations giving $t_F = -10^\circ\text{C}$, the viscosity of glycerol is even lower than that of ethyl alcohol.

What type of flow can we expect for the example of a domestic heat pump using circular collector tubes with an inner diameter of $d_i = 0.035\text{ m}$ and a fluid velocity $w = 0.6\text{ m/s}$.

Figure 5.3 gives the Reynolds number, Re [-], as function of the fluid temperature for the given example. The figure shows that we at operation temperatures near 0°C can expect laminar flow (below $Re = 2300$) for propylene glycol and glycerol and freezing point $t_F = -20^\circ\text{C}$. For propylene glycol and $t_F = -15^\circ\text{C}$ as well as for ethyl alcohol and $t_F = -15^\circ\text{C}$ and -20°C , we may get transitional flow ($2300 < Re < 3000$) but not for $t_F = -10^\circ\text{C}$. For the other freezing points of the aqueous solutions we can expect turbulent flow.

5.3.4 Friction coefficient, pressure drop and pumping power

In order to be able to use a pump with moderate pumping power the fluid should have low viscosity giving small friction coefficient and pressure drop for the system fluid flow. The pressure drop due to friction, Δp_f , for flow in a tube (in which no consideration is taken for inlet and exit losses or pressure drop in bends and valves etc) are estimated with equations 5.2-4 and 5.6.

Figure 5.4 gives pumping power as function of fluid temperature for the four concentrations of the two aqueous solutions. From the charts we see that the values go down 30-40% when the flow turns laminar but increases quickly again with lower temperature that gives higher viscosity. As we here consider only the pressure drop due to friction, both pressure drop and pumping power are proportional to the friction coefficient. The pumping power is only between 45 and 75 W with the chosen pump efficiency.

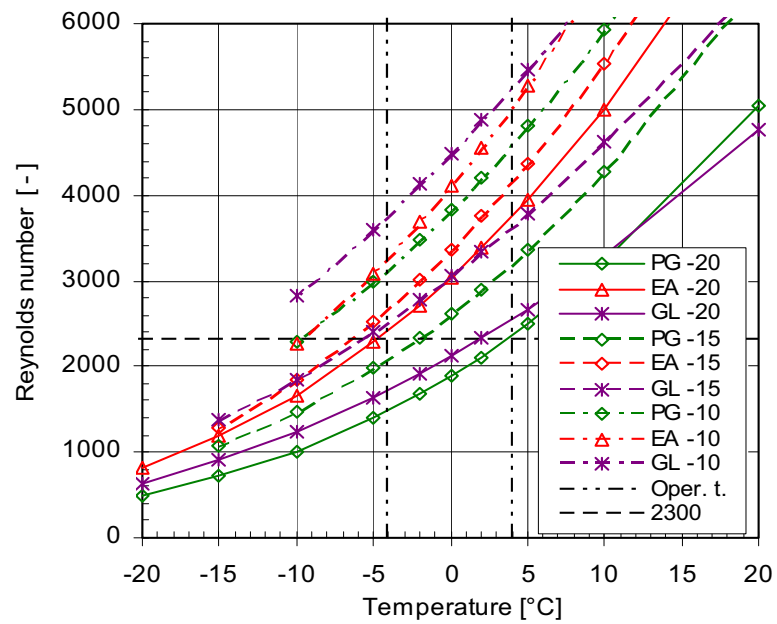


Figure 5.3. Reynolds number as function of fluid temperature

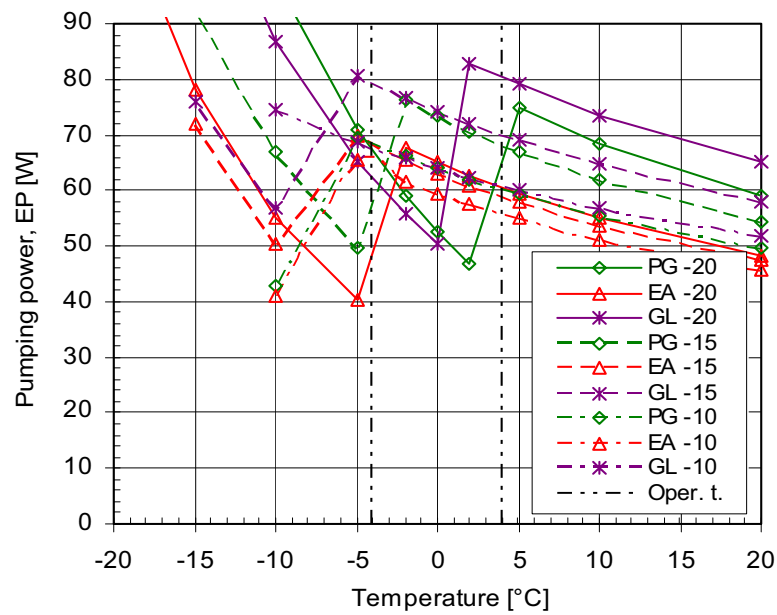


Figure 5.4. Pumping power as function of fluid temperature

5.3.5 Heat transfer and temperature differences

Figures 5.5 and 5.6 give heat transfer coefficient, α , and temperature difference between the liquid and inner tube wall in the collector tube, ϑ_m , estimated with equations 5.7-8 and 5.11 for the example in 5.3.1.

Figures 5.5 and 5.6 give heat transfer coefficient and temperature difference as function of fluid temperature for the three concentrations of the three aqueous solutions.

From Figures 5.5 and 5.6 we see that laminar flow should be avoided as it would make the temperature difference between the liquid and inner tube wall in the collector tube more than 2K larger which would be detrimental to the heat pump system. Transitional flow, with Reynolds number up to $Re = 3000$ or even more could have a similar effect though it is not so easy to know how large the temperature difference then will be.

5.4 Comparison for a supermarket cooling application

5.4.1 Example used to compare properties of aqueous solutions for cooling

The following fluids are used in this comparison: Aqueous solutions of ethylene glycol, ethyl alcohol, potassium acetate and potassium formate with concentrations giving the freezing point $t_F = -15^\circ\text{C}$ and $t_F = -20^\circ\text{C}$. A non aqueous heat transfer fluid that has been suggested for supermarket cooling, a hydrofluoroether with low viscosity (HFE-7100) is here included in the comparison (Horton and Groll 2003). Operating temperatures are somewhere between -4 to -12°C . See also [PAPER 7](#) (Melinder 2005).

In order to illustrate how the fluids differ with temperature, the following example is used. Cooling cabinets with the cooling capacity $\dot{Q} = 2000 \text{ W}$ have a total circular tube length $L_{\text{tot}} = 20 \text{ m}$, the inner tube diameter $d_i = 0.015 \text{ m}$, each straight tube length $L_{\text{str}} = 1.5 \text{ m}$ and the fluid velocity is $w = 0.8 \text{ m/s}$. The pumping efficiency is set to $\eta_p = 0.4$.

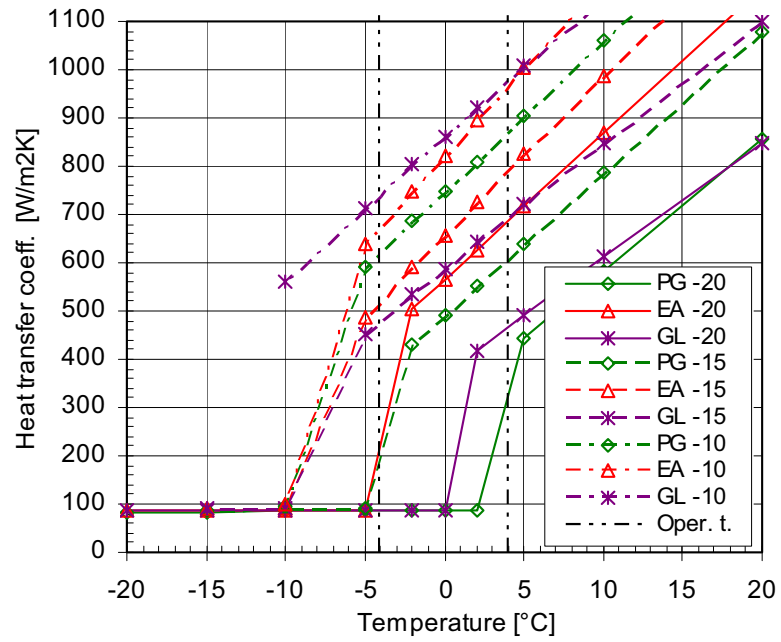


Figure 5.5. Heat transfer coefficient as function of fluid temp.

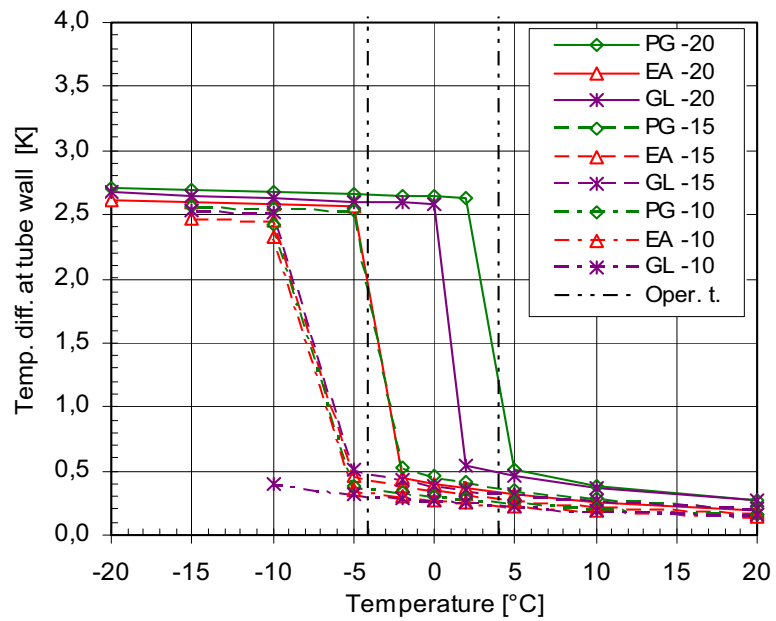


Figure 5.6. Temperature difference as function of fluid temp.

5.4.2 Heat transport ability and volumetric heat capacity

How do the two concentrations or freezing points of the aqueous solutions and the non aqueous fluid compare as to volumetric heat capacity that determines the heat transport ability of the fluid? Figure 5.7 shows that ethyl alcohol gives 5-10 % higher ρc_p values than the other aqueous solutions and the volumetric heat capacity, ρc_p , decreases with higher concentrations of each fluid but the differences are not too big. However, ρc_p of the non aqueous fluid in the comparison, hydrofluoroether, is less than half of ρc_p of the aqueous solutions. This means that more than twice as big volume flow is needed to get the same cooling capacity.

5.4.3 Viscosity and type of flow

The aqueous solution with lowest viscosity in Figure 5.2, ethyl alcohol, has the highest viscosity of the fluids compared in Figure 5.8, followed by ethylene glycol. The salts, especially potassium formate, have low viscosity. However, hydrofluoroether has very low viscosity, which is its best property.

What type of flow can we expect for the example of circular tubes in a cooling cabinet? Figure 5.9 gives Reynolds number, Re [-], as function of the fluid temperature for the given example.

Figure 5.9 indicate that we at operation temperatures can expect laminar flow (below $Re = 2300$) for ethylene glycol and ethyl alcohol for both freezing points. With potassium acetate we may get transitional flow ($2300 < Re < 3000$) while potassium formate and hydrofluoroether give turbulent flow.

5.4.4 Friction coefficient, pressure drop and pumping power

Friction coefficient, pressure drop and pumping power are proportional to each other. How do the fluids compare?

Figure 5.10 gives pumping power as function of fluid temperature for the four concentrations of the two aqueous solutions. From the charts we see that the values go down 30-40% when the flow turns laminar but increases quickly again with lower temperature that gives higher viscosity. As we here consider only the pressure drop due to friction, both pressure drop and pumping power are

proportional to the friction coefficient. The pumping power is only between 5 and 8 W with the chosen pump efficiency.

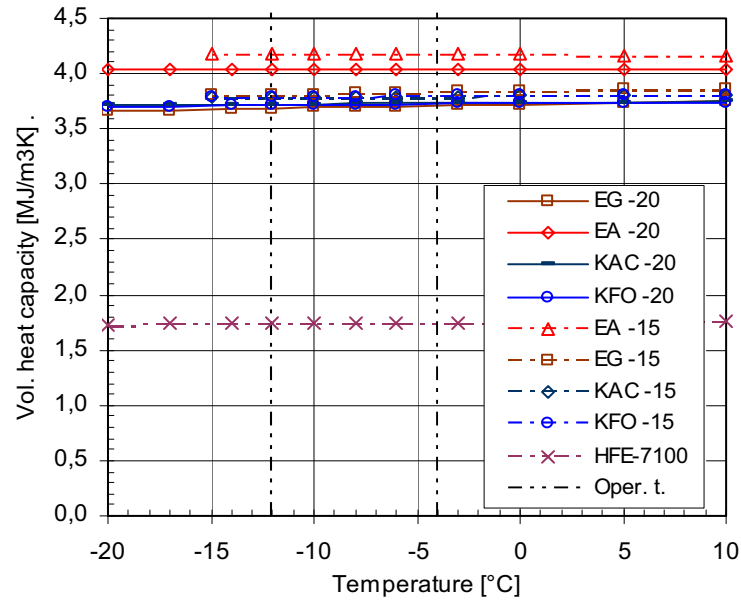


Figure 5.7. Volumetric heat capacity, ρc_p [kJ/(m³·K)] as function of fluid temperature

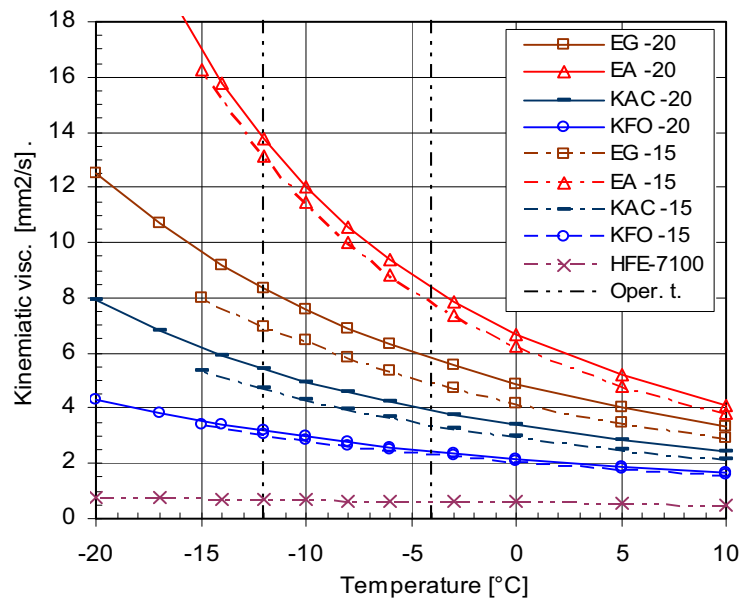


Figure 5.8. Kinematic viscosity as function of fluid temperature

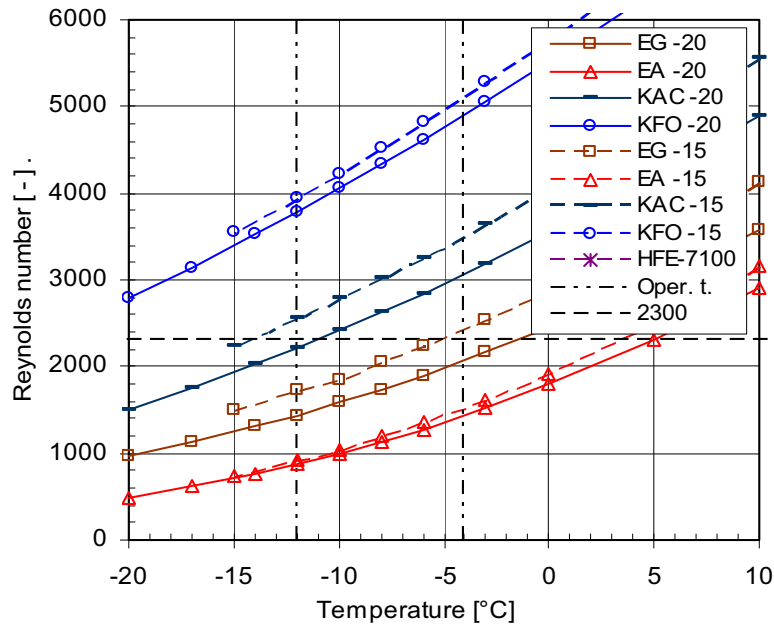


Figure 5.9. Reynolds number as function of fluid temperature

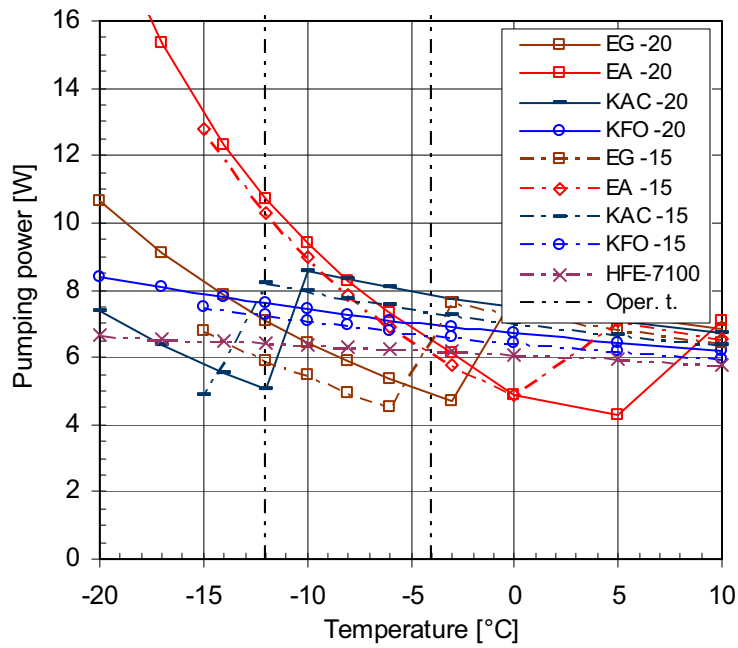


Figure 5.10. Pumping power as function of fluid temperature

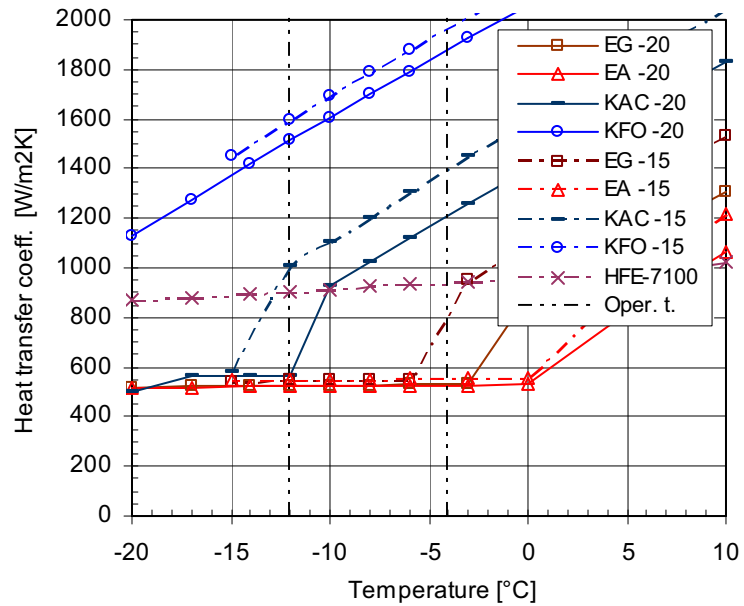


Figure 5.11. Heat transfer coefficient as function of fluid temp.

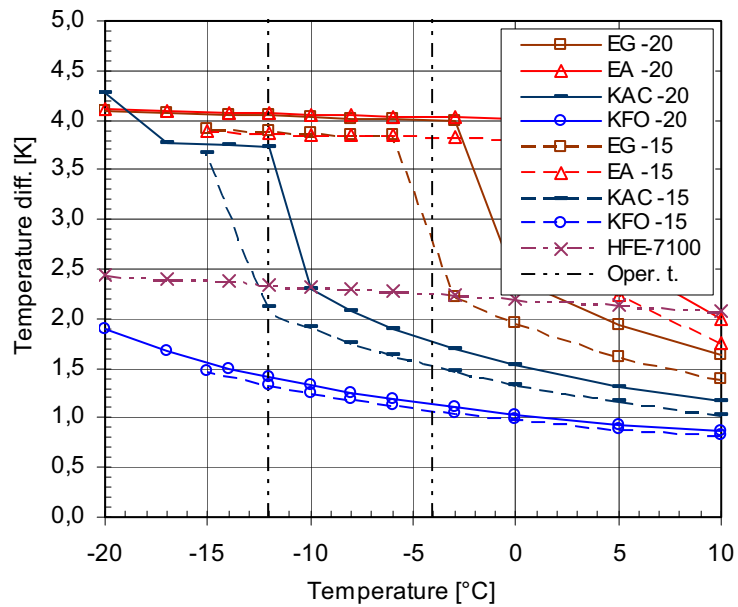


Figure 5.12. Temperature difference as function of fluid temp.

5.3.5 Heat transfer and temperature differences

Figures 5.11 and 5.12 give heat transfer coefficient, α , and temperature difference between the liquid and inner tube wall in the collector tube, ϑ_m , estimated with equations 5.7-8 and 5.11 for the example in 5.3.1.

Figures 5.11 and 5.12 give heat transfer coefficient and temperature difference as function of fluid temperature for the four concentrations of the two aqueous solutions.

From the charts we see that laminar flow should be avoided as it would make the temperature difference between the liquid and inner tube wall in the collector tube more than 2K larger which would be detrimental to the heat pump system. Transitional flow, with Reynolds number up to $Re = 3000$ or even more could have a similar effect though it is not so easy to know how large the temperature difference then will be.

5.5 Discussion

For the chosen example we see that laminar flow can be expected for the highest concentrations (lowest freezing point) of propylene glycol. We may get transitional flow for both types of fluids except for freezing points $t_F = -5^\circ\text{C}$ and -10°C , when we can expect turbulent flow. The influence on friction coefficient, pressure drop and pumping power is not crucial with this example. However, laminar and also transitional flow should be avoided because of the added temperature difference between the liquid and inner tube wall in the collector tube.

5.6 Nomenclature

The following nomenclature is used in chapter 5.

c_p	specific heat [J/kgK]
d	tube diameter [m]
f_l	friction factor [-]
k	thermal conductivity [W/mK]

Nomenclature, cont.

t	temperature [°C]
w	fluid flow velocity [m/s]
A	heat transfer area [m ²]
$\dot{E}_p$	pumping power [W]
$\dot{Q}$	cooling capacity [W]
$\dot{V}$	volume flow [m ³ /s]
α	heat transfer coefficient [W/m ² K]
Δt	temperature change [K]
μ	dynamic viscosity; $\mu = \nu \cdot \rho$ [Pa·s]
ν	kinematic viscosity [m ² /s]
ρ	density [kg/m ³]
$\mathcal{G}_m$	temperature difference [K]
Nu	Nusselt number; $Nu = \alpha \cdot d / k$ [-]
Pr	Prandtl number; $Pr = \mu \cdot c_p / k$ [-]
Re	Reynolds number; $Re = (w \cdot d) / \nu$ [-]

6. Basic thermophysical properties of aqueous solutions used as ice slurry

6.1. Introduction

The most basic form of ice slurry is pure water with ice which is an excellent secondary working fluid. However, when temperatures below 0°C are required, freezing point depressant additives are used. Most aqueous solutions used as single-phase secondary fluids can also be used as ice slurry, a fluid consisting of liquid and ice where small ice crystals are produced, usually with some type of ice generator. The ice crystals are then transported to the cooling object from which heat is removed when some or all of the ice crystals melt. The great enthalpy change when ice melts requires less temperature change in the cooling object and makes it possible to reduce the volume flow rate for a given cooling capacity and also to use smaller pipe dimensions in the system.

Some pioneers in the field of ice slurries focused on getting ice slurry equipment working for various applications, such as air conditioning, cooling cabinets in supermarket, district cooling, preserving fish, shrimps and vegetables. (Paul 1995, Snoek 1993, Wang and Goldstein 1996) A lot of research has since then been made, focusing also on various aspects and details of ice slurry properties, control, equipment, measurements etc. (Bel 1999, Ben Lakhdar 1998, Egolf et al 1996, Fournaison et al 2001, Guilpart et al 2005, Meewisse 2004, Melinder and Hägg 2005, Pronk et al 2005). Much of this research has been carried out in connection with an IIR Working party that since 1999 has arranged workshops that from 2006 were transferred into conferences on Phase Change Materials and Slurries for Refrigeration and Air Conditioning. This work has resulted in an IIR Handbook on Ice Slurries (Kauffeld, Egolf and Kawaji 2005).

The requirements that have to be fulfilled by an ideal secondary ice slurry fluid are basically the same as for single phase fluids as outlined in chapter 1.3.2, except for the fact that ice slurries operate a few degrees below the freezing point of the solution while single

phase fluids operate several degrees above the freezing point to give sufficient freezing protection.

Ice slurry may be produced at nighttime and stored in an ice slurry tank. The ice slurry may then be introduced in the secondary fluid loop at daytime and ice may be melted when there is high need for cooling. If lower electric cost is available at night time, this could lead to significantly lower operation costs. For design purposes it is of interest to estimate ice mass fraction and enthalpy values and enthalpy-phase diagrams and related tables can serve that purpose. In order to choose a secondary fluid for ice slurry use and to make correct technical calculations of the ice slurry system there is a need to examine and evaluate thermophysical properties and other aspects of ice and of the aqueous solution used.

6.2 Thermophysical properties of ice

In order to predict the thermophysical properties of ice slurries, it is necessary to know the properties of the aqueous solution and of ice by combining these properties in the correct ratio. The ice mass fraction or ice concentration is here a crucial parameter. Also viscosity, thermal conductivity and enthalpy of the ice slurry need to be determined carefully as described in the following sections.

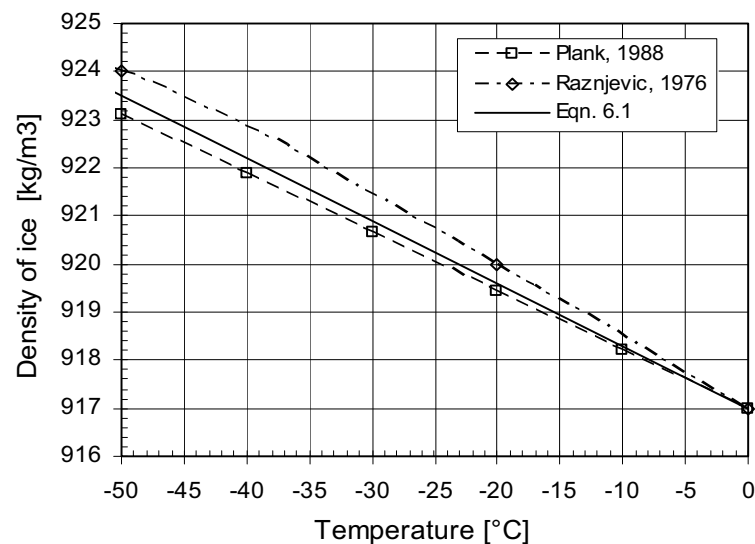


Figure 6.1. Density of ice as function of temperature

Linear equations for density, thermal conductivity and specific heat have been chosen based on data in handbooks (Plank 1988, Raznjevic 1976) and some other sources. See Figures 6.1-3.

The **density of ice** is lower than the density of water in the liquid phase (ice floats on the liquid surface) but increases with lower temperature. In Figure 6.1, the following equation for density of ice, ρ_1 [kg/m³], as function of temperature ϑ_1 [°C] is based on data from Plank (1988) and Raznjevic (1976):

$$\rho_1 = 917 - 0.13 \cdot t_1 \quad (6.1)$$

The **specific heat of ice** is much lower than the specific heat of water in the liquid phase. Measured values of specific heat of ice by Dickinson-Osborne, published in CRC Handbook of Chemistry & Physics (1988-89) and a series of measurements by Fransson (2002) are plotted besides data in Plank (1988) and Raznjevic (1976). Note that data from Plank (1988) show a much steeper slope than the rest and there is a spread of values at or near 0°C.

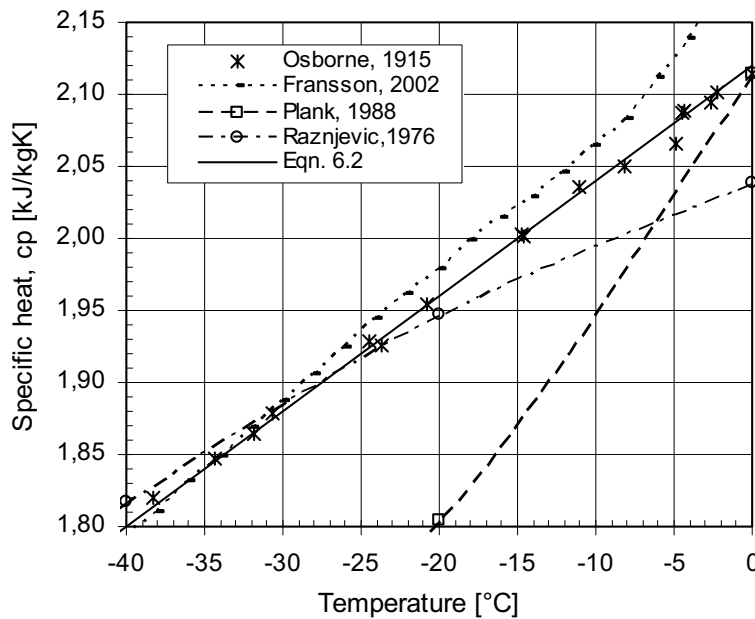


Figure 6.2. Specific heat of ice as function of temperature

Based on these data, the following equation for specific heat of ice, c_{pI} [kJ/kgK], as a function of temperature can be described by:

$$c_{pI} = 2.12 + 0.008 \cdot t_1 \quad (6.2)$$

The enthalpy of ice, h_I [kJ/kg], at the temperature, ϑ_I [°C], can then be calculated with the relation, where the heat of melting of ice is set to 332.4 kJ/kg:

$$h_I(t_I) = -332.4 + t_I (2.12 + 0.008 \cdot t_I) \quad (6.3)$$

This equation has been used to construct enthalpy phase diagrams for a number of aqueous solutions.

The **thermal conductivity of ice** is much higher than that of water in the liquid phase. This feature has a good influence on the performance of ice slurries.

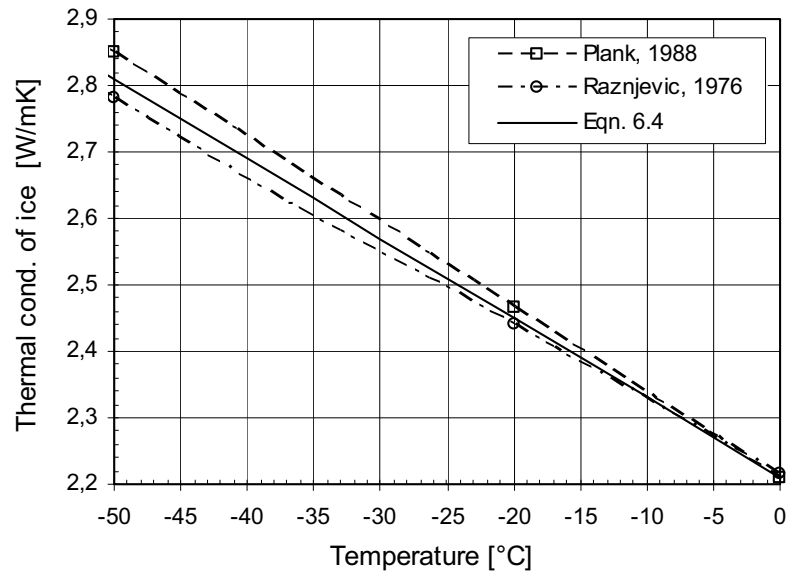


Figure 6.3. Thermal conductivity of ice as function of temp.

Based mainly on data from Plank (1988) and Raznjevic (1976) as shown in Figure 6.3, the following equation for thermal conductivity of ice, k_I [W/mK], a function of temperature, t_I [°C], is used:

$$k_I = 2.21 - 0.012 \cdot t_I \quad (6.4)$$

Note that the equation by Bobkov (1977) $k_I = 2.22(1 - 0.0015 \cdot t_I)$ gives much lower increase (of k_I) with lower temperatures.

6.3 Freezing point, ice fraction and density of ice slurries in equilibrium

6.3.1 Freezing point and ice mass fraction

Figure 6.4 shows how the freezing point changes with the composition of the mixture when in equilibrium. The diagram can be used for any aqueous solution (compare Figure 2.1). An increased concentration of the freezing point depressant additive lowers the freezing point temperature. This continues until we reach the concentration where the additive and water forms a eutectic solution. With even higher concentration the freezing point is raised again.

Let us assume a solution with the additive mass fraction c_0 and note what happens when it is cooled down from $\underline{A}$ to the freezing point $\underline{F}$ (c_0, t_F) and below that temperature. If the temperature decreases below the freezing point and the process takes place in equilibrium, a separation of the mixture takes place. Ice crystals that ideally consist of pure water freeze out and are separated from the rest of the solution and the additive mass fraction in the remaining solution rises as a consequence.

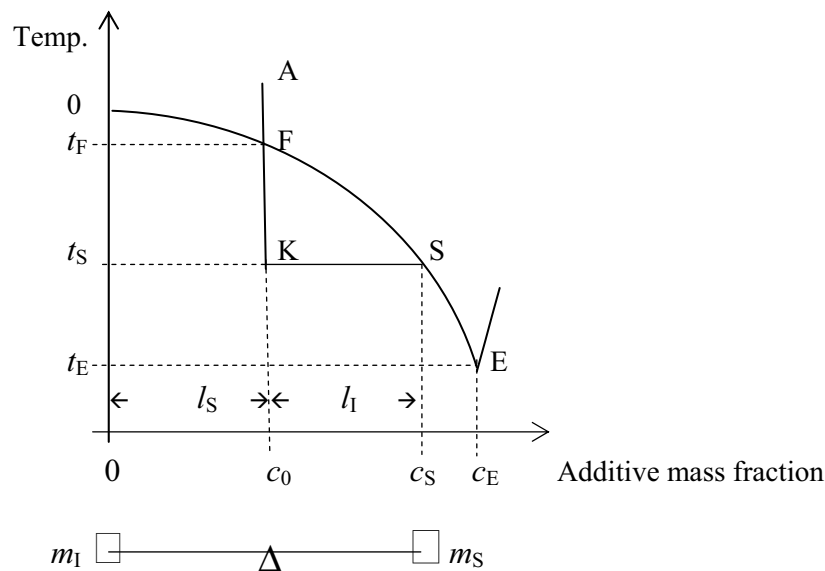


Figure 6.4. Freezing point diagram with seesaw balance

When the temperature has been lowered so that it corresponds to $\underline{K}$ in Figure 6.4, there are ice crystals with the condition $(c = 0, t_S)$ as well as solution with the condition (c_S, t_S) .

The solution does not freeze to a solid mass when the freezing point is passed, but ice slurry is formed. Only when the temperature goes down to the eutectic temperature will the solution fully solidify. We should also keep in mind that sub-cooling phenomena often occur in practice. The first ice crystals are often not formed until the solution temperature falls to one or two degrees below the equilibrium freezing point, but when some ice crystals have been formed, the continuation of the process seems easier to take place in equilibrium.

The concentration of the remaining solution is given by a mass balance based on the condition that the mixture has an unchanged total amount of additive. Let us refer to the mass of pure ice by m_I and the mass of solution by m_S , the concentration of which is c_S . Then the amount of additive is $m_S c_S$. In the original solution the additive amount was $c_0(m_I + m_S)$. As the amount of additive has not changed, it follows that $m_S c_S = c_0(m_I + m_S)$, which also can be written as $m_S(c_S - c_0) = m_I c_0$ (Granryd and Melinder 2005).

The **mass fraction of ice** or ice concentration, c_I , can then easily be calculated from

$$c_I = m_I / (m_I + m_S) = (c_S - c_0) / c_S \quad (6.5)$$

The ice mass fraction is closely linked to the slope of the freezing point curve between the temperatures considered. The amounts $(c_S - c_0)$ and c_0 are in Figure 6.4 illustrated as the lengths l_I and l_S . The correlations for mass balance and ice mass fraction can then be expressed as

$$m_I / m_S = l_I / l_S \quad \text{or} \quad c_I = l_I / (l_I + l_S) \quad (6.6)$$

These correlations follow the so called **seesaw principle**. Imagine that we place the mass m_I at $c_A = 0$, the mass m_S at $c_A = c_S$ and that these represent weights on a seesaw that should be in balance when supported at $c_A = c_0$. This helps us see that eqn. 6.6 is valid (Granryd 2005, Melinder and Hägg 2005).

Extensive studies on ice slurry properties, such as those by Bel et al (1996) and Guilpart et al (1999), present ice mass fraction as a function of the temperature for various additive concentrations of

the freezing point depressant substance, usually ethyl alcohol-water. Suitable equations for ice slurries are also given in Ben Lakhdar (1998), Kauffeld (1999) and Lottin and Epiard (2000).

Figure 6.5 gives ice mass fraction as a function of temperature for various additive concentrations of aqueous solutions of ethyl alcohol. Property values are taken from CRC Handbook of Chemistry and Physics (1988-89) and the Matlab and Excel programs described in chapter 4.

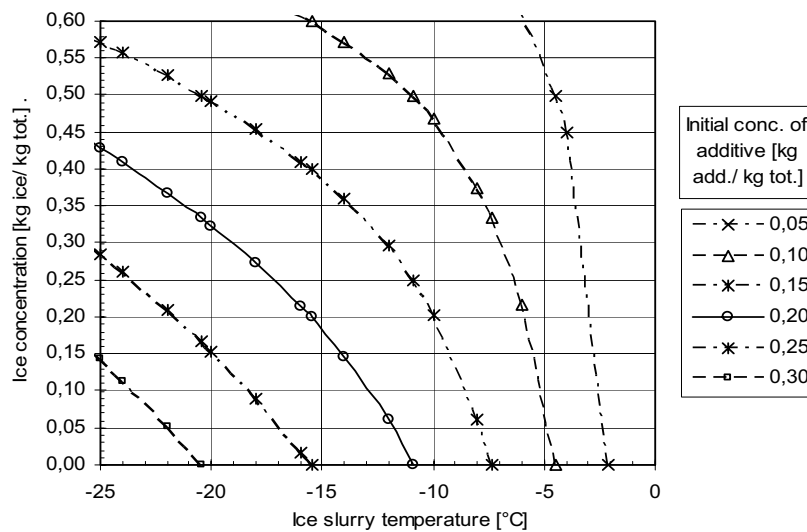


Figure 6.5. Ice mass fraction as function of temperature for aqueous solutions of ethyl alcohol

6.3.2 Density and volumetric ice fraction

The density of ice slurries at equilibrium, ρ_{IS} , may be estimated with a mixing equation where ρ_{CS} is the density of the concentrated aqueous solution (Melinder 2003a):

$$\rho_{IS} = 1 / [(c_I / \rho_I) + (1 - c_I) / \rho_{CS}] \quad (6.7)$$

In Figure 6.6 the density of ice slurry based on eqn. 6.7 is given as function of temperature for six types of aqueous solutions. Density of ice and three of the aqueous solution types as single phase are also given in the figure. The aqueous solutions of salts have at single phase density so high that they are outside the figure. The density of ice slurries is generally lower than the density of single-

phase fluids because of lower additive concentrations and the low density of ice.

Aqueous solutions of ethyl alcohol have the lowest density, quite near that of the ice particles, which may prevent stratification of the fluid caused from the buoyancy force on the ice crystals. The high density of the salts may cause separation between ice particles and the rest of the solution. Besides those aspects that need consideration for ice slurry applications, a high density generally improves transport and transfer abilities of a fluid.

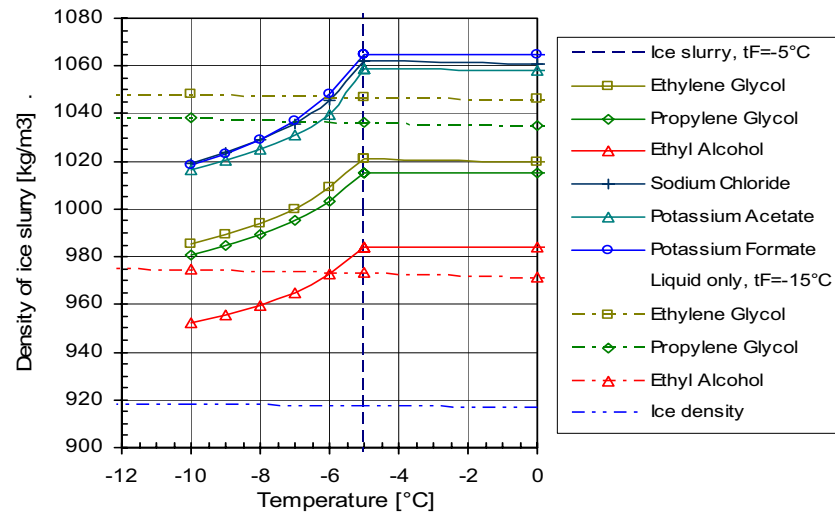


Figure 6.6. Density as function of temperature for ice slurry and single phase of some aqueous solutions

A comparison of some properties of aqueous solutions for liquid only and ice slurry is made in [PAPER 4](#) (Melinder 2003a). This comparison includes density, volumetric enthalpy change, dynamic viscosity and thermal conductivity.

In some equations for viscosity and thermal conductivity the volume fraction of particles is introduced, corresponding to a **volume fraction of ice**, c_{IV} defined as

$$c_{IV} = c_I / ((c_I + (1 - c_I)\rho_I) / \rho_{CS}) \quad (6.8)$$

with symbols or nomenclature as earlier given in this chapter part.

6.4 Heat transport ability and enthalpy change of aqueous solutions

6.4.1 Heat transport ability, enthalpy change and apparent specific heat

The secondary fluid flow rate required to achieve a cooling capacity of $\dot{Q}$ in an indirect system with a single-phase fluid can be calculated by the following relation:

$$\dot{Q} = \dot{m} c_p \Delta t = \dot{V} \rho c_p \Delta t \quad (6.9)$$

For an ice slurry system, it is more suitable to apply the relation:

$$\dot{Q} = \dot{m} \Delta h = \dot{V} \rho \Delta h \quad (6.10)$$

Both sensible heat and latent heat of melting are included in the enthalpy difference, Δh . In these relations, $\dot{m}$, is the fluid mass flow rate, $\dot{V}$, is the volumetric flow rate, Δt is the temperature change between the fluid entering and exiting the cooling unit, ρ is the fluid density and c_p is its specific heat.

The heat transport ability of ice slurries in equilibrium is proportional to the **enthalpy difference**, Δh , obtained for a certain temperature change between the fluid entering and exiting. It is much higher for ice slurry than for mono-phase aqueous solutions, as there is a benefit from the latent heat of melting of ice crystals, which reduces the fluid flow rate required for a given cooling capacity and also the temperature change. This heat transport ability depends partly on the specific heat of the fluid, but more on the ice fraction of the ice slurry that enters the cooling unit, on the amount of ice that melts (Melinder, 2002b).

This heat transport ability can also be expressed as an **apparent specific heat**, $c_{p,a}$ that can be defined as $c_{p,a} = \Delta h / \Delta t$.

This apparent specific heat includes sensible heat of the fluid and latent heat of melting of the ice.

Figure 6.7 gives apparent specific heat of ethyl alcohol. Just below the freezing point there are very high values especially at low additive fractions. Some works on ice slurry properties give apparent specific heat as a function of the temperature for various concentrations of the freezing point depressant substance, usually for

ethyl alcohol – water (Bel and Lallemand 1999, Guilpart et al 1999). See also [PAPER 6](#) (Melinder and Granryd 2005).

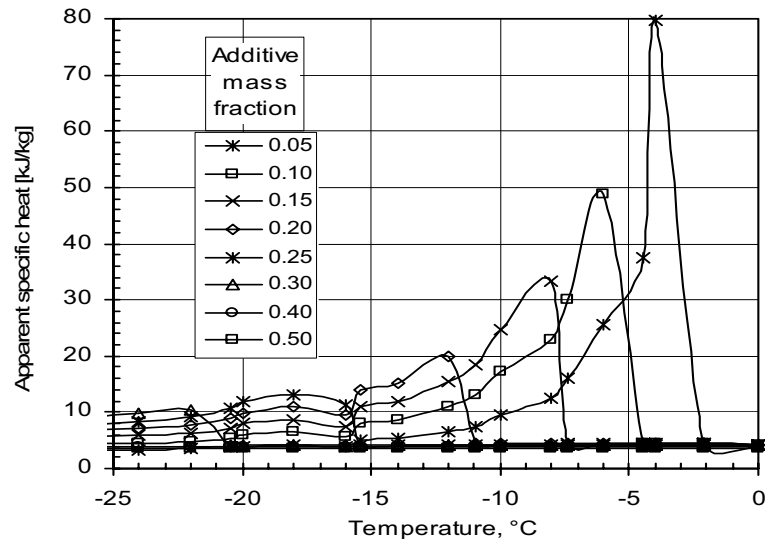


Figure 6.7. Apparent specific heat as function of temperature and additive mass fraction of ethyl alcohol - water

6.4.2 Heat transport ability of ice slurries and the volumetric enthalpy change

The volumetric heat transport ability of a fluid, the cooling capacity divided with the fluid volume flow rate, $\dot{Q}/\dot{V}$ is equal to the estimated **volumetric enthalpy change** for a certain temperature change between the fluid entering and exiting the cooling object, $\rho\Delta h$ for ice slurries and $\rho c_p \Delta t$ for single-phase fluids (Compare eqns. 6.9 and 6.10).

In [Figure 6.8](#) a comparison is made of this volumetric enthalpy change from $t = -5^\circ\text{C}$ for ice slurries with concentrations giving the freezing point $t_F = -5^\circ\text{C}$ and for single-phase fluids with concentrations giving the freezing point $t_F = -15^\circ\text{C}$. The figure shows that the volumetric enthalpy change obtained for a certain temperature change for cooling applications can be 7 to 10 times higher for ice slurries than for single phase aqueous solutions, because of the large benefit from the latent heat of melting of ice crystals.

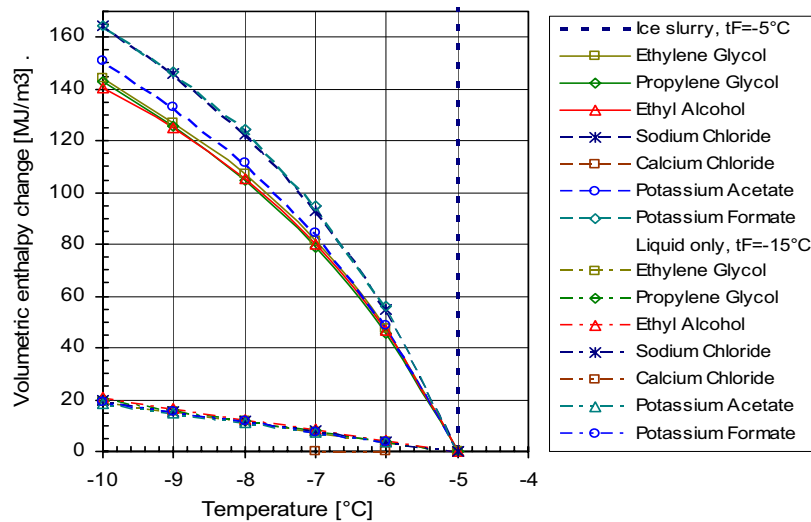


Figure 6.8. Volumetric enthalpy change from $t = -5^{\circ}\text{C}$ of ice slurries compared to that of single-phase fluids

This volumetric enthalpy change depends only a little on the sensible heat of the fluid and much on the latent heat, the amount of ice that will melt, i.e. on the ice fraction of the ice slurry that enters the cooling object. This high volumetric heat transport ability of ice slurries for cooling applications makes it possible to reduce the size of piping in the system.

The enthalpy values used to construct the curves in Fig. 6.8 for ice slurries are obtained from the type of enthalpy phase diagrams introduced in 6.5 (Compare Kauffeld et al 2005, Guilpart et al 2006). See also [PAPER 2](#) (Melinder 2002a).

6.5 Enthalpy phase diagrams of aqueous solutions

6.5.1 Constructing enthalpy phase diagrams of aqueous solutions

A way to visualize the benefits from the latent heat or enthalpy difference in the melting process of ice may be achieved by constructing an enthalpy-phase diagram of the aqueous solution at atmospheric pressure in a similar way as Bošnjaković (1961) and Rant (1961). The amount of heat exchanged in the freezing process can be estimated from such enthalpy-phase diagrams with

some added features. Such diagrams can then help us estimate what additive concentrations and temperatures give most benefit.

For calculation purposes we choose to view an ice slurry solution as consisting of i. small ice crystals of pure water that freeze out, and ii. the rest of the solution with a higher concentration of the freezing point depressant additive. In order to estimate enthalpy values of an aqueous solution with ice particles we need freezing point data, values of specific heat as well as enthalpy values of ice and heat of mixing of the solution. The heat of mixing can, in the diagrams, be seen as the variation in enthalpy values with the concentration of a reference isothermal line in the liquid phase (such as $t_R = 0^\circ\text{C}$).

Let us first as an example consider a basic enthalpy-phase diagram of sodium chloride – water (Figure 6.9) that mainly considers the area of interest for an ice slurry application. Besides isotherms it gives lines for ice mass fraction and it is extended beyond the eutectic point. An enthalpy-phase diagram of ethyl alcohol –water is then presented that gives further added features (Figure 6.10).

The **enthalpy of ice slurry**, h_{IS} , at point K in Figure 6.9, can be expressed by the following equation:

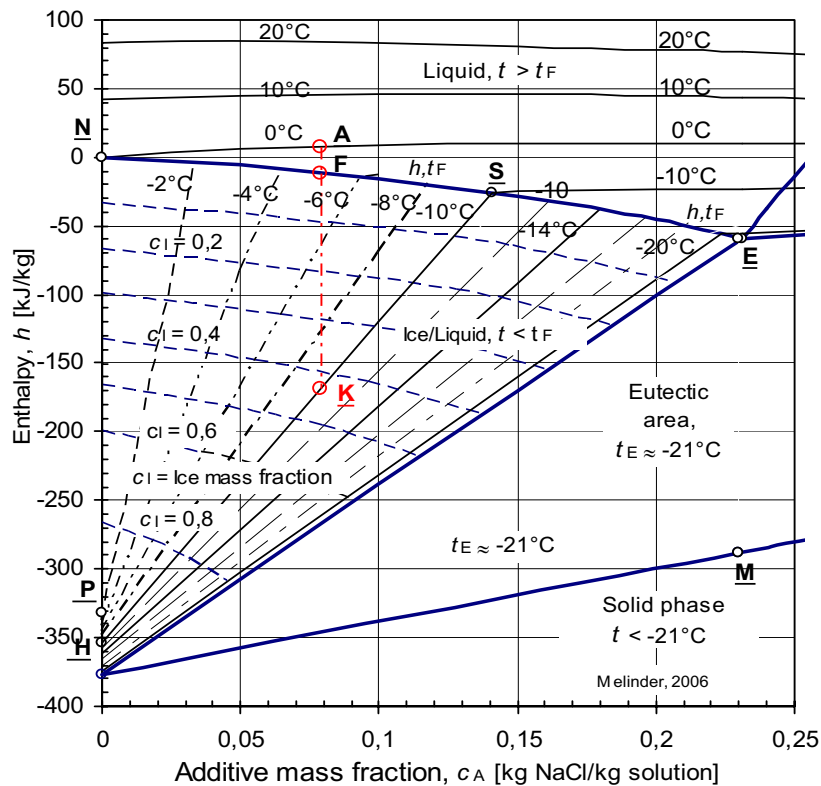
$$h_{IS} = h_{(K)} = h_I(t_{CS})c_I + h_{CS}(c_{CS}, t_{CS})(1-c_I) \quad (6.11)$$

where $h_I(t_{CS}) = h_H - h_N = h_H - h_P + (h_P - h_N)$ and

$$h_{CS}(c_{CS}, t_{CS}) = h_{0,R} + \Delta h_M(c_{CS}, t_R) + \int_{t,R}^{t,CS} (c_P \cdot dt) \quad (6.12)$$

The enthalpy difference between N and P, the heat of melting of water, is $h_N - h_P = 332.4 \text{ kJ/kg}$. The enthalpy of ice at various temperatures has been estimated using Equation 6.3.

The enthalpy of water at the reference temperature $t_R = 0^\circ\text{C}$, is defined as: $h_{0,R}(0^\circ\text{C}) = 0 \text{ kJ/kg}$. The enthalpy values for different additive fractions for the reference temperature corresponds then to the heat of mixing value, $\Delta h_M(c_{CS}, t_{CS})$. The sources of heat of mixing values, Δh_M , and the reference temperature, t_R , for each diagram are as follows: Sodium chloride ($t_R = 0^\circ\text{C}$), Δh_M from Bošnjakovic (1961); Ethyl alcohol ($t_R = 0^\circ\text{C}$), Δh_M from Landolt-Börnstein, (1976). Heat of mixing values for other aqueous solutions have also been taken from Christensen C. et al (1982) and Christensen J. et al (1986).



Figures 6.9 Enthalpy phase diagram of sodium chloride

Added features in Figure 6.10 compared to Figure 6.9 are the enthalpy lines connecting the temperatures t_{F-1} , t_{F-2} , etc. to t_{F-10} . These lines correspond to 1, 2, etc. to 10K below the freezing point curve. These enthalpy lines that are marked h, t_{F-1} , h, t_{F-2} , etc. to h, t_{F-10} , in the diagrams are made as an aid to estimate the heat transport ability of each aqueous solution when used as ice-slurry.

6.5.2 Example of simple adiabatic ice slurry processes in equilibrium

With an enthalpy-phase diagram with some added features it is possible to estimate what happens during ice slurry processes, especially those that take place adiabatically in equilibrium. The way these enthalpy-phase diagrams are constructed makes it also possible to generate the following basic diagrams, all as function of the additive mass fraction of ethyl alcohol. A simple example of the process in an ice generator is drawn in each of the figures.

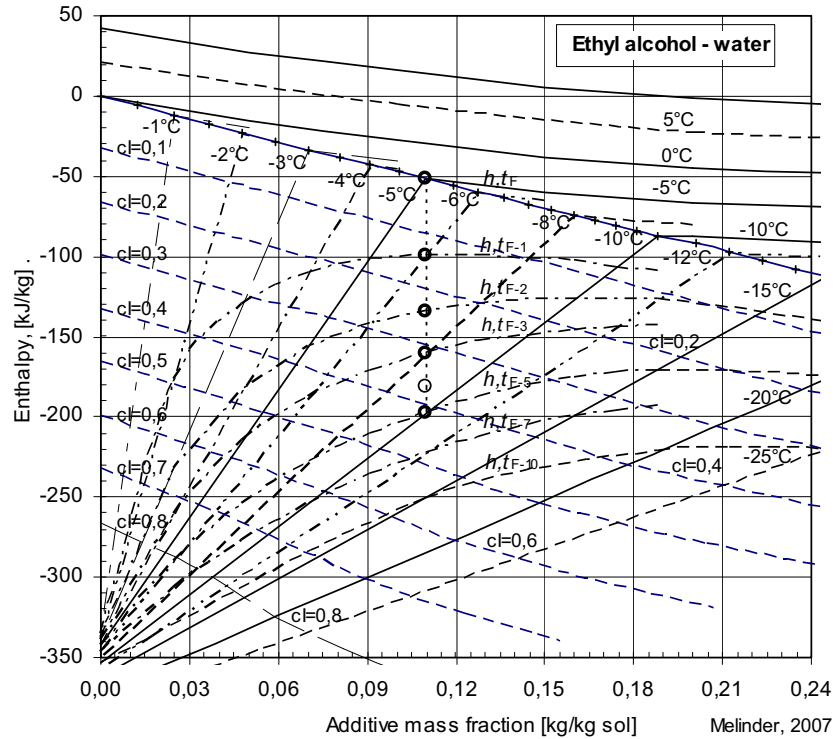


Figure 6.10. Enthalpy phase diagram of ethyl alcohol with added enthalpy lines

Figure 6.11 gives first an enthalpy phase diagram. Figure 6.12 gives ice slurry enthalpy, figure 6.13 gives ice mass fraction based on eqn. 6.5, figure 6.14 gives ice slurry density based on eqn. 6.7. All figures are given for additive mass fractions $c_A = 0.0$ to 0.21 , corresponding to freezing points $t_F = 0^\circ\text{C}$ to $t_F \approx -12^\circ\text{C}$. Each figure gives values for freezing point and for some temperatures below the freezing point curve, $t_{F-0.5}$, t_{F-1} , $t_{F-1.5}$ to t_{F-7} , etc.

Processes in an ice slurry plant where ice is neither removed nor added (as from an external ice storage tank) ideally takes place vertically in an enthalpy-phase diagram. In an ice generator heat is removed from the fluid, the enthalpy decreases and the ice fraction increases as the process goes vertically down in the diagram. In the user unit (ie. a cooling cabinet battery) heat is taken up by the fluid, the enthalpy increases and the ice fraction decreases as the process goes vertically up in the diagram. The enthalpy change is obtained by the refrigerating effect $\Delta h = Q/m$ applied to the fluid.

Example 1: Let us as an example assume that an aqueous solution of 11 mass-% ethyl alcohol, $c_A = 0.11$ with 5 mass-% ice, $c_I = 0.05$, entering the ice generator that has the refrigerating effect $Q/m = -90$ kJ/kg applied to the fluid. What freezing point, temperature, enthalpy, ice mass fraction and density can we expect to get? (Compare also Guilpart et al 2005). The entering condition corresponds to point **A** in Figure 6.11 and this point is indicated also in Figures 6.12-14. The process in the diagrams is vertical as the additive mass fraction is the same after the ice generator. Note first from Figure 6.11 that the freezing point temperature of the solution, $t_F = -5.0^\circ\text{C}$, is obtained by going vertically up to the freezing point curve, marked h, t_F .

Figure 6.14 gives the enthalpy value of the mixture when entering the ice generator to $h_{in} = -70$ kJ/kg. When exiting the ice generator at **B** it is $h_{ex} = -70 - 90 = -160$ kJ/kg and $c_I = 0.31$. From isotherms in the two-phase region we find the temperature to be $t_B = -8.0^\circ\text{C}$ (In Figure 6.14, **B** is on the line h, t_{F-3} , hence $t = -5 - 3 = -8^\circ\text{C}$. Figure 6.13 gives densities as: $\rho_{IS,in} \approx 979.5$ kg/m³, $\rho_{IS,ex} \approx 959.5$ kg/m³.

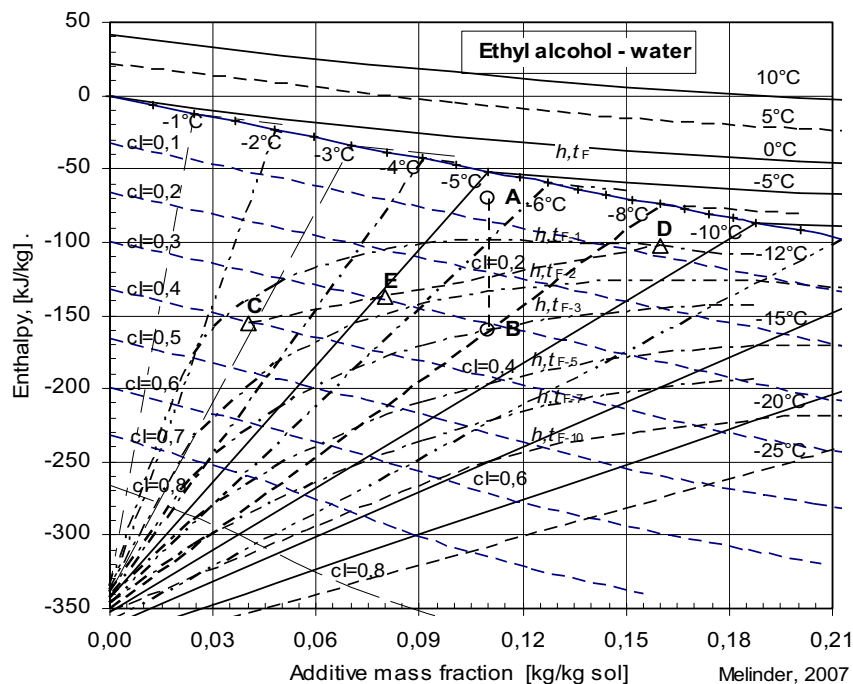


Figure 6.11. Enthalpy phase diagram of ethyl alcohol – water for examples 1 and 2

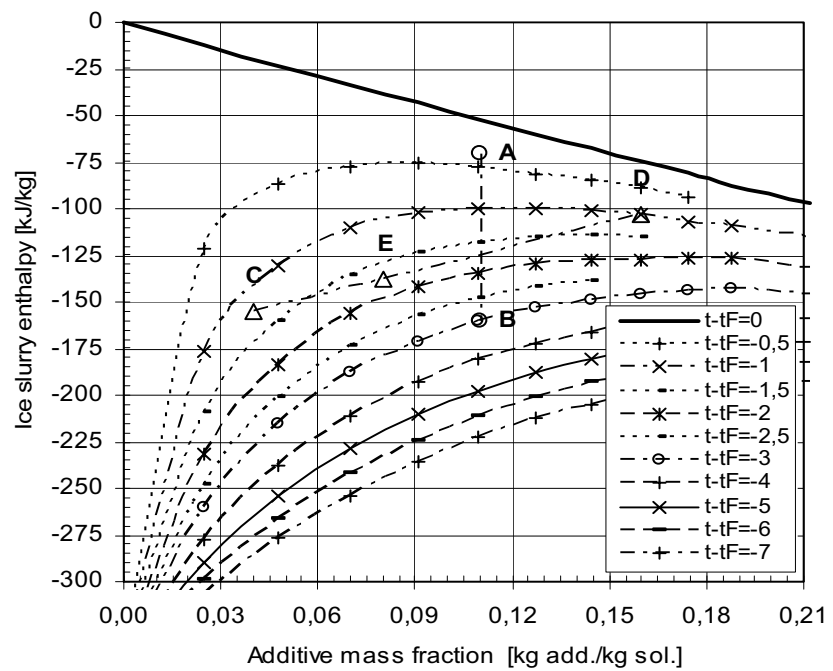


Figure 6.12. Ice slurry enthalpy of ethyl alcohol - water

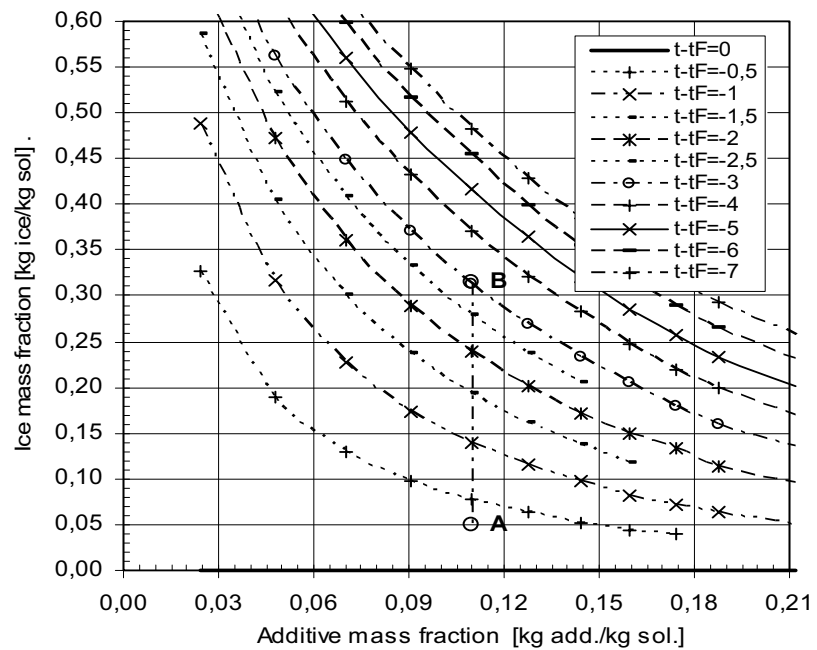


Figure 6.13 Ice mass fraction of ethyl alcohol - water

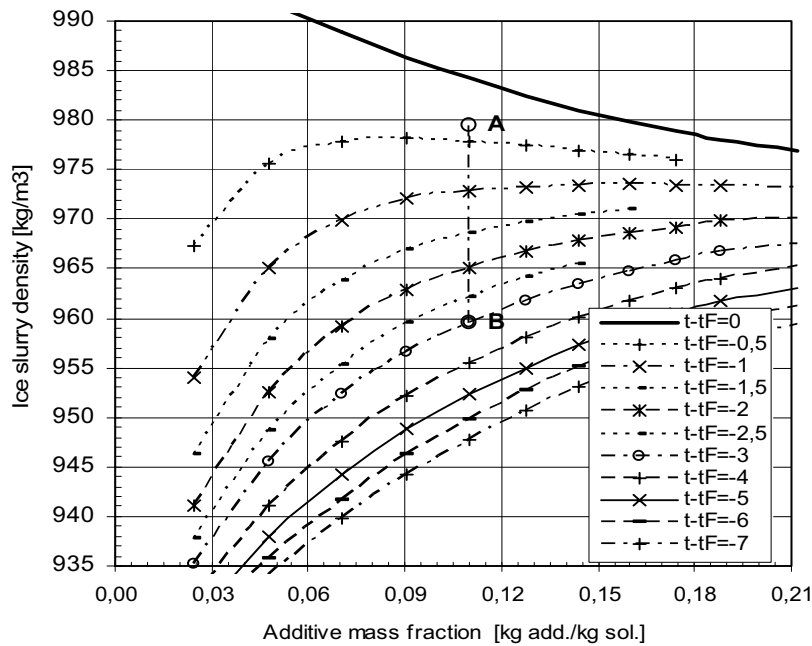


Figure 6.14. Ice slurry density of ethyl alcohol - water

6.5.3 Examples of simple adiabatic ice slurry mixing processes

What happens if solutions with different additive concentrations are mixed? If we can assume that the mixing takes place under adiabatic conditions, then (what has been called) the seesaw principle can be applied in an enthalpy-phase diagram and the mixing will ideally take place on a straight line between the two conditions. See Melinder and Hägg (2005), Granryd and Melinder (2005).

In Example 2 two ice slurry solutions with different additive concentrations are mixed. An ice slurry solution of ethyl alcohol – water with $c_A = 0.16$; $t = -9^\circ\text{C}$ ($t_F \approx -8^\circ\text{C}$; $c_I \approx 0.09$) is mixed with twice as much of an ice rich solution with: $c_A = 0.04$; $c_I = 0.40$ ($t \approx -2.9^\circ\text{C}$; $t_F \approx -1.7^\circ\text{C}$). See points **D** and **C** in Figs. 6.11 and 6.12. The mixture will then get the condition in **E**: $c_A = (2 \cdot 0.04 + 1 \cdot 0.16) / 3 = 0.08$; $t_F \approx -3.5^\circ\text{C}$; $t \approx -5.2^\circ\text{C}$; $c_I = 0.29$.

Assume in Example 3 that 25% ethanol – water ($c_A = 0.25$) at $t = -12^\circ\text{C}$ point **G** (above the freezing point curve in Figure 6.15) is mixed with the same mass amount of ice – water ($c_A = 0$; $t = 0^\circ\text{C}$) with the ice mass fraction $c_I = 0.6$, in **F**. See Figure 6.15. What will

the ideal mixing condition be? According to the seesaw principle, the mixing condition will be on a straight line between **G** and **F**, at point **H** with $c_H = (0.25-0)/(1+1) = 0.0125$. The freezing point of this mixture ($c_A = 0.125$) is $t_F \approx -5.8^\circ\text{C}$ and the ice mass fraction is $c_I \approx 0.26$. The enthalpy of the mixture, h_H , can be estimated by the seesaw principle as $h_H \approx 1 \cdot (-332.4 \cdot 0.6) + 1 \cdot (-100) \approx -150 \text{ kJ/kg}$ (see Figure 6.15). This example illustrates how ice – water can be mixed with a single-phase secondary fluid (that could be produced without ice generator) at or near the freezing point ideally giving ice slurry of suitable composition as a result.

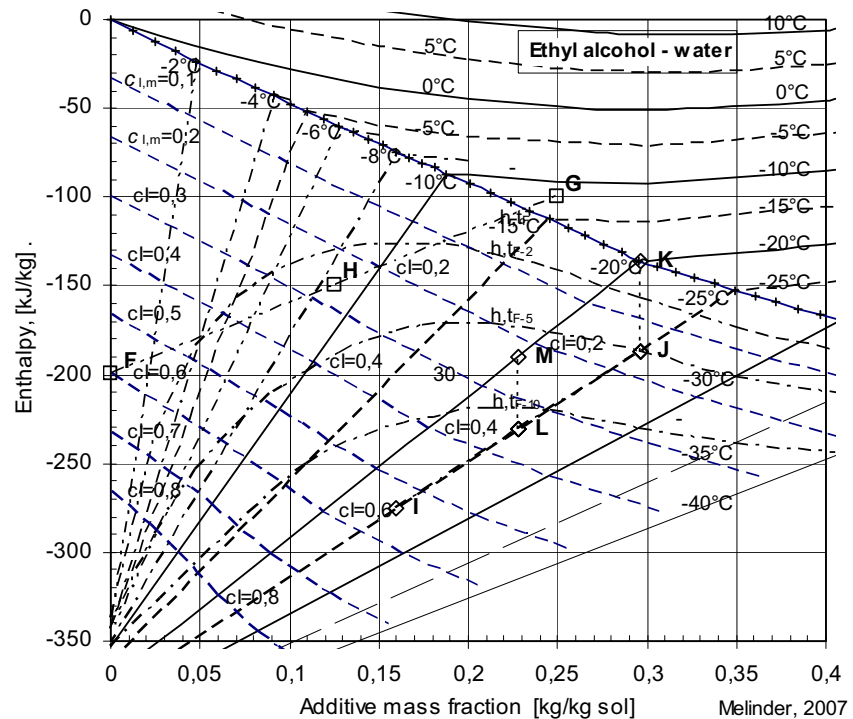


Figure 6.15 Enthalpy-phase diagram of ethyl alcohol – water for examples 3 and 4.

In Example 4 ice slurry solutions with different additive concentrations are mixed. An ice slurry solution of ethanol – water, $c_A \approx 0.296$; $t \approx -25^\circ\text{C}$; $t_F \approx -20^\circ\text{C}$; $c_I \approx 0.15$ is mixed with the same mass of an ice rich solution with: $c_A = 0.16$; $t \approx -25^\circ\text{C}$; $c_I = 0.54$. See points **J** and **I** in Figure 6.15. The mixture has the condition in point **L**: $c_A = (1 \cdot 0.296 + 1 \cdot 0.16) / 2 = 0.228$; $t \approx -25^\circ\text{C}$; $c_I = 0.34$.

This is a higher mass ice fraction than in point **J**, but Figure 6.15 shows that in this case it is not possible to have larger enthalpy change between the temperatures $t \approx -25^\circ\text{C}$ and $t \approx -20^\circ\text{C}$ from point **L** (to **M**) than it is from point **J** (to **K**). In other words, it is not possible to melt more ice between the given temperatures.

The enthalpy-phase diagram and the seesaw-principle show that even if one mixes in ice or an ice rich solution, as from a storage tank, it is not possible to melt more ice than at thermal equilibrium between given temperatures, t and t_F . Even though someone might think that ice slurry becomes more effective if more ice is mixed in than what is obtained at thermal equilibrium, this example shows that this is not always the case. All these examples given show that an enthalpy-phase diagram and other charts and tables may be of value to visualize both possibilities and limitations of ice slurries at various temperatures. See also [PAPER 9](#) (Melinder 2006b).

6.6 Viscosity of ice slurries in equilibrium

The shear viscosity of a concentrated suspension, taking into account the hydrodynamic interaction of particles, particle rotation, collision between particles, agglomerate formation, and so on, can be expressed by a theoretical equation in a power series (Thomas, 1965):

$$\eta^* / \eta_l = 1 + a_1 c + a_2 c^2 \pm a_3 c^3 \pm \dots \quad (6.13)$$

The constant a_1 is assumed to have the value $a_1 = 2.5$ determined by Einstein (Kauffeld et al 2005). Based on many experimental investigations of the viscosity of suspensions it was shown that the term $1 + 2.5 \cdot c + 10.05 \cdot c^2$ accounts for over 97.5% of the values of the relative viscosity η^* / η_l for volume fractions of particles, c , below 25% ($c < 0.25$). For values of $c > 0.25$ satisfactory results were obtained by Thomas (1965):

$$\eta^* / \eta_l = 1 + 2.5 \cdot c + 10.05 \cdot c^2 + \Delta \quad (6.14)$$

where $\Delta = A \cdot e^{B \cdot c}$, $A = 0.0273$, $B = 16.6$, η^* is shear viscosity of the suspension, and for two-phase mixtures like ice-slurry the background shear viscosity η_l is the viscosity of the liquid solution.

The Thomas equation that assumes a spherical form of particles and Newtonian behavior has been used extensively as a reference in ice slurry research, especially for low or moderate ice concentrations. It can for ice slurries be written as follows

$$\mu_{IS} = \mu_{CS} (1 + 2,5 c_{I,v} + 10,05 c_{I,v}^2 + 0,00273 e^{16,6 c_{I,v}}) \quad (6.15)$$

where $c_{I,v}$ is the volumetric ice fraction as given in equation 6.7.

Viscosity values of many aqueous solutions can be obtained from Melinder (1997). Some literature data for the shear viscosity of solutions are also given in Tvsetkov (2001) and Vargaftic (1972).

Figure 6.16 shows the dynamic viscosity of ice slurry in equilibrium (based on eqn. 6.15) as function of temperature for six types of aqueous solutions for ice slurries with concentrations giving the freezing point $t_F = -5^\circ\text{C}$ and for single-phase fluids with concentrations giving the freezing point $t_F = -15^\circ\text{C}$.

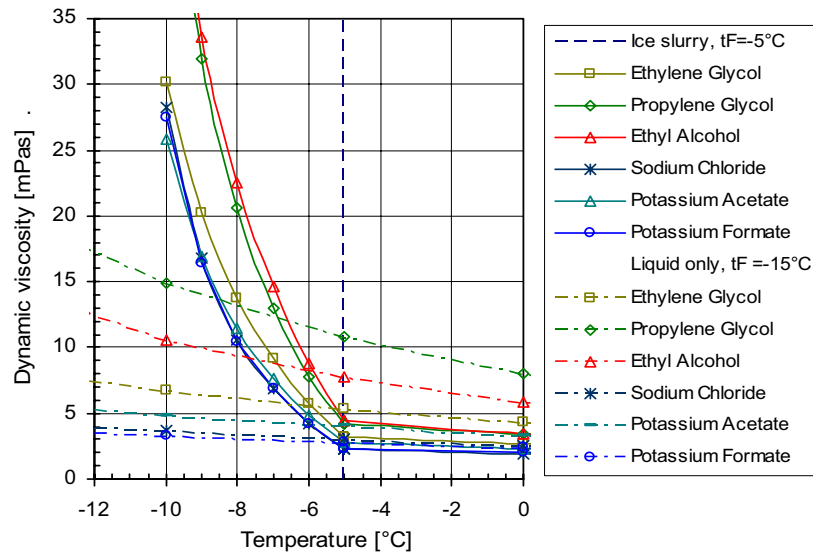


Figure 6.16. Dynamic viscosity as function of temperature for ice slurry and single phase of some aqueous solutions

The figure shows how the viscosity of ice slurry increases rapidly below the freezing point temperature as ice crystals are formed and the ice fraction increases rapidly. The aqueous solutions with highest viscosity are in this case ethyl alcohol and propylene glycol.

Their viscosity is about double that of the salts, still not viewed as too high to be used.

6.7 Thermal conductivity of ice slurry

The theory of the heat conduction phenomena through a random suspension of spherical particles in a matrix of uniform thermal conductivity has suggested that the volume fraction of the particles is small enough to make all interactions between the spheres negligible. For this kind of suspension an effective thermal conductivity k^* in the Maxwell approach can be represented by

$$k^*/k_1 = 1 + 3\beta c \quad (6.16)$$

where $\beta = (\alpha - 1)/(\alpha + 2)$, $\alpha = k_2/k_1$, c is the volume fraction of the particles (spheres), k_1 is the thermal conductivity of a matrix, and k_2 is the thermal conductivity of the spheres.

However, the validity of equation (6.16) is restricted to a very small volume fraction of particles.

To represent the actual behaviour of the thermal conductivity including the effect of interactions between spheres, the following approximation of k^* has been proposed (Jeffrey, 1972).

$$k^*/k_1 = 1 + 3\beta c + 3\beta^2 c^2 (1 + 2\delta \frac{\alpha - 1}{\alpha + 2}) \quad (6.17)$$

The information about the coefficients δ , β and α is summarized by Oleg B. Tsvetkov in chapter 2 of IIR Handbook on Ice Slurries (Kauffeld et al 2005). For the ice-slurry models the background conductivity k_1 is the thermal conductivity of the liquid water solution, and k_2 is the thermal conductivity of ice.

In [Figure 6.17](#) the overall thermal conductivity of ice slurries in equilibrium, k_{IS} , is estimated by the following mixing equation (assuming spherical form of the ice slurry particles) given by Perry (1973)

$$k_{IS} = k_{CS}(1 + 2c_{IV}y)/(1 - c_{IV}y) \quad (6.18)$$

where

$$y = (1 - k_{CS}/k_I)/(2 \cdot k_{CS}/k_I + 1) \quad (6.19)$$

where k_{CS} is the thermal conductivity of the concentrated aqueous solution, k_I is the thermal conductivity of ice and c_{IV} is the volumetric ice fraction as given in equation 6.8.

Figure 6.17 shows the overall thermal conductivity of ice slurry in equilibrium (based on eqn. 6.18) as function of temperature for six types of aqueous solutions for ice slurries with concentrations giving the freezing point $t_F = -5^\circ\text{C}$ and for single-phase fluids with concentrations giving the freezing point $t_F = -15^\circ\text{C}$. The figure shows that the thermal conductivity increases rapidly below the freezing point temperature, as ice crystals are formed. This is due to the high thermal conductivity of ice. The figure shows that the thermal conductivity obtained for ice slurries is much higher than for single phase aqueous solutions. Sodium chloride and potassium formate have highest values as ice slurries with the freezing point partly because of the high values of the single phase aqueous solutions.

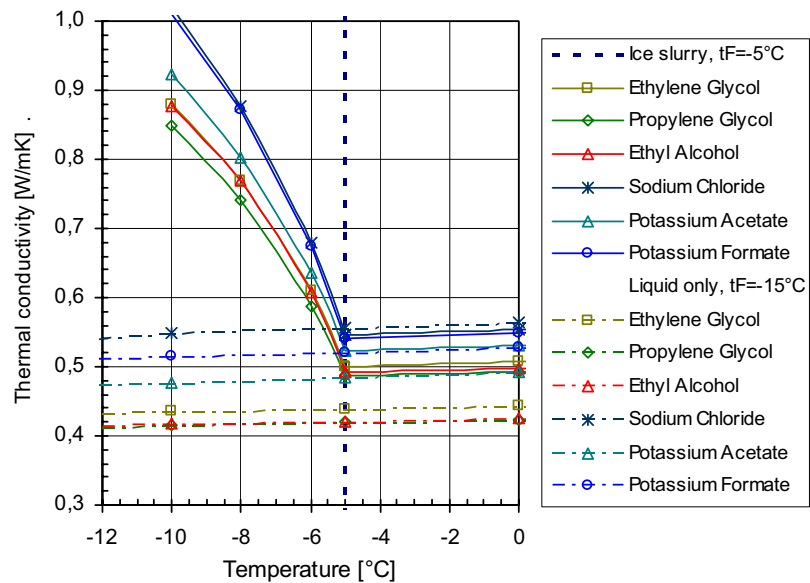


Figure 6.17. Thermal conductivity as function of temperature for ice slurry and single phase of some aqueous solutions

6.8 Nomenclature

The following nomenclature is used in chapter 6.

c_A	additive mass fraction [kg add./kg sol.]
c_p	specific heat [J/(kg·K)]
c_I	ice mass fraction [kg ice/kg sol.]
c_{Iv}	ice volume fraction [m ³ ice/ m ³ sol.]
h	enthalpy [J/kg]
k	thermal conductivity [W/m K]
m	mass [kg]
$\dot{m}$	mass flow rate [kg/s]
t	temperature [°C]
$\dot{Q}$	cooling capacity [W]
$\dot{V}$	volume flow rate [m ³ /s]
Δh	enthalpy difference [J/kg]
Δh_M	heat of mixing [J/kg]
Δt	temperature change [K]
η^*	shear viscosity of the suspension [Pa s]
η_l	is shear viscosity of liquid solution [Pa s]
μ	dynamic viscosity [Pa s]
ρ	density [kg/m ³]

Index:

A	additive
CS	concentrated solution
F	freezing point
I	ice
IS	ice slurry
0	initial
R	reference
in	inlet
ex	exit

Summary and conclusions

Aqueous solutions have long been used as single phase secondary working fluids for cooling in supermarkets, ice rinks, heat recovery systems, heat pumps and other applications. They are increasingly used also for freezers in supermarkets and other applications in low temperature refrigeration.

When comparing different secondary working fluids, reliable data are needed of basic thermo-physical properties in order to choose suitable fluid and to make technical calculations of an indirect refrigeration or heat pump system.

The main focus of this work has been to select property data of freezing point, density, viscosity, specific heat, and thermal conductivity with good or acceptable technical accuracy of a number of aqueous solutions used by the refrigeration and heat pump industry, rather than focusing on a limited number of property values or scientifically very accurate measuring techniques.

A thorough literature search was made to find the most reliable property values of aqueous solutions. Detailed literature references are given for thermo-physical properties of the following aqueous solutions: ethylene and propylene glycol, ethyl and methyl alcohol, glycerol, ammonia, potassium carbonate, calcium, lithium, magnesium and sodium chlorides as well as potassium acetate and potassium formate. Some laboratory measurements were made when literature values were incomplete or deemed unreliable. Methods used are briefly described and results are given.

The material in the Eng. Licentiate Thesis: *Thermo-physical properties of liquid secondary refrigerants, A Critical Review on Literature References and Laboratory Measurements* (Melinder 1998a) forms the basis for the charts and tables used in the IIR-publication *Thermo-physical properties of liquid secondary refrigerants* (Melinder, 1997). The present Doctoral thesis reports on the update made since 1998, including review work done on two additional fluids not covered in Melinder (1998a).

The thesis describes how the selection of property values results in tables and charts intended for the industry. Coefficients for polynomial equations are generated from these property values using a Matlab program and this material is intended as a useful tool for computer treatment. Aqueous solution of ethyl alcohol is used as

example to see how this process is made. This choice of fluid can also be seen as a test of this method, as the basic thermo-physical properties of aqueous solutions of ethyl alcohol present more challenges than the other fluids examined.

A comparison is made of a few types of aqueous solutions used as secondary working fluids for two types of applications. The first example is bedrock heat pumps and the second is cooling cabinets in a supermarket. An effort is made to see how the additive concentration affects the thermal performance.

Most aqueous solutions used as single-phase secondary fluids can also be used as ice slurry, a fluid consisting of liquid and ice where small ice crystals are produced, usually with some type of ice generator. The ice crystals are then transported to the cooling object from which heat is removed when ice crystals melt. This results in less temperature change in the cooling object and makes it also possible to reduce the volume flow rate and to use smaller pipe dimensions in the system.

In order to choose a secondary fluid for ice slurry use and to make correct technical calculations of the ice slurry system there is a need to examine and evaluate thermo-physical properties and other aspects of ice and of the aqueous solution used. For dimensioning purposes it is of interest to estimate ice mass fraction and enthalpy values and enthalpy-phase diagrams can serve that purpose. This thesis presents enthalpy-phase diagrams made by author that besides isotherms contain lines with ice fraction and lines connecting enthalpies at freezing point and 1, 2, etc. to 10K below the freezing point curve. Examples are given to see how the diagrams can be used .

The intention of the author is now to finalize the updated publication with *Thermophysical properties of secondary working fluids* and make it ready for the industry and others who may want to use it. At the same time, with some funds from the Swedish Energy Agency and support from many companies, work to produce a handbook on indirect systems with secondary working fluids has also recently started.

References

- Aittomäki, A., Lahti, A., 1997, Potassium formate as a second refrigerant, *Int. Journal of Refrigeration*, vol. 20, No 4, p.276-282
- Aittomäki, E.A., Karkkiainen, S.V., 1999, *Lowering temperature by brines in ice banks*, 20th Int. Congress of Refrigeration, IIF/IIR, Sydney.
- Aittomäki, A., Kianta, J., 2003, Indirect Refrigeration Systems Design Guide Book, Tampere Univ. of Technology, Finland.
- American Institute of Physics Handbook, 1972, 3rd edition
- Andersson, O., 1996, 2002, Internal reports of measurements of thermal conductivity and volumetric heat capacity with transient hot wire technique at Umeå univ.
- Arias, J., Lundqvist, P., 2005, *Modelling supermarkets energy usage*, IIF/IIR, Commercial refrigeration, Vicenza
- ASHRAE Handbook, 2001, Fundamentals
- Bel, O. et al.; 1996, *Thermal study of an ice slurry used as refrigerant in a cooling loop*; Applications for Natural Refrigerants, Aarhus
- Bel, O., Lallemand, A., 1999, *Study of a two-phase secondary refrigerant. 1. Intrinsic thermophysical properties of an ice slurry*, Int. Journal of Refrigeration, IIR/IIF, Paris, p.164-174
- Ben Lakhdar, M.A., 1998, Thermal hydraulic behavior of a two phase mixture: The ice slurry. Theoretical and experimental studies (French), *Doctoral Thesis*, Lyon, France.
- Ben Lakhdar, M.A., Melinder, Å., 2002, *Facing the challenge to produce ice slurry for freezer application*, 5th IIR Workshop on Ice Slurries, Stockholm
- Bobkov V.A., 1977, *Production and Application of Ice*, Foods Industry Publ. House, Moscow, p. 232.
- Bošnjaković, F., 1961, Technische Thermodynamik, II Teil, Band 12, in *Wärmelehre und Wärmewirtschaft in Einzeldarstellung*, Dresden/Leipzig
- Bäckström, M., Emblik, E., 1965, Kältetechnik
- Carrier, Handbook of Air Conditioning System Design, 1965
- Christensen, C., Hanks, R.W., Izatt, R.M., 1982, Handbook of Heats of Mixing
- Christensen, J., Gmehling, J., Rasmussen, P., Weidlich, U., 1986. Heats of Mixing Data Collection, 1. Binary Systems

- Conde, M.R., 2004, *Properties of aqueous solutions of lithium and calcium chlorides: formulations for use in air conditioning equipment design*, Int. Journal of Thermal Sciences, Volume 43, p.367-383.
- CRC Handbook of Chemistry and Physics, 69th edit., Weast, 1988-89
- DOW, Organic Chemicals / The Glycols, 1974
- Egolf, P. W., Brühlmayer, J., Özvegyi, F., Abächerli, F., Renold, P., 1996, *Properties of ice slurry*, IIRConf., Applications for Natural Refrigerants, Aarhus, p. 12.1-10.
- Egolf, P.W., Kauffelt, M., 2005, *From physical properties to industrial ice slurry applications*, Int. Journal of Refrigeration, Volume 28, Number 1, p.4-12, Paris.
- Evenmo, K., 1998, *A New Secondary Fluid*, Natural Working Fluids'98, IIR – Gustav Lorentzen Conf., Oslo.
- Fennema, O.R. Powrie, W.D., Marth, E.H., 1973, Low-temperature preservation of foods and living matter, Wisconsin
- Field, B.S., Kauffeld M., 2003, *Use of Ice Slurry in a Supermarket Display Case*, 21st Int. Congress of Refrigeration, IF/IIR, Washington DC
- Florian, O., 1997, Heat Transfer and Pressure drop of Liquid Secondary refrigerants - Experiments and theory; Engineering Thesis work, Energy Technology, KTH.
- Fournaison, L., Chourot, J.M., Guilpart, J., 2001, *Different ice mass fraction measurement methods*, Fourth IIR Workshop on Ice Slurries, Osaka, Japan
- Fransson, Å., 1996, 2002, Internal reports of measurements of specific heat with differential scanning calorimetry at Umeå univ.
- Fukusako, S., Kozawa, Y., Yamada, M., Tanino, M., 1999, *Research and Development Activities on Ice Slurries in Japan*, First IIR Workshop on Ice Slurries, Yverdon-les-Bains, Switzerland
- Geiringer, P.L., 1962, Handbook of Heat Transfer Media, Reinhold Publ. Co., New York
- Gmelins Handbuch der Anorganischen Chemie, 1960, Lithium, 20
- Gnielinski, V., 1976, *New equations for heat and mass transfer in turbulent pipe and channel flows*, Int. Chem. Eng., 16, 359-368.
- Granryd, E., 2002, *Perspective on ice slurry*, Fifth IIR Workshop on Ice Slurries, Stockholm, May
- Granryd, E., Melinder, Å., 2005, Secondary refrigerants for indirect systems (pp.6:2-6), in *Refrigerating Engineering*, Granryd, E. et al, KTH, Stockholm.

- Guilpart J., Fournaison L., Ben Lakhdar, M.A., 1999, *Calculation method of ice slurries thermophysical properties-application to water/ethanol mixture*, Int. Congress of Refrigeration, IIR/IIF, Sydney
- Guilpart J., Stamatiou, E., Fournaison L., 2005, *The control of ice slurry systems: an overview*, Int. Journal of Refrigeration, Volume 28, Number 1, p. 98-107, Paris.
- Guilpart J., Stamatiou, E., Delahaye, A., Fournaison L., 2006, *Comparison of the performance of different ice slurry types depending on the application temperature*, Int. Journal of Refrigeration, Volume 29, Number 15, p. 781-788, Paris.
- Hofmann, E., 1959, *Verfahren der Kälteerzeugung und Grundlagen der Wärmeübertragung*,.
- Horton, W.T., Groll, E.A., 2003, *Secondary Loop Refrigeration in Supermarket Applications: A Case Study*, 21st Int. Congress of Refrigeration, IIR, Washington DC
- Hägg, C, Melinder, Å., 2002, *Selecting fluids suitable for ice slurry applications at low Temperatures, Zero leakage – Minimum Charge*, IIF/IIR, Stockholm
- International Critical Tables of Numerical Data, Physics, Chemistry and Technology, 1928-29.
- Int. Journal of Thermophysics, 1987, Vol. 8, p.351
- Int. Journal of Thermophysics, 1994, Vol. 15, p.165
- Jeffrey D.J., 1972, *Conduction through a random suspension of spheres*, Proc. Roy. Soc., London A, vol. 335: p. 355-367.
- Jokinen, J., 2003, *Betaine based heat transfer fluid as a solution for toxicity and corrosion problems in heating and cooling systems*, 21st Int. Congress of Refrigeration, IF/IIR, Washington D.C.
- Jokinen, J., Willems, B., 2004, *Betaine based heat transfer fluid as a natural solution for environmental, toxicity and corrosion problems in heating and cooling systems*, Sixth Gustav Lorenzen Conference on Natural Working Fluids, IIF/IIR, Glasgow, UK.
- Kauffeld M., Christensen K.G., Lund, S., Hansen, T. M., 1999, *Experience with ice slurry*, 20th Int. Congress of Refrigeration, IIR/IIF, Sydney
- Kauffeld M., Kawaji M., Egolf P.W., 2005, *Handbook on Ice Slurries – Fundamentals and Engineering*, IIF/IIR, Paris
- Kruse, H., 2005, *Commercial refrigeration – on the way to sustainability*, IIF/IIR, Commercial refrigeration, Vicenza
- Landolt-Börnstein, *Eigenschaften der Materie in Ihren Aggregatzuständen*, 1960-1971.

- Landolt-Börnstein, Numerical Data and Functional Relationships in Science and Technology; 1977, Vol. 1b
- Landolt-Börnstein, Numerical Data and Functional Relationships in Science and Technology, 1976, Group IV, Vol. 2. Heats of Mixing and Solution
- Lange's Handbook of Chemistry, 1985, 13th edition
- Lindborg, A, 2003, *Indirect refrigeration Systems*, 21st Int. Congress of Refrigeration, IIF/IIR, Washington DC
- Lottin, O., Epiard, C. Thermodynamic properties of some currently used water - antifreeze mixtures when used as ice slurries, *Eighth Int. Refrigeration*, Purdue, USA, p.391-398, 2000
- Lundqvist, P.G., 2000. *Recent refrigeration equipment trends in supermarkets: Energy efficiency as leading edge?*, Bulletin No 2000.5 of the Int. Institute of Refrigeration, IIR, Paris
- Macmillan's Chemical and Physical Data, 1992
- Malter, L., 1996, *Binary-ice generation and applications of pumpable ice slurries for indirect cooling*, Application for Natural Refrigerant, Aarhus
- Mao, Y., Hrnjak, P.S., 1999, *Defrost issues of display cabinets with evaporating R404A and heat exchangers with secondary refrigerants*, 20th Int. Congress of Refrigeration, IIF/IIR, Sydney.
- Meewisse, J.W., 2004, Fluidized bed Ice Slurry Generator for Enhanced Secondary Cooling Systems, *Doctoral Thesis*, Delft, The Netherlands.
- Melinder, Å., Berendson, J., Granryd, E., Kyrk, B., 1989, Köldbärare för värmepumpstillämpningar. Termodynamiska egenskaper och korrosion för kaliumkarbonat samt jämförelse med andra köldbärare, Report R18:1989, Swedish Council for Building Research
- Melinder, Å., 1997, Thermo-physical properties of liquid secondary refrigerants. Tables and Diagrams for the Refrigeration Industry, IIF/IIR, Paris
- Melinder, Å., 1998, *Thermophysical properties of liquid secondary refrigerants, A Critical Review on Literature References and Laboratory Measurements*, (Engineering Licentiate Thesis), KTH, Stockholm
- Melinder, Å., 1998, *A Critical Review on Thermophysical Properties of Liquid Secondary Refrigerants*, Natural Working Fluids'98, IIR – Gustav Lorentzen Conf., Oslo
- Melinder, Å., 2002a, *Enthalpy-phase diagrams of aqueous solutions for ice slurry applications*, Fifth IIR Workshop on Ice Slurries, Stockholm

- Melinder, Å., 2002b, *Influence of temperature on latent heat benefits in ice slurry applications*, Zero leakage – Minimum Charge, IIF/IIR, Stockholm
- Melinder, Å., 2003a, *Comparing Properties of Aqueous Solutions for Liquid Only and Ice slurry Applications of Indirect Systems*, Proc. 21st Int. Congress of Refrigeration, IIR/IIF, Washington
- Melinder, Å., 2003b, *Update on indirect systems and secondary fluids with focus on supermarket refrigeration*, Int. Conf. on Fluid and Thermal Conversion, FTEC2003, Bali, Indonesia
- Melinder, Å., 2004, *Secondary fluids for low operating temperatures*, Sixth Gustav Lorenzen Conference on Natural Working Fluids, IIF/IIR, Glasgow, UK
- Melinder, Å., 2005, *Properties of various secondary fluids for supermarket applications*, Thermophysical Properties and Transfer Processes of Refrigerants, IIF/IIR, Vicenza, Italy
- Melinder, Å., Granryd, E., 2005, *Using property values of aqueous solutions and ice to estimate ice concentrations and enthalpies of ice slurries*, Int. Journal of Refrigeration, Volume 28, Number 1, p.13-19, Paris
- Melinder, Å., Hägg, C., 2005, *Enthalpy-phase diagrams and the see-saw-principle – aids to describe various ice slurry conditions*, Thermophysical Properties and Transfer Processes of Refrigerants, IIF/IIR, Vicenza, Italy.
- Melinder, Å., 2006a, *Updating and extending secondary fluid data for various refrigeration and heat pump applications*, Seventh Gustav Lorenzen Conference on Natural Working Fluids, IIF/IIR, Trondheim
- Melinder, Å., 2006b, *Equilibrium properties of ice slurry based on aqueous solutions of ethyl alcohol*, Seventh Gustav Lorenzen Conference on Natural Working Fluids, IIF/IIR, Trondheim
- Melinder, Å., 2006c, *Equilibrium properties of ice slurry based on two chosen types of aqueous solutions*, Seventh IIR Conference on Phase Change Materials and Slurries for Refrigeration and Air Conditioning, Dinan, France
- Melinder, Å., 2006d, *Updating properties of secondary working fluids for various refrigeration systems*, Int. Conf. on Fluid and Thermal Conversion, FTEC2006, Jakarta
- Mellan, 1970, *Industrial Solvents Handbook*, New Jersey: Noyes Data Corporation, 133 p.
- Paul, J., 1995, *Binary Ice as a secondary refrigerant*, 19th Int Congress of Refrigeration, Vol.4b, Den Hague

- Paul, J., Jahn, E., 1997, *Binary-ice generation and applications of pumpable ice slurries for indirect cooling*, The Inst. of Refrigeration, UK
- Paul, J., Malter, L., Münster, U., 1999, *Generation and utilization of liquid ice (binary ice)*, 20th Int. Congress of Refrigeration, IIR/IIF, Sydney.
- Pearson, S. F., 1993, *Development of Improved Secondary Refrigerants*; Int. Institute of Refrigeration, UK
- Perry, R.H., Chilton, C.H., 1973, *Chemical Engineers' Handbook*, 5th edition, New York
- Plank, R., 1988, *Handbuch der Kältetechnik*, VI/B (edited by Steimle, F.).
- Pronk, P., Hansen, T.M., Infante Ferreira, C.A., Witkamp G.C., 2005, *Time dependant behaviour of different ice slurries during storage*, Int. Journal of Refrigeration, Volume 28, Number 1, p.27-36, 2005, Paris.
- Pronk, P., 2006, *Fluidized bed heat exchangers to prevent Fouling in ice slurry systems and industrial crystallizers*, *Doctoral Thesis*, Delft, The Netherlands.
- Rant, Z. 1961, Diagramm-Mappe zu Band 13, Verdampfen in Theorie und Praxis, in *Wärmelehre und Wärmewirtschaft*
- Raznjevic, K. 1976, *Handbook of Thermodynamic Tables and Charts*.
- Schmidt, G.H., *Design experience with Flo-Ice and binary ice as a secondary refrigerant for refrigeration installations*, First IIR Workshop on Ice Slurries, Yverdon-les-Bains, Switzerland
- Sherwood, G., 1995, *Secondary Heat Transfer Systems and the Application of a New Hydrofluoroether*; The 1995 International CFC and Halon Alternatives Conference
- Sieder, E.N., Tate, G.E., 1956, *Heat transfer and pressure drop of liquids in tube*, Int. Chem. Eng., 28, p.1429-1435
- Snoek, C.V., Hampton, D., 1993, *Ice slurry based district cooling systems*, Proceedings, Int District Heating and Cooling Ass. 8th Ann. Cooling Conference, Tulsa, Oklahoma
- Snoek, C.W., 1995, *District cooling using ice slurry as the heat transfer fluid*, Nouvelles technologies et sécurité dans le froid, Journées Européennes du froid, p.157-184, AFF-Interclima, Paris.
- Stakelbeck, H., Plank, R., 1929, *Ueber die Zähigkeit von Chlornatrium-, Chlorkalzium- und Chlormagnesium-lösungen in Abhängigkeit von Temperatur und Konzentration*, Zeitschrift für Die Gesamte Kälte-Industrie, p.105-112

- Stamatiou, E., Meewisse, J., Kawaji, M., *Ice slurry generation involving moving parts*, Int. Journal of Refrigeration, Volume 28, Number 1, p. 60-72, 2005, Paris
- Svensk Kylnorm, 1992, Faktablad 7, Etanolbaserade köldbärarsystem
- Thomas, D.G., 1965, *Transport characteristics of suspension*, Journal of Colloid Science Vol. 20, p. 267-277
- Timmermans, J., 1960, The Physico-chemical Constants of Binary Systems in Concentrated Solutions, Vol. 3 and 4
- Tsvetkov O.B., Laptev Ju. A., Kolodiaznaia V.S., 2001, *One-and two-phase refrigerating liquid media*, Kholodilnaja Teknika, no. 10: p. 8-12.
- Ure, Z., 2003, *Secondary refrigeration developments and experiences in Europe*, 21st Int. Congress of Refrigeration, IIF/IIR, Washington DC
- Vargaftic, N.B., 1972: Thermophysical properties of gases and liquids. Handbook, Nauka Publ. House, Moscow
- VDI Heat Atlas, 1993
- VDI Wärmeatlas, 2002, Berechnungsblätter für den Wärmeübergang, VDI-GVC
- Vervoerd, M., Liem, N., van Gerwen, R.J.M., 1999, *Performance of a frozen food display cabinet with potassium formate as secondary refrigerant*, 20th Int. Congress of Refrigeration, IIF/IIR, Sydney.
- Wang, M.J., Hansen, T., Kauffelf, M., Christensen, K.C., Goldstein, V., 1999, *Application of ice slurry technology in fishery*, 20th Int. Congress of Refrigeration, IIF/IIR, Sydney
- Wang, M.J., Goldstein, V., 1996, *A novel ice slurry generation system and its applications*. 2nd Conf. on the use of non-artificial substances, Aarhus, DK, p. 12-19-12-23
- Wimby, J.M., Berntsson, T.S., 1994, *Viscosity and density of aqueous solutions of LiBr, LiCl, ZnBr₂, CaCl₂ and LiNO₃*, 1. Single salt solutions, J. Chem. Engineering Data 39, p.68-72
- Yang Y., 2001, Investigation of Deep-freeze Refrigeration Systems in Supermarket Application, Masters Thesis, Energy Technology, KTH, Stockholm.

APPENDIX

Table 2.3. Conversion chart of additive concentrations

Additive conc., % by weight	Additive conc., % by volume			
	Ethylene glycol, EG	Propylene glycol, PG	Ethyl alcohol, EA	Glycerol, GL
0	0	0	0	0
5	4.4	4.8	6.24	4.2
10	8.9	9.6	12.39	8.5
15	13.6	14.5	18.47	12.7
20	18.1	19.4	24.47	17.0
25	22.9	24.4	30.39	21.2
30	27.7	29.4	36.18	25.5
35	32.6	34.4	41.83	30.2
40	37.5	39.6	47.33	35.0
45	42.5	44.7	52.66	39.5
50	47.6	49.9	57.83	44.1
55	52.7	55.0	62.84	49.3
60	57.8	60.0	67.69	54.5
65	62.8	65.0	72.38	59.5
70	68.3	70.0	76.91	64.5
75	73.6	75.0	81.27	69.7
80	78.9	80.0	85.46	75.0
85	84.3	85.0	89.46	81.2
90	89.7	90.0	93.25	87.4
95	95	95.0	96.79	93.7
100	100	100	100	100

Table 2.4. Index of refraction of aqueous solutions at 20°C, $n_{20^\circ\text{C}}$, as function of the additive concentration

wt-%	EG	PG	EA	MA	GL	NH3
0	1,333	1,333	1,333	1,333	1,333	1,333
2	1,335	1,335	1,334	1,333	1,335	1,334
5	1,338	1,338	1,336	1,334	1,339	1,335
10	1,343	1,344	1,340	1,335	1,345	1,338
15	1,347	1,349	1,343	1,337	1,351	1,341
20	1,352	1,355	1,347	1,338	1,357	1,344
24	1,356	1,360	1,350	1,339	1,362	1,345
28	1,361	1,364	1,352	1,340	1,368	1,347
32	1,365	1,369	1,355	1,341	1,373	1,348
36	1,369	1,374	1,357	1,342	1,379	1,349
40	1,373	1,378	1,358	1,343	1,384	1,350

wt-%	EG	PG	EA	MA	GL	NH3
44	1,377	1,382	1,360	1,343	1,390	
48	1,381	1,387	1,361	1,343	1,395	
52	1,385	1,391	1,362	1,343	1,401	
56	1,389	1,395	1,363	1,343	1,407	
60	1,393	1,400	1,364	1,343	1,413	

wt-%	K2CO3	CaCl2	MgCl2	NaCl	LiCl
0	1,333	1,333	1,333	1,333	1,333
2	1,337	1,338	1,338	1,337	1,335
5	1,342	1,345	1,346	1,342	1,344
10	1,350	1,358	1,359	1,351	1,354
15	1,359	1,371	1,372	1,359	1,365
20	1,367	1,384	1,386	1,368	1,376
22	1,371	1,390	1,392	1,372	1,380
24	1,374	1,395	1,398	1,376	1,384
26	1,377	1,401	1,403		1,389
28	1,381	1,407	1,409		1,394
30	1,384	1,412	1,415		1,398
32	1,387	1,418			
36	1,394				
40	1,401				

Table 2.5. Thermal volume expansion, TVE , of aqueous solutions with concentrations giving $t_F = -15^\circ\text{C}$ and water.

Fluid symbol	Temperature of fluid[$^\circ\text{C}$]						
	-15	-10	0	10	20	30	40
EG	0,000	0,100	0,347	0,654	1,017	1,434	1,902
PG	0,000	0,120	0,420	0,795	1,237	1,738	2,291
EA	0,000	0,110	0,410	0,808	1,291	1,849	2,469
MA	0,000	0,060	0,238	0,493	0,820	1,218	1,683
GL	0,000	0,119	0,403	0,739	1,115	1,521	1,947
NH3	0,000	0,034	0,179	0,416	0,733	1,118	
K2CO3	0,000	0,100	0,356	0,675	1,042	1,445	1,867
CaCl2	0,000	0,085	0,318	0,612	0,941	1,277	1,618
NaCl	0,000	0,163	0,506	0,872	1,261	1,672	2,106
KAC	0,000	0,113	0,368	0,663	0,995	1,364	1,767
KFO	0,000	0,077	0,287	0,552	0,852	1,167	1,476
LiCl	0,000	0,020	0,116	0,279	0,497	0,760	1,059
H2O			0,000	0,000	0,015	0,165	0,415

Results of measurements of viscosity

CALCIUM CHLORIDE - WATER (Melinder; 1983-1996)

Concentration (wt-%)	28	28	28	28	28
Number of measurem.	2	3	3	3	4
Mean temperature (°C)	-18,9	-9,0	2,0	13,0	24,4
Max temp variation (°C)	0,1	0,1	0,2	0,1	0,1
Density (kg/m ³)	1277	1273	1268	1263	1257
Ball constant	0,0700	0,0700	0,0700	0,0700	0,0700
Ball density (kg/m ³)	2228	2228	2228	2228	2228
Aver falling time (s)	143,17	97,27	68,48	50,40	38,75
Dyn. visc. (mNs/m³)	9,53	6,50	4,60	3,40	2,63
Kin. visc. (mm ² /s)	7,46	5,11	3,63	2,70	2,10
Concentration (wt-%)	28	28	28	28	28
Number of measurem.	3	4	5	5	5
Mean temperature (°C)	-18,6	-9,1	0,4	9,9	20,0
Max temp variation (°C)	0,2	0,1	0,2	0,1	0,1
Density (kg/m ³)	1277	1273	1269	1264	1259
Ball constant	0,0718	0,0718	0,0718	0,0718	0,0718
Ball density (kg/m ³)	2411	2411	2411	2411	2411
Aver falling time (s)	113,87	78,98	58,44	45,67	36,09
Dyn. visc. (mNs/m³)	9,27	6,45	4,79	3,76	2,99
Kin. visc. (mm ² /s)	7,26	5,07	3,78	2,98	2,37
Concentration (wt-%)	22	22	22	22	22
Number of measurem.	5	5	5	5	5
Mean temperature (°C)	-21,4	-15,0	-6,8	-0,6	5,0
Max temp variation (°C)	0,1	0,1	0,1	0,1	0,1
Density (kg/m ³)	1212	1211	1209	1207	1205
Ball constant	0,0718	0,0718	0,0718	0,0718	0,0718
Ball density (kg/m ³)	2411	2411	2411	2411	2411
Aver falling time (s)	82,38	64,45	48,68	40,35	34,77
Dyn. visc. (mNs/m³)	7,09	5,55	4,20	3,49	3,01
Kin. visc. (mm ² /s)	5,85	4,59	3,47	2,89	2,50
Concentration (wt-%)	22	18	18	18	
Number of measurem.	6	1	2	3	
Mean temperature (°C)	10,0	-6,95	-4,50	1,28	
Max temp variation (°C)	0,1	0,1	0,1	0,1	
Density (kg/m ³)	1204	1168	1167	1165	
Ball constant	0,0718	0,0718	0,0718	0,0718	
Ball density (kg/m ³)	2411	2411	2411	2411	
Aver falling time (s)	30,04	39,92	36,89	30,87	
Dyn. visc. (mNs/m³)	2,60	3,56	3,29	2,76	
Kin. visc. (mm ² /s)	2,16	3,05	2,82	2,37	

LITHIUM CHLORIDE - WATER (Melinder, 1996)

Concentration (wt-%)	20	20	20	20	20
Measuring serie	1	2	3	4	5
Number of measurem.	4	4	4	4	4
Mean temperature (°C)	-18,8	-13,4	-5,0	2,3	9,8
Max temp variation (°C)	0,1	0,1	0,1	0,1	0,1
Density (kg/m ³)	1124	1123	1122	1120	1118
Ball constant	0,0700	0,0700	0,0700	0,0700	0,0700
Ball density (kg/m ³)	2228	2228	2228	2228	2228
Aver falling time (s)	92,60	73,85	54,18	42,74	34,57
Dyn. visc. (mNs/m³)	7,16	5,71	4,20	3,31	2,69
Kin. visc. (mm²/s)	6,37	5,09	3,74	2,96	2,40

Concentration (wt-%)	20	20	20
Measuring serie	6	7	8
Number of measurem.	5	3	4
Mean temperature (°C)	15,0	22,3	32,5
Max temp variation (°C)	0,1	0,1	0,1
Density (kg/m ³)	1117	1115	1111
Ball constant	0,0700	0,00918	0,00918
Ball density (kg/m ³)	2228	2228	2228
Aver falling time (s)	30,19	198,45	165,92
Dyn. visc. (mNs/m³)	2,35	2,03	1,70
Kin. visc. (mm²/s)	2,10	1,82	1,53

Concentration (wt-%)	15	15	15	15	15
Measuring serie	1	2	3	4	5
Number of measurem.	4	4	5	5	2
Mean temperature (°C)	-16,8	-13,0	-5,0	5,0	21,9
Max temp variation (°C)	0,1	0,1	0,1	0,1	0,1
Density (kg/m ³)	1092	1091	1090	1089	1085
Ball constant	0,0700	0,0700	0,0700	0,0700	0,00918
Ball density (kg/m ³)	2228	2228	2228	2228	2228
Aver falling time (s)	68,05	57,75	43,11	31,26	153,85
Dyn. visc. (mNs/m³)	5,41	4,60	3,43	2,49	1,62
Kin. visc. (mm²/s)	4,96	4,21	3,15	2,29	1,49

POTASSIUM FORMATE (HYCOOL) (Melinder, 1999)

Concentration (vol-%)	45	45	45	40	40
Density at 20°C (kg/m ³)	1318	1318	1318	1294	1294
Concentration (wt-%)	47	47	47	44	44
Number of measurements	4	4	5	4	4
Mean temperature (°C)	5,8	10,5	20,3	2,8	7,1
Max temp variation (°C)	0,1	0,1	0,1	0,2	0,1
Density (kg/m³)	1326	1323	1318	1303	1301
Ball constant	0,0700	0,0700	0,0700	0,0700	0,0700
Ball density (kg/m ³)	2228	2228	2228	2228	2228
Aver falling time (s)	51,34	45,29	36,27	48,92	43,56
Dyn. visc. (mNs/m³)	3,24	2,87	2,31	3,17	2,83
Kin. visc. (mm²/s)	2,44	2,17	1,75	2,43	2,17

Concentration (vol-%)	40	30	30
Density at 20°C (kg/m ³)	1294	1248	1248
Concentration (wt-%)	44	38	38
Number of measurem.	5	5	5
Mean temperature (°C)	19,4	5,24	11,5
Max temp variation (°C)	0,1	0,1	0,1
Density (kg/m ³)	1294	1255	1252
Ball constant	0,0700	0,0700	0,0700
Ball density (kg/m ³)	2228	2228	2228
Aver falling time (s)	32,60	36,62	31,47
Dyn. visc. (mNs/m³)	2,13	2,49	2,15
Kin. visc. (mm²/s)	1,65	1,99	1,72

APPENDIX PAPERS:

- PAPER 1:** Melinder, Å., 1998, *A Critical Review on Thermophysical Properties of Liquid Secondary Refrigerants*, Natural Working Fluids'98, IIR – Gustav Lorentzen Conf., Oslo
- PAPER 2:** Melinder, Å., 2002, *Enthalpy-phase diagrams of aqueous solutions for ice slurry applications*, Fifth IIR Workshop on Ice Slurries, Stockholm
- PAPER 3:** Melinder, Å., 2003, *Update on indirect systems and secondary fluids with focus on supermarket refrigeration*, Int. Conference on Fluid and Thermal Conversion, FTEC2003, Bali, Indonesia
- PAPER 4:** Melinder, Å., 2003, *Comparing Properties of Aqueous Solutions for Liquid Only and Ice slurry Applications of Indirect Systems*, Proc. 21st Int. Congress of Refrigeration, IIR/IIF, Washington
- PAPER 5:** Melinder, Å., 2004, *Secondary fluids for low operating temperatures*, Sixth Gustav Lorentzen Conference on Natural Working Fluids, IIF/IIR, Glasgow
- PAPER 6:** Melinder, Å., Granryd, E., 2005, *Using property values of aqueous solutions and ice to estimate ice concentrations and enthalpies of ice slurries*, Int. Journal of Refrigeration, Volume 28, Number 1, p.13-19, Paris
- PAPER 7:** Melinder, Å., 2005, *Properties of various secondary fluids for supermarket applications*, Thermophysical Properties and Transfer Processes of Refrigerants, IIF/IIR, Vicenza
- PAPER 8:** Melinder, Å., 2006, *Updating and extending secondary fluid data for various refrigeration and heat pump applications*, GLC 2006, Trondheim
- PAPER 9:** Melinder, Å., 2006, *Equilibrium properties of ice slurry based on two chosen types of aqueous solutions*, 7th Conference on Phase Change Materials and Slurries for Refrigeration and Air Conditioning, 2006, Dinan, France